

Supplemental Material:

Time-reversal inference with uncertain observation channels

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Relation to the main argument

The main paper asks when recorded asymmetry requires an irreversible source. Sections S1–S2 establish why unrestricted observation channels prevent that conclusion, even with exact knowledge of a finite-record law. The calibrated construction in Sec. S3 restricts the detector contribution. Section S4 then turns that population tolerance and the spectral cycle identity into tests by controlling finite-record errors.

The remaining proofs address whether those tests can be informative. Section S5 propagates a reference confidence set through every allowed persistence value before bounding a detection probability. Section S6 isolates the recording cost of dependence when response and source information are already known. Section S7 derives the needed bounds from the physical models and proves the continuous-time sampling ambiguity. The study sections report how coverage, threshold size,

and numerical availability affect the complete decisions. The table locates each main result within these arguments.

Main-text result	Supporting material
Theorem 1: finite-law ambiguity	Section S1, Theorem S1 and Proposition S1; real-delay extension in Sec. S2, Theorem S3.
Calibrated population tolerance	Section S3, including response confidence and supplied covariance information.
Theorems 2–3: finite-record tests	Section S4: Theorem S7 gives the calibrated rule; Theorem S5 gives the cycle rule.
Theorem 5 and the detection example	Section S5, Complete definitions and statements precede Sec. S5.1. Proofs of the hull bound and detection example are in Secs. S5.4 and S5.9–S5.10.
Theorem 4: recording cost	Section S6, Theorems S8 and S9.
Physical examples	Section S7.
Synthetic performance	Sections S8–S11 give design, results, diagnostics, and fixed comparison tables. Section S12 summarizes earlier comparisons that use different designs.

Notation

A covariance envelope bounds the covariance of the complete record in every linear direction. A confidence hull is the smallest closed interval that contains a confidence set. A filter bank is a finite list of common filters declared before observation.

Symbol	Meaning
$C_{ij}(k)$	Observed covariance $\text{Cov}\{Y_i(t+k), Y_j(t)\}$. Source covariances are identified explicitly.
F, S, μ	Observed spectral matrix, source spectral matrix, and source spectral measure.
f_n	Frobenius norm of the two-time reflection coefficient matrix. It differs from the spectral matrix F .
K_B	Cumulative-sum matrix for centered candidate innovations.
$\omega, \mathcal{S}$	Frequency and source spectral ceiling-to-floor ratio.
q, K_i	Standardized full covariance envelope and channel-scaled diagonal envelope entries.
B_{ij}, L	Cycle bias allowance and lag cutoff.
$r, \hat{r}$	True phase tolerance and calibrated upper bound.
V_i, D_i, D_g	Centered recorded variance, its upper confidence bound, and the geometric mean of the two bounds.
ϕ, ψ, a_j	True source persistence, candidate persistence, and fixed filter coefficient.
$[\ell, u], \Lambda(x)$	Source confidence hull and $\Lambda(x) = (1+x)/(1-x)$.
d_{mom}	Dimension of the finite retained-covariance space.
$D(\psi), Q(\psi), R_n(\psi)$	Centered innovation energy, cumulative innovation energy, and their pivot ratio.
N, n, M	Record length, number of innovations or adjacent pairs, and bank size. Each construction gives their exact relation. The delay construction uses M as its integer separation parameter.
η_i, η_H	Per-sample recording and calibration error bounds.
ξ, a, w	Physical channel correlation, rotational drift decay, and rotational frequency coefficient. The main text writes the last two as λ, ν .
$\log, \log_2$	Natural logarithm and base-two logarithm.

The reflection sign convention is $U_t = Y_1(t) + Y_1(t+1)$ and $W_t = Y_2(t) - Y_2(t+1)$. Its uncentered mean is $C_{12}(1) - C_{12}(-1)$. All tests use its absolute value. Spectral densities on the discrete-time torus use $d\omega/(2\pi)$. Continuous-time densities use Lebesgue measure as stated in their construction.

S1 Channel-specific delays for finite Gaussian records

A finite Gaussian law is determined by its retained means and covariances. The construction below matches those quantities exactly with a reversible source and channel-specific delays. The source and its unobserved extension can depend on the target law.

The difficulty is to satisfy reversibility and exact covariance matching at the same time. A real source spectrum has no complex channel phases of its own. Separated delays can approximate the phases of each rank-one spectral mass, but an approximation does not give the same finite law. We therefore apply the approximation to nearby valid covariance arrays that surround the target. Small changes preserve this containment, so a positive mixture of their outputs matches the target exactly. This also permits smoothing the masses into a strictly positive density before taking the final mixture. The notation below specifies the finite set of moments that must be preserved.

Let $m, N \geq 1$ be integers. A retained covariance array consists of real numbers $C_{ij}(k)$, $1 \leq i, j \leq m$, $|k| \leq N-1$, with $C_{ji}(-k) = C_{ij}(k)$. Its block Toeplitz matrix is $\Gamma_N(C) = (C_{ij}(t-u))_{(t,i),(u,j)}$, $0 \leq t, u < N$. The cone $\mathcal{C}_{m,N}$ contains the arrays for which $\Gamma_N(C) \succeq 0$. Use the independent coordinates $C_{ii}(k)$, $0 \leq k < N$, and $C_{ij}(k)$, $i < j$, $|k| < N$. Their real dimension is

$$d_{\text{mom}} = mN + \binom{m}{2}(2N-1) = m^2N - \frac{m(m-1)}{2}. \quad (\text{S1})$$

Throughout, $\|C\|_{\text{coord}}$ is the maximum absolute value of these independent coordinates. Each entry of $\Gamma_N(C)$ is one coordinate, possibly repeated. In particular

$$\|\Gamma_N(C)\|_{\text{op}} \leq mN\|C\|_{\text{coord}}. \quad (\text{S2})$$

Spectral densities are with respect to $\nu(d\omega) = d\omega/(2\pi)$ on $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, with moment convention $C_{ij}(k) = \int e^{ik\omega} \mu_{ij}(d\omega)$. For integer channel delays $d = (d_1, \dots, d_m)$ put

$$D_d(\omega) = \text{diag}(e^{-id_1\omega}, \dots, e^{-id_m\omega}).$$

Thus $Y_i(t) = X_i(t - d_i)$ has spectral measure $D_d \mu D_d^*$ and

$$C_{ij}^Y(k) = C_{ij}^X(k - d_i + d_j). \quad (\text{S3})$$

Theorem S1 (Common hierarchical delays). *Let $C^{(1)}, \dots, C^{(s)} \in \mathcal{C}_{m,N}$ be a nonempty finite collection with $\Gamma_N(C^{(c)}) \succ 0$. Write $\lambda_c = \lambda_{\min}(\Gamma_N(C^{(c)}))$ and $v_c = \max_i C_{ii}^{(c)}(0)$. If $m \geq 2$, every integer $M \geq 2$ satisfying*

$$M-1 > \pi(N+1)(2d_{\text{mom}}-1) \max_{1 \leq c \leq s} \left(\frac{2mNv_c}{\lambda_c} + 1 \right) \quad (\text{S4})$$

admits the following simultaneous realizations with the shared delay vector

$$(d_1, \dots, d_m) = (M, M^2, \dots, M^{m-1}, 0). \quad (\text{S5})$$

For each c there is a real stationary reversible Gaussian process $X^{(c)}$, whose spectral density S_c is a real, even, symmetric matrix trigonometric polynomial, such that $S_c(\omega) \succeq \epsilon I_m$ for all ω and

a common $\epsilon > 0$. The delayed process $Y_i^{(c)}(t) = X_i^{(c)}(t - d_i)$ has retained covariance exactly $C^{(c)}$. The density degrees have a common finite upper bound L , and each source is L -dependent. For every coordinate,

$$\text{Var}(X_i^{(c)}(t)) = C_{ii}^{(c)}(0). \quad (\text{S6})$$

Consequently these are exact realizations of the retained Gaussian laws, with zero added errors. Arbitrary constant mean vectors are preserved by assigning the same mean vectors to the sources. Each delay response has unit coefficient norm.

For $m = 1$ the same conclusion holds with $d_1 = 0$, without a restriction on M . For $N = 1$ it holds for every prescribed vector of nonnegative integer delays, without condition (S4).

Proof. We first reduce the retained law to finitely many rank-one spectral masses. We then move their frequencies so the prescribed delays approximate their phases. Finally, convex mixtures correct the retained moments without a signed spectral perturbation that could destroy positivity.

Finite spectral representation. Every $C \in \mathcal{C}_{m,N}$ has a finite positive semidefinite matrix spectral measure obeying $\mu(-A) = \overline{\mu(A)}$ and the prescribed retained moments. To see this, take a real Gram representation $C_{ij}(t - u) = \langle v_{t,i}, v_{u,j} \rangle$ in a real space E of dimension $\text{rank } \Gamma_N(C)$. The map $v_{t,i} \mapsto v_{t+1,i}$ on the span with $0 \leq t < N - 1$ is well defined and isometric: Toeplitz equality preserves the squared norm of every linear combination, including a zero combination. The domain and range have equal dimension. Thus extend this map to a real orthogonal map $U : E \rightarrow E$. When $N = 1$, one may simply take $U = I_E$. Then $v_{t,i} = U^t v_{0,i}$ for the retained times. The spectral decomposition of the complexification of U supplies the measure. More explicitly, with a complex inner product linear in its first argument, let P_ω be the eigenspace projection for $e^{i\omega}$ and set $W_{\omega,ij} = \langle P_\omega v_{0,i}, P_\omega v_{0,j} \rangle$. These are Hermitian positive semidefinite matrices and $C_{ij}(k) = \sum_\omega e^{ik\omega} W_{\omega,ij}$. Realness of U pairs conjugate eigenspaces. The endpoint matrices at 0 and π are real. There are at most $\dim E \leq mN$ distinct nodes.

Equivalently, write the real moments as a finite sum

$$C_{ij}(k) = \sum_\ell \text{Re}(e^{ik\omega_\ell} v_{\ell,i} \overline{v_{\ell,j}}). \quad (\text{S7})$$

Here a conjugate pair of matrix masses has first been combined into its half-circle mass and then split into rank-one matrices; endpoint masses can be split into real rank-one matrices. Each term in (S7) represents the full-circle measure $(vv^* \delta_\omega + \overline{v}v^T \delta_{-\omega})/2$. In particular,

$$\sum_\ell |v_{\ell,i}|^2 = C_{ii}(0), \quad \sum_\ell |v_{\ell,i}| |v_{\ell,j}| \leq \sqrt{C_{ii}(0) C_{jj}(0)}. \quad (\text{S8})$$

Simultaneous phase approximation. The retained moments change little under a small frequency displacement. The hierarchical delays make such a displacement sufficient to set one channel phase at a time, with decreasing effects on phases already set. Fix $m \geq 2$, $M \geq 2$, an arbitrary real representative ω of a frequency, and phases $p_1, \dots, p_{m-1}$ on the unit circle. Starting with $\theta_0 = \omega$, choose θ_j nearest to θ_{j-1} among the real solutions of $e^{-iM^j \theta_j} = p_j$. Their spacing is $2\pi/M^j$, so $|\theta_j - \theta_{j-1}| \leq \pi/M^j$. Put $\theta = \theta_{m-1}$. Then

$$|\theta - \omega| \leq \sum_{j=1}^{m-1} \frac{\pi}{M^j} \leq \frac{\pi}{M-1}, \quad (\text{S9})$$

$$|e^{-iM^i \theta} - p_i| \leq M^i |\theta - \theta_i| \leq \pi \sum_{r=1}^{m-1-i} M^{-r} \leq \frac{\pi}{M-1}, \quad 1 \leq i < m. \quad (\text{S10})$$

The sum in the last line is empty for $i = m - 1$, whose phase is exact. Frequencies are reduced modulo 2π only after this construction; no restriction to a half-circle is imposed on θ .

Apply this construction separately to each vector v in (S7). Multiplying v by a global unit complex number does not change vv^* . If $v_m \neq 0$, use this freedom to make v_m real nonnegative. If $v_m = 0$, use any global phase. Set $a_i = |v_i|$ and write $v_i = p_i a_i$, taking $p_m = 1$ and assigning any unit phase to a zero coordinate. Thus $v = \text{diag}(p)a$ with $a \in [0, \infty)^m$ even when some coordinates vanish. Give the source the real-even measure

$$\frac{1}{2}aa^\top\delta_\theta + \frac{1}{2}aa^\top\delta_{-\theta}.$$

Its delayed moment contribution is $\text{Re}(e^{ik\theta} q_i \bar{q}_j) a_i a_j$, where $q_i = e^{-id_i\theta}$ and $q_m = 1$. At a coincident reflected pair ($\theta = 0$ or π modulo 2π), the two half masses add to one mass, and every q_i is real because d_i is integral. The output mass there is therefore real and admissible. At every other frequency the reflected output mass is the conjugate of the first one. These observations also cover a frequency displaced across an endpoint.

For $|k| \leq N - 1$, equations (S9)–(S10) give

$$\begin{aligned} |e^{ik\theta} q_i \bar{q}_j - e^{ik\omega} p_i \bar{p}_j| &\leq |k||\theta - \omega| + |q_i - p_i| + |q_j - p_j| \\ &\leq \frac{\pi(N+1)}{M-1}. \end{aligned}$$

Combining all the real source atoms gives an output $\tilde{C}$ with

$$\|\tilde{C} - C\|_{\text{coord}} \leq \frac{\pi(N+1)V}{M-1}, \quad \tilde{C}_{ii}(0) = C_{ii}(0), \quad V = \max_i C_{ii}(0), \quad (\text{S11})$$

by (S8). The same delay vector works for every such array. Splitting into rank-one terms before moving frequencies is essential; no simultaneous diagonal dephasing of a general matrix mass has been assumed.

A simplex with an explicit perturbation margin. The approximate output need not equal the target. To correct this error while retaining positive source measures, we approximate several arrays around the target. A simplex is their convex hull. The margin below ensures that the target stays inside that hull after the approximation. For a target $C^{(c)}$, use the natural coordinate basis $E_1, \dots, E_{d_{\text{mom}}}$ and write

$$a_c = \frac{\lambda_c}{2mN}, \quad B_c = v_c + a_c, \quad V_{c,0} = C^{(c)} - a_c \sum_{j=1}^{d_{\text{mom}}} E_j, \quad V_{c,j} = C^{(c)} + a_c E_j \quad (1 \leq j \leq d_{\text{mom}}).$$

Every matrix perturbation has operator norm at most $mNa_c = \lambda_c/2$ by (S2); hence all vertices are positive definite. Their maximal coordinate variances are at most B_c . This simplex contains the coordinate cube with center $C^{(c)}$ and radius

$$\rho_c = \frac{a_c}{2d_{\text{mom}} - 1}. \quad (\text{S12})$$

Indeed the barycentric coordinates of displacement x are

$$\alpha_0 = \frac{1 - \sum_j x_j/a_c}{d_{\text{mom}} + 1}, \quad \alpha_j = \alpha_0 + x_j/a_c.$$

For $\|x\|_\infty \leq \rho_c$, the numerator of α_0 is at least $1 - d_{\text{mom}}/(2d_{\text{mom}} - 1) \geq 0$, and the numerator of α_j is at least $1 - (2d_{\text{mom}} - 1)/(2d_{\text{mom}} - 1) = 0$. These inequalities also hold for $d_{\text{mom}} = 1$.

If each vertex of this simplex moves by at most $\eta < \rho_c$ in coordinate norm, its new convex hull contains the cube of radius $\rho_c - \eta$ about $C^{(c)}$. For each linear functional u , the hull's support function drops by at most $\eta\|u\|_1$. The original cube provides support $u \cdot C^{(c)} + \rho_c\|u\|_1$. The support-function characterization of containment proves the claim. In particular the perturbed hull is full dimensional. Since it has $d_{\text{mom}} + 1$ vertices, they are affinely independent and $C^{(c)}$ has strictly positive barycentric coordinates.

Apply (S11) to the vertices. The coordinate errors are at most $\eta_c = \pi(N + 1)B_c/(M - 1)$. Condition (S4) is exactly $\eta_c < \rho_c$ for each c . Thus every target belongs to the strict interior of a simplex of outputs of finite reversible source measures under the prescribed delay vector. The strict inequality leaves a positive margin for the next step. Without that margin, smoothing could move the output vertices so that their convex hull no longer contains the target.

Positive polynomial source densities at fixed delays. The convex hull currently contains outputs of atomic source measures. Smoothing gives polynomial densities and adding white power makes them strictly positive. Both operations change the moments, so they must fit inside the remaining containment margin before the exact mixture is chosen. Now hold M and the delay vector fixed. Denote the source measure at a perturbed vertex by $\mu_{c,j}$. For $L \geq 0$ let

$$F_L(\omega) = \frac{1}{L+1} \left| \sum_{r=0}^L e^{ir\omega} \right|^2 = \sum_{|r| \leq L} \left(1 - \frac{|r|}{L+1} \right) e^{ir\omega}.$$

This is a real, even, nonnegative trigonometric polynomial of integral one against ν . For $\epsilon > 0$ define

$$S_{c,j}^{L,\epsilon}(\omega) = \int F_L(\omega - \theta) \mu_{c,j}(d\theta) + \epsilon I_m. \quad (\text{S13})$$

These are real-even symmetric polynomial densities bounded below by ϵI_m . For each fixed integer r , convolution multiplies the r th source moment by $1 - |r|/(L+1)$ when $L \geq |r|$. Added white power changes only the zero-lag diagonal moments. By (S3), only source lags of absolute value at most

$$Q = N - 1 + M^{m-1} \quad (\text{S14})$$

enter the retained output moments. Thus the smoothed output vertices converge to their unsmoothed counterparts as $L \rightarrow \infty$ and $\epsilon \rightarrow 0$. There are finitely many vertices, so finite common L and positive common ϵ preserve all strict simplex containments. With their resulting positive barycentric coefficients $\beta_{c,j}$, put $S_c = \sum_{j=0}^{d_{\text{mom}}} \beta_{c,j} S_{c,j}^{L,\epsilon}$. The coefficients sum to one, so $S_c \succeq \epsilon I_m$, and linearity of the output moment map gives exactly $C^{(c)}$.

These matrix densities define stationary real Gaussian processes. Their source covariance matrices are real symmetric and even in the lag, which makes their Gaussian finite-dimensional distributions invariant under time reversal. Polynomial degree at most L makes all source covariances at $|k| > L$ vanish. Joint Gaussianity then gives independence for any finite sets of source variables separated in time by more than L ; a monotone-class argument gives independence of the corresponding generated sigma fields. Thus the sources are L -dependent. At every frequency, D_d is diagonal unitary and hence leaves the diagonal spectral measures unchanged. Exact matching of the retained zero-lag diagonal moments therefore proves (S6).

Scalar records and records of length one. For $m = 1$, the finite spectral representation has an even scalar measure and already gives a reversible source at every simplex vertex without a phase approximation or delay. The same smoothing and simplex correction complete the argument. For $N = 1$ and any fixed prescribed integer delay vector, represent each positive definite simplex vertex by its matrix mass at frequency zero. All delay factors are one there, so the vertex is already represented exactly by a real-even source measure. Holding the delays fixed, smoothing

and simplex correction again give the claimed conclusion. For zero delays alone, white sources suffice directly. $\square$

Proposition S1 (Uniform bounds at a fixed chosen delay). *Fix $m \geq 2$, $N \geq 1$, $\lambda_* > 0$ and $v_* < \infty$. The preceding conclusion applies to every target satisfying $\Gamma_N(C) \succeq \lambda_* I_{mN}$ and $\max_i C_{ii}(0) \leq v_*$, using a common delay parameter M whenever*

$$M \geq 2, \quad M - 1 > \pi(N + 1)(2d_{\text{mom}} - 1) \left(\frac{2mNv_*}{\lambda_*} + 1 \right).$$

The targets may be an infinite family; their sources may differ. In particular this applies uniformly to every compact subset of the positive definite moment cone. For each fixed such M , the sources can also have common finite polynomial degree and a common positive spectral lower bound. One sufficient explicit choice is as follows:

$$a = \frac{\lambda_*}{2mN}, \quad B = v_* + a, \quad \rho = \frac{a}{2d_{\text{mom}} - 1}, \quad \eta = \frac{\pi(N + 1)B}{M - 1}, \quad b = \rho - \eta > 0, \\ Q = N - 1 + M^{m-1}, \quad \epsilon = \frac{b}{3}, \quad L \geq Q, \quad L + 1 > \frac{3QB}{b}.$$

Every source obtained by the construction then obeys $S_C(\omega) \succeq \epsilon I_m$ and has degree at most L . These are sufficient bounds, not minimum resource requirements; they depend on the fixed chosen M .

Proof. Use the same a, B, ρ for the simplex about every target. The atomic approximation preserves each source diagonal mass, so the source measure at every vertex has coordinate variances at most B . For any integer r , its matrix moment satisfies

$$|\hat{\mu}_{ij}(r)| \leq \sqrt{\hat{\mu}_{ii}(0)\hat{\mu}_{jj}(0)} \leq B.$$

For the finite real source measures constructed above, this follows directly by summing the absolute values of their rank-one contributions and applying Cauchy–Schwarz. At a required source lag $|r| \leq Q$, Fejér smoothing therefore changes a moment by at most $QB/(L + 1)$. The added white density changes each output coordinate by at most ϵ ; for diagonal delays it affects only a zero-lag diagonal coordinate. The total smoothed vertex displacement from the original simplex is consequently at most

$$\eta + \frac{QB}{L + 1} + \epsilon < \eta + \frac{b}{3} + \frac{b}{3} < \rho.$$

The simplex argument applies separately to every target with the same M, L, ϵ . Compactness is not otherwise needed: a compact subset of the positive definite cone supplies the displayed uniform eigenvalue and variance bounds by continuity. $\square$

Scope. The theorem concerns exact finite Gaussian record laws. It permits the source law to vary with the target and with the selected delays. It does not assert equality to a prescribed full-process law, a smallest delay, a source range independent of M , or equality of non-Gaussian laws from covariance matching. Its maximal constructed delay M^{m-1} is a sufficient choice. Polynomial source spectra make every covariance vanish beyond a finite lag.

S1.1 Limits of the finite-record construction

The rank-one decomposition cannot in general be replaced by dephasing a whole matrix mass. For example, with $r = 2/5$ the Hermitian matrix

$$W = \begin{pmatrix} 1 & r & -ir \\ r & 1 & r \\ ir & r & 1 \end{pmatrix}$$

has least eigenvalue at least $1 - 2r = 1/5$. Its cycle product $W_{12}W_{23}W_{31} = ir^3$ is invariant under diagonal unitary congruence. A real symmetric matrix has a real cycle product, so no such congruence makes W real. Moving its rank-one pieces separately avoids this obstruction.

Strict positive definiteness of the retained target is necessary for the asserted positive source floor. If $S_X \succeq \epsilon I_m$, then $D_d S_X D_d^* \succeq \epsilon I_m$. For any block vector u , integration of its Fourier polynomial and Parseval's identity give $u^\top \Gamma_N(C^Y) u \geq \epsilon \|u\|_2^2$. A singular retained covariance therefore cannot have this type of realization. This does not exclude realizations with singular spectra.

A prescribed full-process spectrum imposes further restrictions. Consider a bivariate spectrum with unit diagonal and cross entry

$$f(\omega) = \frac{1}{4} \left(1 + \frac{i}{2} \sin \omega \right).$$

It has the real-process conjugation symmetry and is uniformly positive, since $|f(\omega)| \leq \sqrt{5}/8 < 1$. A reversible-source explanation by integer delays would require $e^{i\Delta\omega} f(\omega)$ to be real, where $\Delta = d_1 - d_2 \in \mathbb{Z}$. The derivative of its imaginary part at zero is $(\Delta + 1/2)/4$, which never vanishes for integer Δ . Hence this full-process law has no such explanation, although every finite retained covariance is positive definite and satisfies Theorem S1. The source and the unobserved output extension may depend on the chosen finite record.

The source-degree bound in Proposition S1 also must depend on the selected delay. For $m \geq 2$, fix a temporally white positive definite target whose contemporaneous covariance is the identity except for $C_{1m}(0) = C_{m1}(0) = r$, with $0 < |r| < 1$. Under the hierarchical delays, exactness requires $C_{1m}^X(-M) = r$. A polynomial source spectrum of degree L has zero moments at all lags of magnitude greater than L , so necessarily $L \geq M$. Thus no bound independent of M is possible, even for a singleton compact family of targets. The bounds uniform over targets in Proposition S1 hold at each fixed M .

S1.2 A linear-delay construction under an observable covariance margin

A stronger condition on the retained covariances permits an explicit construction with linearly spaced delays. The comparison matrix below combines lower bounds for the auto spectra with absolute bounds for the cross spectra. Its positive definiteness is sufficient to keep the constructed source spectrum positive. This gives a direct construction for a smaller covariance class than the universal theorem.

For each condition c , put $H = N - 1$, $v_{ci} = C_{ii}^{(c)}(0)$ and define

$$a_{ci} = v_{ci} - 2 \sum_{k=1}^H |C_{ii}^{(c)}(k)|, \quad (B_c)_{ij} = 2 \sum_{k=-H}^H |C_{ij}^{(c)}(k)| \quad (i \neq j),$$

with $(B_c)_{ii} = 0$. Let

$$Q_c = \text{diag}(a_{c1}, \dots, a_{cm}) - B_c.$$

Assume $N, m \geq 2$ and $Q_c \succ 0$ in every condition. This sufficient condition uses only retained covariances. Positive definiteness of the observed record alone does not imply this condition. The simpler inequalities $a_{ci} > \sum_{j \neq i} (B_c)_{ij}$ imply $Q_c \succ 0$ by strict diagonal dominance.

Theorem S2 (Direct positive spectra with linearly spaced delays). *For any nonempty finite family satisfying the displayed condition, the common channel delays*

$$d_i = (m - i)(N - 1), \quad 1 \leq i \leq m,$$

realize all retained laws exactly with zero errors and reversible Gaussian sources whose real symmetric even trigonometric polynomial spectral densities have degree at most $m(N-1)$ and are bounded below by γI_m , where $\gamma = \min_c \lambda_{\min}(Q_c) > 0$. Source coordinate variances equal the observed variances. In particular the maximum delay is $(m-1)(N-1)$, linear in the retained horizon at fixed channel count.

Proof. It suffices to construct one condition; omit its superscript. Put $H = N-1$ and $\Delta_{ij} = d_i - d_j = (j-i)H$ for $i < j$. Define

$$S_{ii}(\omega) = v_i + 2 \sum_{k=1}^H C_{ii}(k) \cos(k\omega).$$

For $i < j$, the numbers $n_k = \Delta_{ij} - k$, $-H \leq k \leq H$, are distinct nonnegative integers. Define

$$S_{ij}(\omega) = S_{ji}(\omega) = \sum_{k=-H}^H (2 - \mathbf{1}_{n_k=0}) C_{ij}(k) \cos(n_k \omega).$$

Fourier orthogonality against $d\omega/(2\pi)$ gives source moment $R_{ij}(k - \Delta_{ij}) = C_{ij}(k)$ for each retained k . The coefficient one at frequency zero is required because its integral is one, whereas $\cos^2(n\omega)$ integrates to $1/2$ for $n > 0$. The diagonal source moments also match the observed auto moments, which pure delays preserve. Symmetry and stationarity give the corresponding j, i output entries.

Pointwise, $S_{ii}(\omega) \geq a_i$ and $|S_{ij}(\omega)| \leq B_{ij}$. For every real vector u , write $|u|$ for its component-wise absolute value. Then

$$u^\top S(\omega) u \geq \sum_i a_i u_i^2 - \sum_{i \neq j} B_{ij} |u_i| |u_j| = |u|^\top Q |u| \geq \lambda_{\min}(Q) \|u\|_2^2.$$

This positive real-even density defines a stationary reversible Gaussian source. Its degree is at most $\max_{i < j} (\Delta_{ij} + H) = mH$. Thus its covariances vanish beyond mH , and its Gaussian process is mH -dependent. All zero-lag diagonal moments equal v_i . Apply this construction independently to each condition, preserving the same delays and taking the minimum floor. $\square$

The comparison-matrix condition is unchanged by nonzero rescaling of individual channels. If D is the diagonal scaling matrix, the transformed comparison matrix is $|D|Q_c|D|$, so positive definiteness is equivalent. This condition can hold when a row-dominance inequality fails. For example, white auto-covariances with variances $(1, 9)$ and cross lags $(C_{12}(-1), C_{12}(0), C_{12}(1)) = (0.02, 1, 0.01)$ give $Q = \begin{pmatrix} 1 & -2.06 \\ -2.06 & 9 \end{pmatrix} \succ 0$, but its first row is not diagonally dominant. The explicit construction then also proves validity of this retained covariance law. The result concerns this specified sufficient class; the universal theorem also covers targets outside it.

Corollary S1 (Exact minimum delay span under the comparison-matrix condition). *In addition to $Q_c \succ 0$ in every condition, suppose that for every channel pair $i < j$ and every integer $|\Delta| \leq N-2$, some condition c has*

$$C_{ij}^{(c)}(\Delta - 1) \neq C_{ij}^{(c)}(\Delta + 1).$$

Then the minimum span $\max_i d_i - \min_i d_i$ among pure-delay explanations that use one vector across conditions is exactly $(m-1)(N-1)$. The same minimum is attained by zero-error strictly positive polynomial sources. For $m = 2$, the minimum absolute relative delay is exactly $N-1$. The additional inequalities hold outside a finite union of proper linear subspaces of the retained-moment coordinate space; thus they hold for Lebesgue-almost-every tuple in this open comparison-matrix class.

Proof. A compatible vector cannot have $|d_i - d_j| \leq N - 2$, since the specified condition would violate its retained reflection equation. Hence every pair of delay values is separated by at least $N - 1$. Sort the m delays and add the $m - 1$ adjacent gaps to obtain span at least $(m - 1)(N - 1)$. The explicit linearly spaced construction attains it. Each exceptional case fixes a channel pair and relative delay and requires the same nontrivial linear equality in all conditions. There are finitely many such cases. Their union has Lebesgue measure zero in the independent retained coordinates. The positive-definiteness conditions define an open set, so restricting to it preserves this almost-everywhere statement. This is a statement about this specified covariance class and coordinate measure, not a frequency claim about vector autoregressive (VAR) models, physical processes, or the supplied numerical sample. $\square$

S2 Fixed real skew and finite-law nonidentifiability

Large adjustable integer delays are not the only source of finite-law ambiguity. With unrestricted continuous-time frequencies, prescribed real delays can also give exact representations under the independence condition below. Frequencies that alias to the same sampled frequency supply different channel phases. The delay vector stays fixed; the source spectrum changes with the target law. Sample spacing is one. For a real stationary m -channel Gaussian record at times $0, \dots, N - 1$, write

$$C_{ij}(k) = \text{Cov}(Y_i(t + k), Y_j(t)), \quad \Gamma_N(C) = (C_{ij}(t - u))_{(t,i),(u,j)}.$$

The independent real coordinates are $C_{ii}(k)$ for $0 \leq k < N$ and $C_{ij}(k)$ for $i < j$, $|k| < N$; their number is $d_{\text{mom}} = mN + \binom{m}{2}(2N - 1)$. Source spectra below are integrated against ordinary Lebesgue measure $d\omega$ on $\mathbb{R}$, with covariance convention $R(t) = \int_{\mathbb{R}} e^{it\omega} S(\omega) d\omega$. A real symmetric even matrix spectrum gives a reversible real Gaussian source, with the same reversal parity for all channels.

Theorem S3 (Fixed incommensurate skew). *Let $m \geq 2$, $N \geq 1$, and fix $d = (d_1, \dots, d_{m-1}, 0) \in \mathbb{R}^m$ such that $1, d_1, \dots, d_{m-1}$ are linearly independent over $\mathbb{Q}$. For every nonempty finite collection $C^{(1)}, \dots, C^{(s)}$ with $\Gamma_N(C^{(c)}) \succ 0$, there are stationary continuous-time reversible Gaussian sources $X^{(c)}$ such that*

$$Y_i^{(c)}(k) = X_i^{(c)}(k - d_i), \quad 0 \leq k < N,$$

has exactly the prescribed retained law in each condition, with zero added errors. Each source has an integrable real symmetric even matrix spectral density $S_c(\omega) \succ 0$ at every $\omega \in \mathbb{R}$, continuous covariance, and a common finite dependence range. Each coordinate variance is exactly $C_{ii}^{(c)}(0)$. No positive uniform spectral floor on $\mathbb{R}$ and no bounded frequency support is asserted. The prescribed delay vector is the same for every condition and is not selected anew for each target.

Lemma S1 (Dense alias phases). *Under the theorem's independence hypothesis, for every $\vartheta \in \mathbb{R}$ the sequence*

$$\left(e^{-i(\vartheta + 2\pi n)d_1}, \dots, e^{-i(\vartheta + 2\pi n)d_{m-1}} \right), \quad n = 1, 2, \dots,$$

is dense in the product of $m - 1$ unit circles.

Proof. For each nonzero $h \in \mathbb{Z}^{m-1}$, rational independence implies $h \cdot d \notin \mathbb{Z}$. The geometric-series formula gives

$$\frac{1}{Q} \sum_{n=1}^Q e^{-2\pi i n h \cdot d} \longrightarrow 0.$$

The same averages for any trigonometric polynomial therefore converge to its torus integral. Uniform approximation of continuous functions by trigonometric polynomials gives the statement for

continuous functions. A nonnegative continuous function supported in an arbitrary nonempty open set, with positive integral, must be positive at some orbit point. The orbit is dense, and multiplying its coordinates by $e^{-i\vartheta d_j}$ preserves density. This is the classical Kronecker–Weyl argument [1]. $\square$

Proof of the theorem. The integer-time moments cannot distinguish frequencies separated by a sampling alias. The fixed real delays can distinguish their channel phases. We use this freedom to approximate each spectral mass, then use the same convex-containment argument as for integer delays to obtain an exact law. The continuous-time source also needs an integrable density with finite covariance range, which requires a different smoothing kernel. *Finite atomic representation.* Every real PSD stationary retained array has a finite spectral representation

$$C_{ij}(k) = \sum_{\ell} \operatorname{Re}(e^{ik\vartheta_{\ell}} v_{\ell,i} \overline{v_{\ell,j}}), \quad 0 \leq \vartheta_{\ell} \leq \pi. \quad (\text{S15})$$

Here the masses at $0, \pi$ may be split with real vectors. To recall a construction, represent the retained Toeplitz matrix by real Gram vectors $g_{t,i}$. The shift $g_{t,i} \mapsto g_{t+1,i}$ on the span of the first $N-1$ blocks is well-defined and isometric by the Toeplitz identities. Extend it to a real orthogonal operator on the whole finite Gram space. Its complex spectral decomposition yields conjugate PSD masses at opposite frequencies, with real endpoint masses. Combining each conjugate pair and splitting each PSD mass into rank-one terms gives (S15). For $N=1$ the identity operator suffices. In particular,

$$\sum_{\ell} |v_{\ell,i}|^2 = C_{ii}(0), \quad \sum_{\ell} |v_{\ell,i}| |v_{\ell,j}| \leq \sqrt{C_{ii}(0) C_{jj}(0)}. \quad (\text{S16})$$

Approximation with the fixed skew. For each vector v in (S15), choose its global phase so that v_m is real nonnegative when nonzero. Write $v_i = a_i p_i$, $a_i = |v_i|$, $|p_i| = 1$, and take $p_m = 1$; zero coordinates may be assigned any phase. For any $\delta > 0$, the lemma supplies an integer $n \geq 1$ such that, with $\omega = \vartheta + 2\pi n$, $q_i = e^{-i\omega d_i}$ satisfies $|q_i - p_i| < \delta$ for $i < m$; $q_m = p_m = 1$. Use the source measure

$$\frac{1}{2} a a^{\top} \delta_{\omega} + \frac{1}{2} a a^{\top} \delta_{-\omega}.$$

Its delayed integer-lag contribution is

$$a_i a_j \cos((k - d_i + d_j)\omega) = a_i a_j \operatorname{Re}(e^{ik\vartheta} q_i \overline{q_j}),$$

because $e^{2\pi i k n} = 1$ for every integer k . The difference from the desired term is at most $2\delta a_i a_j$. Summing and using (S16) gives coordinate error at most $2\delta \max_i C_{ii}(0)$. Diagonal source masses, and hence all coordinate variances, are unchanged. This is valid also for discrete endpoint frequencies $0, \pi$: their positive aliases are ordinary real frequencies, whose reflected partners are distinct because $n \geq 1$. Thus every retained positive-semidefinite covariance array can be approximated arbitrarily closely by a finite real-even source measure under this one fixed d . The approximation preserves all coordinate variances. The next step turns this approximation property into exact matching for positive-definite targets.

An interior simplex and its stability. Density of the approximate outputs does not by itself give exact equality. Positive definiteness supplies nearby valid arrays around the target. Approximating these arrays preserves a convex representation of the target with positive weights. Fix a positive-definite target C . Let $E_1, \dots, E_{d_{\text{mom}}}$ be the independent coordinate basis, and choose $a > 0$ sufficiently small that

$$V_0 = C - a \sum_{j=1}^{d_{\text{mom}}} E_j, \quad V_j = C + a E_j \quad (1 \leq j \leq d_{\text{mom}})$$

all have positive-definite retained matrices. This is possible since positive definiteness is open. They are affinely independent, and their barycenter is C with all weights $1/(d_{\text{mom}}+1)$. Barycentric coordinates are continuous functions of the vertices while the associated affine matrix is invertible. Consequently a positive number τ exists such that moving every vertex by less than τ in coordinate norm preserves both affine independence and strictly positive barycentric coordinates of C . For finitely many conditions, choose such a simplex and tolerance for each condition. Then use the smallest tolerance. Approximate all their vertices as above, within one third of that tolerance. There are finitely many resulting atomic source measures $\mu_{c,j}$.

Integrable spectra, strict positivity, and finite dependence. We now replace the atomic spectra without losing that convex representation. The smoothing kernel below has a compactly supported covariance. A second kernel with different zero locations supplies strict spectral positivity without a constant floor, which would not be integrable on the real line. For $L > 0$ define

$$\kappa_L(\omega) = \frac{L}{2\pi} \left(\frac{\sin(L\omega/2)}{L\omega/2} \right)^2, \quad h_L(t) = (1 - |t|/L)_+.$$

The value at zero is defined by continuity. The triangular function is L^{-1} times the convolution of the indicators of $[-L/2, L/2]$ with itself. The Fourier convolution identity therefore shows that $\kappa_L \geq 0$, $\int \kappa_L = 1$, and $\int e^{it\omega} \kappa_L(\omega) d\omega = h_L(t)$ under the stated convention. Put

$$g_L = \frac{1}{2}(\kappa_L + \kappa_{\sqrt{2}L}), \quad S_{c,j}^{L,\varepsilon}(\omega) = \int \kappa_L(\omega - u) \mu_{c,j}(du) + \varepsilon g_L(\omega) I_m.$$

The nonzero zeros of the two kernels are $2\pi n/L$ and $2\pi n/(\sqrt{2}L)$ for nonzero integers n ; these sets are disjoint. Both kernels are positive at zero, so $g_L(\omega) > 0$ everywhere. Thus each displayed density is integrable, real symmetric, even, and strictly positive definite at every frequency.

Its covariance is

$$R_{c,j}^{L,\varepsilon}(t) = h_L(t) R_{c,j}(t) + \frac{\varepsilon}{2} \{h_L(t) + h_{\sqrt{2}L}(t)\} I_m.$$

This is continuous and zero for $|t| \geq \sqrt{2}L$. Only the finitely many real arguments $k - d_i + d_j$, $|k| < N$, enter the retained outputs. At each such argument $h_L(t) \rightarrow 1$ as $L \rightarrow \infty$, and the added diagonal covariance has magnitude at most ε . There are only finitely many source measures. Thus a common sufficiently large L and sufficiently small $\varepsilon > 0$ make every regularized output differ from its atomic output by less than one third of the chosen tolerance. This bound is on moments, not on the locations of the carrier frequencies.

Exact matching and the Gaussian realization. Each target remains strictly inside the simplex of its regularized output vertices. Let $\beta_{c,j} > 0$ be its barycentric weights, summing to one, and set $S_c = \sum_j \beta_{c,j} S_{c,j}^{L,\varepsilon}$. Linearity gives exactly the target retained moments. Each S_c is integrable, real symmetric and even, and satisfies $S_c(\omega) \succeq \varepsilon g_L(\omega) I_m \succ 0$. The covariance kernel defines a real stationary Gaussian process (by its PSD finite-dimensional covariance matrices and consistency). Its even symmetric cross covariances imply reversibility. Its covariance vanishes at distances at least $\sqrt{2}L$, so Gaussian finite-dimensional vectors on time sets separated by more than this range are independent. The same holds for their generated sigma fields, by the monotone-class extension from finite cylinder events. Thus the sources share a finite dependence range and have continuous covariance. Pure delays preserve source coordinate variances. Exact matching at zero lag gives the asserted variances, and Gaussianity gives equality of retained laws. Constant channel means can be copied to the sources. $\square$

Corollary S2 (Open calibration sets and independent replication). *Every nonempty open subset of $\mathbb{R}^{m-1}$ contains a fixed relative-delay vector for which the preceding theorem holds. Therefore the*

union model over that set contains every nonsingular finite stationary Gaussian record-law tuple, even with continuous source covariance, finite source dependence and everywhere positive integrable spectra. Every test with size at most α uniformly over this model has power at most α at every such tuple under arbitrary finite independent replication of the retained records.

Proof. Vectors violating rational independence belong to the countable union of proper affine hyperplanes $h_0 + \sum_{i=1}^{m-1} h_i d_i = 0$, with integer coefficients and $(h_1, \dots, h_{m-1}) \neq 0$. This union has Lebesgue measure zero, so it cannot cover a nonempty open set. Choose an admissible fixed vector and apply the theorem. Independent products of equal retained Gaussian laws remain equal, which gives the testing consequence. $\square$

For nonnegative delays with a fixed zero reference, apply this corollary to a nonempty open set contained in the positive orthant. If arbitrary signed relative delays are allowed, a common shift makes all absolute delays nonnegative without changing any stationary law, but the designated reference need no longer have absolute delay zero. No claim is made for calibration sets confined to lower-dimensional rational-relation sets. Even an exactly known skew vector satisfying the theorem is universal: uncertainty is not essential to its aliasing mechanism.

Proposition S2 (A rational-delay obstruction). *Not every fractional skew is universal. At relative delay $1/2$, reversibility requires $C_{12}(0) = C_{12}(1)$. A stationary output with white unit auto covariances and $C_{12}(k) = r \mathbf{1}_{k=0}$, $0 < r < 1$, violates this equality and has positive-definite records at every finite horizon. Thus its two-time law cannot be explained at the exact half-sample delay, even with orthogonal scalar errors.*

Proof. The two cross covariances would be $R_{12}(-1/2)$ and $R_{12}(1/2)$, which coincide. The specified output is an independent sequence of bivariate Gaussian vectors with covariance $\begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix} \succ 0$. Orthogonal scalar errors contribute no cross covariance. $\square$

Proposition S3 (A supplied spectral envelope controls skew error). *Let μ be a finite real-even PSD source spectral measure, and let $v_i = \mu_{ii}(\mathbb{R})$. Suppose the true relative delay is $\Delta = \Delta_0 + e$. For any real lag k , put $F = C_{ij}(k) - C_{ij}(2\Delta_0 - k)$. If $b_i(\Omega) = \mu_{ii}(\{|\omega| > \Omega\})$, then*

$$|F| \leq 2 \min(1, \Omega|e|) \sqrt{v_i v_j} + 2 \sqrt{b_i(\Omega) b_j(\Omega)}.$$

In particular, a source supported in $[-\Omega, \Omega]$ obeys $|F| \leq 2\Omega|e| \sqrt{v_i v_j}$.

Proof. Evenness and the cosine difference identity give

$$F = 2 \int \sin((k - \Delta_0)\omega) \sin(e\omega) \mu_{ij}(d\omega).$$

On $|\omega| \leq \Omega$ the absolute integrand multiplier is at most $2 \min(1, \Omega|e|)$; on its complement it is at most two. PSD matrix-measure Cauchy–Schwarz bounds the cross total variation on each set by the geometric mean of the diagonal masses on that set. Bounding the low-frequency masses by v_i, v_j proves the inequality. $\square$

The envelope is independent information about the continuous-time source, not an estimate supplied by its aliased finite record. Finite dependence without a quantitative spectral envelope does not provide this control.

Interpretation and scope. The theorem concerns a finite sampled law, with the unobserved extension and source allowed to depend on that law and its horizon. It does not match an arbitrary preassigned full infinite sampled process with one source, establish resource bounds, or cover non-Gaussian laws. The finite range and spectral shape can change as the targets or delays change. A constant positive floor on all of $\mathbb{R}$ is impossible for any integrable nonzero spectral density. The irrationality condition is sufficient, not a characterization of every universally realizing delay vector. An exact rational restriction and a nonzero-width real interval are different model classes. Integer-grid calibration results remain valid in their original model; without a spectral restriction they do not extend to open real-valued skew sets.

S3 Calibrated channel-specific linear filters

The delay constructions leave source phase and detector phase indistinguishable at the retained times. Calibration restricts the detector part. After removing the anchor delays, we take the real part of the filtered spectrum. This operation gives a reversible source spectrum and preserves the observed auto covariances. The removed imaginary part is the quantity that response calibration must bound. The comparator record can still contain unequal anchor delays; reversibility belongs to its source.

This comparator gives a population tolerance. Response confidence regions must then supply a phase allowance for that tolerance and response-shape information for a separate covariance envelope. These are different requirements. No source bandwidth bound is needed when calibration covers all possible source frequencies. When singular spectra are allowed, that coverage must include spectral atoms.

S3.1 Model and phase calibration

Let X be a real stationary reversible Gaussian source with m channels and common reversal parity. Its matrix spectral measure μ is real, symmetric, even, and positive semidefinite. The frequency domain is either $\mathbb{R}$ or the discrete-time torus. On the torus, all delays below are integers. Channel i has a measurable transfer function H_i such that $H_i(-\omega) = \overline{H_i(\omega)}$. Assume that its mean-square output exists, with variance

$$A_i = \int |H_i(\omega)|^2 \mu_{ii}(d\omega) < \infty.$$

At integer times observe $Y_i = (H_i X_i) + E_i$. The stationary scalar errors have zero covariance with the source and with each other at every lag. Their scalar auto covariances can be arbitrary. The finite-sample tests require jointly Gaussian observed records. The population covariance bounds do not require Gaussian errors.

Write $C_{ij}(k) = \text{Cov}(Y_i(t+k), Y_j(t))$. The filtered signal has spectral measure

$$\nu_{ij}(d\omega) = H_i(\omega) \overline{H_j(\omega)} \mu_{ij}(d\omega).$$

Orthogonality gives $A_i \leq C_{ii}(0)$. Thus an observed variance ceiling $D_i \geq C_{ii}(0)$ bounds filtered signal variance. It need not bound the unfiltered source variance. No inverse gain bound is assumed.

Fix an integer anchor vector d and put $\Delta_{ij} = d_i - d_j$. Define

$$s_{ij}(\omega) = \frac{\text{Im}\{e^{i\Delta_{ij}\omega} H_i(\omega) \overline{H_j(\omega)}\}}{|H_i(\omega) H_j(\omega)|}, \quad r_{ij}(u) = \sup_{\omega \in \mathcal{F}} |\sin(u\omega) s_{ij}(\omega)|. \quad (\text{S17})$$

Set $s_{ij} = 0$ at transfer-product zeros. The supplied set $\mathcal{F}$ must contain the source spectral support. Use $\mathcal{F} = \mathbb{R}$ when no support restriction is available. The supremum can also cover a calibrated family of filters. Bounds must hold at spectral atoms. A bound that holds only Lebesgue-almost-everywhere is insufficient when singular source spectra are allowed.

The quantity s_{ij} is the sine of relative phase after anchor delay removal. A phase difference of π has zero error. This convention is necessary because a reversible cross spectrum can have either sign.

S3.2 An exact comparator

Taking the real part must preserve a valid matrix spectrum, not only remove the imaginary entries. Positive semidefiniteness supplies this property and gives a comparator for all channel pairs at once. The reflected-average identity below then expresses its covariances directly in terms of the recorded covariances.

Theorem S4 (Filter comparator). *There is a reversible stationary Gaussian source which, under the anchor delays and the same scalar-error auto covariances, gives a comparator $\tilde{C}$ satisfying*

$$\begin{aligned}\tilde{C}_{ii}(k) &= C_{ii}(k), \\ \tilde{C}_{ij}(k) &= \frac{C_{ij}(k) + C_{ij}(2\Delta_{ij} - k)}{2}, \quad i \neq j.\end{aligned}\tag{S18}$$

For every cross entry,

$$|C_{ij}(k) - \tilde{C}_{ij}(k)| \leq r_{ij}(k - \Delta_{ij})\sqrt{A_i A_j} \leq r_{ij}(k - \Delta_{ij})\sqrt{D_i D_j}.\tag{S19}$$

In particular, the cross covariance at $k = \Delta_{ij}$ is unchanged.

Proof. Dominate μ by its even trace measure θ . Its Radon–Nikodym matrix $M = d\mu/d\theta$ is real, symmetric, even, and positive semidefinite almost everywhere. Put

$$K(\omega) = \text{diag}(e^{id_i\omega} H_i(\omega)), \quad Q(\omega) = K(\omega)M(\omega)K(\omega)^*.$$

The matrix Q is Hermitian positive semidefinite and satisfies $Q(-\omega) = \overline{Q(\omega)}$. Its entrywise real part is real symmetric and positive semidefinite. Indeed, for each real vector v ,

$$v^\top \text{Re } Q v = \text{Re}(v^* Q v) \geq 0.$$

Its diagonal equals the diagonal of Q . Therefore

$$\tilde{\mu}(d\omega) = \text{Re } Q(\omega)\theta(d\omega)$$

is a real-even positive-semidefinite matrix measure. Each diagonal is finite by the assumption $A_i < \infty$. The cross measures are finite by the positive-semidefinite inequality and Cauchy–Schwarz. This measure defines a reversible stationary Gaussian source. Apply the anchor delays and retain the scalar-error auto covariances. Independent stationary Gaussian errors with those scalar covariances give a Gaussian comparator.

The cross difference measure is $ie^{-i\Delta_{ij}\omega} \text{Im } Q_{ij} d\theta$. Its covariance at lag k equals

$$C_{ij}(k) - \tilde{C}_{ij}(k) = - \int \sin((k - \Delta_{ij})\omega) \text{Im } Q_{ij}(\omega) \theta(d\omega).\tag{S20}$$

The imaginary cosine integral vanishes by oddness. The identity $\text{Re } Q = (Q + \overline{Q})/2$ also gives equation (S18).

Since M is real,

$$\text{Im } Q_{ij} = |H_i H_j| s_{ij} M_{ij}.$$

The pointwise inequality $|M_{ij}| \leq \sqrt{M_{ii} M_{jj}}$ implies

$$\int |H_i H_j M_{ij}| d\theta \leq \sqrt{\int |H_i|^2 M_{ii} d\theta \int |H_j|^2 M_{jj} d\theta} = \sqrt{A_i A_j}.$$

This proves equation (S19). The comparator leaves every signal auto spectrum unchanged. The scalar errors add the same auto covariances and no cross covariances. $\square$

The reflected-average construction itself applies to any stationary matrix spectral measure. Source reversibility is needed to bound its imaginary part by the calibrated transfer phase. The comparator preserves channel auto spectra. It can reduce cross-spectrum magnitudes and need not belong to the calibrated filter family. It is not an inverse filter.

S3.3 Reflection with a sharp population constant

For one pair put $\Delta = \Delta_{ij}$. Choose an integer $h \geq 1$ with $|\Delta| + h \leq T - 1$. Let $a = \max(\Delta, 0)$, $e = \max(-\Delta, 0)$ and

$$U = Y_i(a + h) + Y_i(a), \quad W = Y_j(e) - Y_j(e + h).$$

Covariance expansion and equation (S20) give

$$\begin{aligned} \kappa &:= \text{Cov}(U, W) = C_{ij}(\Delta + h) - C_{ij}(\Delta - h) \\ &= -2 \int \sin(h\omega) |H_i H_j| s_{ij} M_{ij} d\theta. \end{aligned}$$

Consequently,

$$|\kappa| \leq b := 2r_{ij}(h) \sqrt{D_i D_j}. \quad (\text{S21})$$

The constant is sharp in the spectral-measure class when only the filters and filtered signal variances are fixed. This class permits line spectra. Suppose ω_* attains the supremum and both gains are nonzero there. Put a real rank-one source measure at $\pm\omega_*$. Assign total diagonal masses $D_i/|H_i(\omega_*)|^2$ and $D_j/|H_j(\omega_*)|^2$ across the pair. Split each mass equally between the two distinct frequencies. Choose either sign for the total cross mass and use zero errors. The observed variances are D_i, D_j , and equality holds in equation (S21). A sequence of atoms approaches the constant if the supremum is not attained. Each source in the sequence has finite variance. A zero bound is trivial, so endpoint cases with zero sine do not require a distinct atom pair.

S3.4 Coefficient confidence and recording error

The population tolerance depends on the true response. A fitted response alone is insufficient because coefficient error can change the relative phase. The coefficient ball below covers that error before any frequency bound is computed. A recording correction enlarges the same ball when the calibration response is rounded. Known-input calibration has model $y = Xh + e$, where X is a deterministic full-rank matrix with m rows and p columns. The coefficient positions are distinct fixed integers. Assume $e \sim N(0, \sigma^2 I_m)$, $\sigma > 0$, and $\nu = m - p > 0$. The response remains unchanged between calibration and testing. If an unknown constant offset is fitted, project X and y onto the orthogonal complement of the constant vector. Require the projected design to have full column rank and use $\nu = m - p - 1 > 0$.

Let $G = X^\top X$, $\hat{h} = G^{-1}X^\top y$, and $\text{RSS} = \|y - X\hat{h}\|^2$. Choose a positive lower bound $\ell_G \leq \lambda_{\min}(G)$. For example, $\ell_G = 1/\|G^{-1}\|_\infty$ is valid because G^{-1} is symmetric. At $t = \log(2/\alpha_H)$, Gaussian projection and chi-square bounds give

$$e_H^2 = \frac{\text{RSS}[p + 2\sqrt{pt} + 2t]}{\ell_G[\nu - 2\sqrt{\nu t}]} \quad (\text{S22})$$

as a coefficient-radius square with failure probability at most α_H , provided the denominator is positive. Indeed, the fitted-error squared norm divided by σ^2 has a chi-square law with p degrees of freedom. The residual squared norm divided by σ^2 has a chi-square law with ν degrees of freedom. Their upper and lower events each fail with probability at most e^{-t} . On their intersection, $\|\hat{h} - h\|_2 \leq e_H$. A union bound suffices. The implementation encloses every numerical quantity in the indicated direction.

For a perturbed calibration response z with $|z_j - y_j| \leq \eta_H$, the design remains exact. Orthogonal projection gives

$$\sqrt{\text{RSS}(y)} \leq \sqrt{\text{RSS}(z)} + \sqrt{m}\eta_H, \quad \|\hat{h}(y) - \hat{h}(z)\|_2 \leq \frac{\sqrt{m}\eta_H}{\sqrt{\ell_G}}.$$

Thus a valid radius about the recorded fit is

$$e_H^+ = \sqrt{\frac{(\sqrt{\text{RSS}(z)} + \sqrt{m}\eta_H)^2[p + 2\sqrt{pt} + 2t]}{\ell_G[\nu - 2\sqrt{\nu t}]}} + \frac{\sqrt{m}\eta_H}{\sqrt{\ell_G}}. \quad (\text{S23})$$

On the ideal calibration coverage event, the enlarged ball about the recorded fit still contains the true coefficients. This implication requires no stochastic model for the perturbation. Finite impulse-response confidence sets and their use for frequency bounds are established methods [2, 3, 4].

S3.5 Phase and response-shape bounds

Phase controls the null contrast, whereas response shape controls how strongly filtering can concentrate the source spectrum. Both bounds must hold over the entire covered coefficient ball. The simple scalar-reference construction bounds them directly. The grid construction below extends phase coverage between tested frequencies. The study has a real scalar reference response and an estimated second response. On a coefficient ball of radius e_H^+ , let b_+ bound the sum of absolute coefficients away from position zero. Let h_- bound the absolute position-zero coefficient from below. If $h_- > b_+$, then

$$|\text{Im } H_2(\omega)| \leq b_+, \quad |H_2(\omega)| \geq h_- - b_+.$$

Consequently $\hat{r} = \min\{1, b_+/(h_- - b_+)\}$ is valid at every frequency. If the denominator is nonpositive, use the universal bound one. The unknown sign of the scalar reference does not change the absolute phase ratio.

A response with p coefficients satisfies $1 \leq C_H := \sup_\omega |H(\omega)|^2/\|h\|_2^2 \leq p$ when $h \neq 0$. Let

$$u_H = \|\hat{h}\|_1 + \sqrt{p}e_H^+, \quad v_H = \|\hat{h}\|_2 - e_H^+.$$

When $v_H > 0$, use $\hat{C}_H = \min\{p, \max\{1, (u_H/v_H)^2\}\}$. Otherwise use p . Triangle inequalities prove $C_H \leq \hat{C}_H$ on the same coefficient event. A zero response uses the universal branch.

For more general response pairs, a frequency grid with a derivative margin provides a direct relative-phase construction. Write the aligned response as $H_i(\omega) = \sum_a h_{ia}e^{-ik_{ia}\omega}$. If $\|h_i - \hat{h}_i\|_1 \leq E_i$ and the grid's maximum nearest-point distance is d_ω , define

$$u_i = E_i + d_\omega \sum_a |k_{ia}\hat{h}_{ia}|.$$

At grid point ω_j , let $p_j = \widehat{H}_1(\omega_j)\overline{\widehat{H}_2(\omega_j)}$ and $\tau_j = u_1|\widehat{H}_2| + u_2|\widehat{H}_1| + u_1u_2$. The true product in that grid cell is within τ_j of p_j . Use

$$a_j = \begin{cases} \min\{1, (|\operatorname{Im} p_j| + \tau_j)/(|p_j| - \tau_j)\}, & |p_j| > \tau_j, \\ 1, & |p_j| \leq \tau_j. \end{cases}$$

Then $\max_j a_j \min\{1, |\sin \omega_j| + d_\omega\}$ is an all-frequency phase bound for the lag-one contrast. The derivative bound and product expansion prove coverage throughout each cell. All cells use one coefficient event, so no further frequency allocation is needed. Transfer zeros use the universal branch or the zero-integrand convention. This construction need not converge at arbitrary common zeros.

S3.6 From spectral information to a full covariance bound

Phase calibration places no bound on temporal dependence. To obtain one, the source spectral floor compares filtered variance with filter energy, and the ceiling bounds the filtered spectral peak. Positive semidefiniteness then controls cross-channel dependence. The error spectrum needs its own ceiling because it is not constrained by the source ratio. Let the normalized source marginal densities satisfy $m_0 \leq g_i \leq M_0$, with $\mathcal{S} = M_0/m_0$. The positive-semidefinite two-channel source spectral matrix is at most twice its diagonal. If $v_{XH,i}$ is the filtered source variance, its marginal spectral peak is at most $\mathcal{S}C_{H,i}v_{XH,i}$. Let mutually orthogonal source-independent errors have variances τ_i^2 and density bounds $f_{E,i} \leq J_i\tau_i^2$, with $J_i \geq 1$. Define

$$q_0 = \max\{2\mathcal{S} \max_i C_{H,i}, \max_i J_i\}.$$

The full observed density is at most $q_0 \operatorname{diag}(v_1, v_2)$ because its diagonal envelope in channel i is

$$2\mathcal{S}C_{H,i}v_{XH,i} + J_i\tau_i^2 \leq q_0(v_{XH,i} + \tau_i^2) = q_0v_i.$$

Fourier integration gives the full recording covariance bound. Replacing each shape factor by its coefficient-event upper bound preserves coverage. A nonzero error spectral atom cannot satisfy this density assumption. White marginal source spectra alone do not imply independent multichannel samples.

S4 Finite-record bounds for calibrated and cycle tests

A population constraint does not by itself bound false rejection from an estimated quantity. For both tests, we must control fluctuations of a centered quadratic form and the difference between its expectation and the target. The same Gaussian concentration result covers the calibrated contrast and the real and imaginary parts of the spectral estimates, including their overlapping samples.

For the cycle, we first bound each spectral edge and include the omitted covariance tail, then propagate the three errors through the product. For the calibrated contrast, the adjacent sum and difference operators permit a sharper matrix bound. Separate variance ceilings supply the scale used by that bound. The last steps enlarge the two constructions for bounded recording error and establish completion of the specified arithmetic. Thus the proof separates a probabilistic error bound from the numerical conditions needed to apply it.

S4.1 Gaussian concentration for a centered quadratic form

The calibrated contrast and the real and imaginary parts of each estimated spectral edge are quadratic forms with no unknown mean term after centering. Their fluctuations depend on the coefficient matrix weighted by the full record covariance. The Frobenius norm controls the square-root tail term, and the signed extreme eigenvalues control the linear term. A trace bound can improve the Frobenius estimate when the covariance ceiling is conservative.

Lemma S2 (Signed Gaussian tails and a variance-sensitive radius). *Let y be Gaussian with covariance Γ , and let a symmetric matrix A annihilate its mean. Put $T = \Gamma^{1/2}A\Gamma^{1/2}$ and $a_+ = \max(\lambda_{\max}(T), 0)$, $a_- = \max(-\lambda_{\min}(T), 0)$. For $t > 0$,*

$$\begin{aligned}\Pr\{y^\top Ay - \mathbb{E}(y^\top Ay) > 2\|T\|_F\sqrt{t} + 2a_+t\} &\leq e^{-t}, \\ \Pr\{\mathbb{E}(y^\top Ay) - y^\top Ay > 2\|T\|_F\sqrt{t} + 2a_-t\} &\leq e^{-t}.\end{aligned}\tag{S24}$$

If $\Gamma \preceq KI$ and $\text{tr } \Gamma \leq NT_*$, then

$$\|T\|_F \leq \min\{K\|A\|_F, \|A\|_{\text{op}}\sqrt{KNT_*}\}, \quad \|T\|_{\text{op}} \leq K\|A\|_{\text{op}}.\tag{S25}$$

Proof. Diagonalization gives $y^\top Ay - \mathbb{E}(y^\top Ay) \stackrel{\text{law}}{=} \sum_j \lambda_j(T)(Z_j^2 - 1)$ for independent standard normal Z_j . For $u > 0$ and $2ua_+ < 1$, its log moment generating function is at most

$$\sum_j \{-u\lambda_j - \frac{1}{2}\log(1 - 2u\lambda_j)\} \leq \frac{u^2\|T\|_F^2}{1 - 2ua_+}.$$

Positive and negative eigenvalues have different roles in this upper tail. Only positive eigenvalues restrict the allowed value of u . For them, expand the logarithm and bound its higher terms by a geometric series. For negative eigenvalues, use $-\log(1+x) + x \leq x^2/2$. Exponential Markov with $u = \sqrt{t}/(\|T\|_F + 2a_+\sqrt{t})$ gives the first tail. Replacing A by $-A$ gives the second. This is the signed quadratic-form form of Gaussian chi-square concentration [5]. Singular cases follow on the covariance range.

The first Frobenius bound and the operator bound follow by multiplication by $\Gamma^{1/2}$. For the other branch,

$$\|T\|_F^2 = \text{tr}(A\Gamma A\Gamma) \leq K \text{tr}(A^2\Gamma) \leq K\|A\|_{\text{op}}^2 \text{tr } \Gamma.$$

The first inequality uses $A\Gamma A \succeq 0$ and $\Gamma \preceq KI$. □

S4.2 Cycle confidence with a diagonal covariance envelope

The source and observation errors are jointly stationary Gaussian processes. The channel identities and even reversal parities are fixed. The responses are real, diagonal and time invariant. Observation errors are source-orthogonal and mutually orthogonal at every lag. The channel means are unknown constants. The observed spectrum has the form $F = D_H S D_H^* + D_E$, with diagonal D_H and D_E . Under a reversible source, S is real symmetric. Therefore

$$F_{12}F_{23}F_{31} = |H_1H_2H_3|^2 S_{12}S_{23}S_{31} \quad \text{is real.}\tag{S26}$$

The identity permits negative real products. To use its realness in a test, we must enclose all three population edges simultaneously. Phase cancellation does not bound fluctuations in their estimates or the contribution of covariances beyond the retained lag window.

Let $N \geq 1$, $0 \leq L < N$ and a set $\mathcal{W}$ of $G \geq 1$ frequencies be fixed before observation. Write $P_N = I_N - \mathbf{1}\mathbf{1}^\top/N$ and

$$(W_\omega)_{st} = \mathbf{1}\{|s-t| \leq L\}e^{-i\omega(s-t)}, \quad \hat{F}_{ij}(\omega) = N^{-1}(P_N y_i)^\top W_\omega(P_N y_j).\tag{S27}$$

The directed edges are 12, 23 and 31. Assume their cross covariances are absolutely summable. Supply positive K_i such that the complete channel-stacked covariance satisfies

$$\Gamma_N \preceq \text{diag}(K_1 I_N, K_2 I_N, K_3 I_N). \quad (\text{S28})$$

For each edge, supply $B_{ij} \geq 0$ with

$$\sum_{|k|>L} |C_{ij}(k)| + \frac{1}{N} \sum_{|k|\leq L} |k| |C_{ij}(k)| \leq B_{ij}. \quad (\text{S29})$$

These bounds apply to the complete observed process, including its responses and errors. The covariance envelope controls the quadratic-form fluctuation. The tail allowance bounds the bias from omitted lags and from the smaller number of pairs at nonzero lags. Sample centering adds a separate bias term.

Theorem S5 (Unequal cycle radii). *For $0 < \delta < 1$, put $h = 2L + 1$, $t_\delta = \log(12G/\delta)$ and*

$$e_{ij,\delta} = B_{ij} + \sqrt{K_i K_j} \left\{ \frac{3h}{N} + 2\sqrt{\frac{ht_\delta}{N}} + \frac{\sqrt{2} ht_\delta}{N} \right\}. \quad (\text{S30})$$

With probability at least $1 - \delta$, every declared frequency and edge satisfies $|\hat{F}_{ij}(\omega) - F_{ij}(\omega)| \leq e_{ij,\delta}$. For nonnegative triples x and e , define

$$\begin{aligned} \mathcal{R}_\times(x, e) &= e_1 x_2 x_3 + e_2 x_1 x_3 + e_3 x_1 x_2 \\ &\quad + e_1 e_2 x_3 + e_1 e_3 x_2 + e_2 e_3 x_1 + e_1 e_2 e_3. \end{aligned} \quad (\text{S31})$$

At level α , reject if any $\omega \in \mathcal{W}$ satisfies

$$\left| \text{Im}\{\hat{F}_{12}\hat{F}_{23}\hat{F}_{31}\} \right| > \mathcal{R}_\times(|\hat{F}_{12}|, |\hat{F}_{23}|, |\hat{F}_{31}|, (e_{12,\alpha}, e_{23,\alpha}, e_{31,\alpha})). \quad (\text{S32})$$

Its rejection probability is at most α under the reversible-source null. Equality does not reject. Missing required bounds cause abstention.

Proof. We first bound the bias of each edge estimate, then its random error. Divide channel i by $\sqrt{K_i}$. The full normalized covariance is at most the identity. Before sample centering, the expected normalized entry is

$$\sum_{|k|\leq L} (1 - |k|/N) e^{-ik\omega} C_{ij}(k) / \sqrt{K_i K_j}.$$

Its bias is at most $B_{ij}/\sqrt{K_i K_j}$. Set $J_N = \mathbf{1}\mathbf{1}^\top/N$. Sample centering replaces W_ω by $P_N W_\omega P_N$, whose difference from W_ω is $-J_N W_\omega - W_\omega J_N + J_N W_\omega J_N$. Each term has rank at most one and nuclear norm at most h . The normalized cross-covariance block has operator norm at most one. Trace duality thus bounds the additional complex bias by $3h/N$.

The remaining error is a fluctuation about this centered expectation. For $A_\omega = \text{Re } W_\omega$ or $\text{Im } W_\omega$, the real statistic has symmetric cross blocks $P_N A_\omega P_N / (2N)$ and its transpose. Its quadratic-form matrix has squared Frobenius norm at most $h/(2N)$ and operator norm at most $h/(2N)$. These bounds follow from at most h unit-magnitude entries in each row and at most Nh such entries in total. Lemma S2 gives the real-component radius $\sqrt{2ht/N} + ht/N$. There are six real components at each frequency. Their two-sided union failure is at most $12Ge^{-t}$. Combining the real and imaginary errors gives $2\sqrt{ht/N} + \sqrt{2}ht/N$. Adding the two biases and rescaling proves (S30).

On that event, write each population entry as its estimate plus an error of modulus at most the corresponding e_i . Expanding the product gives seven nonempty error products. Their total modulus is bounded by (S31). The reversible-source population product is real. Absolute summability makes the cross spectra continuous, so this identity holds at every declared frequency. Rejection therefore requires failure of simultaneous coverage. $\square$

The full covariance bound in (S28) is essential. Two unit marginal covariance blocks with cross block $0.9I_N$ have a full covariance eigenvalue of 1.9. Their separate marginal bounds do not imply a full bound by the identity. From marginal block bounds alone, a valid conservative construction for m channels is

$$\Gamma \preceq m \operatorname{diag}(\Gamma_{11}, \dots, \Gamma_{mm}). \quad (\text{S33})$$

Indeed, positive semidefiniteness bounds each block quadratic term by the geometric mean of its diagonal terms. Their sum is at most $(\sum_i \sqrt{a_i})^2 \leq m \sum_i a_i$.

For the fixed two-frequency set $\mathcal{W} = \{\pi/4, \pi/2\}$, the logarithm is $\log(24/\delta)$. No independence between frequencies is required. Both frequencies remain in this count if one numerical calculation is unavailable. This union permits selection between these two covered frequencies. It does not permit a new frequency or a new lag cutoff chosen from the recording.

A nonzero population cycle alone does not ensure rejection. The next condition requires its imaginary part to exceed both the alternative estimation error and the largest null threshold on the coverage event. It concerns the mathematical rule. Recording errors and numerical completion require the separate conditions below.

Theorem S6 (A sufficient cycle power condition). *Assume jointly stationary Gaussian source and observation errors. Retain the response, error-orthogonality, parity, covariance-envelope, and tail-bound assumptions of Theorem S5, without requiring source reversibility. Let $0 < \alpha, \beta < 1$. At one frequency $\omega \in \mathcal{W}$, put*

$$m = (|F_{12}(\omega)|, |F_{23}(\omega)|, |F_{31}(\omega)|), \quad J = \operatorname{Im}(F_{12}F_{23}F_{31}).$$

Write $e_\delta = (e_{12,\delta}, e_{23,\delta}, e_{31,\delta})$. If

$$|J| > \mathcal{R}_\times(m, e_\beta) + \mathcal{R}_\times(m + e_\beta, e_\alpha), \quad (\text{S34})$$

then the mathematical rule in (S32) has power at least $1 - \beta$. Vector addition is componentwise. The radii retain the full declared frequency count G .

The first term in (S34) bounds the error in the estimated product. The second bounds the rejection threshold using the edge magnitudes allowed by simultaneous β coverage. The strict inequality ensures that the estimated imaginary product exceeds that threshold throughout the coverage event.

Proof. On simultaneous β coverage, expand the estimated product around the population entries. Its error has modulus at most $\mathcal{R}_\times(m, e_\beta)$. Thus the absolute imaginary estimated product is at least $|J| - \mathcal{R}_\times(m, e_\beta)$. Each estimated entry has magnitude at most $m_i + e_{i,\beta}$. Every term in the product radius is nonnegative and increases with these magnitudes. The rejection threshold is therefore at most $\mathcal{R}_\times(m + e_\beta, e_\alpha)$. The strict condition implies rejection on this event, whose failure probability is at most β . $\square$

S4.3 Changes of channel units and recording errors

Proposition S4 (Gain equivariance). *For fixed nonzero real gains a_i and constant offsets b_i , let $y'_i = a_i y_i + b_i \mathbf{1}$. Transport the supplied bounds as*

$$K'_i = a_i^2 K_i, \quad B'_{ij} = |a_i a_j| B_{ij}.$$

The mathematical decision in (S32) is unchanged.

Proof. Centering removes b_i . Bilinearity gives $\hat{F}'_{ij} = a_i a_j \hat{F}_{ij}$, while covariance congruence gives the transported full envelope. Equation (S30) gives $e'_{ij} = |a_i a_j| e_{ij}$. The statistic and every term in its product radius acquire the same positive factor $(a_1 a_2 a_3)^2$. The strict comparison is unchanged, including for negative gains. $\square$

A change of units scales the channel errors as well as the signal. Changing only the signal gain defines a different observation law. A zero gain can remove the witness and is excluded from this equivalence. Finite arithmetic limits can change whether a calculation completes.

Now suppose the ideal Gaussian recording y and the recorded values z obey

$$|y_i(t) - z_i(t)| \leq \eta_i \quad \text{at every retained sample,} \quad \eta_i \geq 0. \quad (\text{S35})$$

The bound is deterministic and can cover errors that depend on y . Define the recorded centered energies $Q_i(z) = N^{-1} \|P_N z_i\|^2$.

Proposition S5 (Cycle transfer through bounded recording errors). *Use the estimates from z and replace each $e_{ij,\delta}$ by*

$$e_{ij,\delta} + \tau_{ij}(z), \quad \tau_{ij}(z) = h \{ \eta_i \sqrt{Q_j(z)} + \eta_j \sqrt{Q_i(z)} + \eta_i \eta_j \}. \quad (\text{S36})$$

The resulting cycle rule has the same level bound for the ideal model. Gain equivariance also holds if $\eta'_i = |a_i| \eta_i$.

Proof. Let $d_i = y_i - z_i$. Orthogonal projection gives $\|P_N d_i\| \leq \sqrt{N} \eta_i$. The lag matrix satisfies $\|P_N W_\omega P_N\|_{\text{op}} \leq h$. Expanding its bilinear form around z bounds the change of each estimate by (S36). On the original Gaussian coverage event, the population entry is therefore within the enlarged radius around the estimate from z . The product argument remains unchanged. The stated transport makes $\tau'_{ij} = |a_i a_j| \tau_{ij}$. $\square$

This result requires a valid uniform error allowance. Saturation without such an allowance causes abstention. At $\pi/4$, outward intervals for $\sqrt{2}/2$ enclose the phase coefficients. A numerical rejection requires a lower bound for the absolute imaginary product to exceed an upper bound for the threshold. An unresolved comparison cannot reject. The same arithmetic rule covers exact quarter-turn coefficients.

A pointwise spectral value cannot replace the full covariance information without an additional leakage bound. For example, take independent channels $Y_i(t) = Z_i(t) + Z_i(t-2)$ with standard Gaussian white innovations. Their marginal densities $2 + 2\cos(2\omega)$ vanish at $\pi/2$. Every finite-record covariance is positive definite because the finite convolution matrix has full row rank. At $N = 3$ and $L = 1$, $P_N W_{\pi/2} P_N$ has a nonzero imaginary part. The corresponding bilinear form between independent channels has positive variance. A radius proportional only to $\sqrt{F_{ii}(\pi/2) F_{jj}(\pi/2)}$ would be zero and cannot cover this finite-record error.

Proposition S6 (Cycle power with a bounded recorder). *Retain the alternative assumptions of Theorem S6 and the recorder contract (S35). Suppose a recording event E_Q has failure probability at most δ_Q and gives deterministic bounds $\tau_{ij}(z) \leq \bar{\tau}_{ij}$. Put $d_\delta = e_\delta + \bar{\tau}$. If*

$$|J| > \mathcal{R}_\times(m, d_\beta) + \mathcal{R}_\times(m + d_\beta, d_\alpha), \quad (\text{S37})$$

then the mathematical rounded rule has power at least $1 - \beta - \delta_Q$. A uniform deterministic bound has $\delta_Q = 0$.

Proof. On E_Q and the ideal β -coverage event, the recorded estimates differ from the population entries by at most d_β . Their threshold radii are at most d_α . Apply the two product bounds in the proof of Theorem S6. The union bound requires no independence between the events. $\square$

For example, bounds $Q_i(z) \leq \bar{Q}_i$ on E_Q give

$$\bar{\tau}_{ij} = h\{\eta_i \sqrt{\bar{Q}_j} + \eta_j \sqrt{\bar{Q}_i} + \eta_i \eta_j\}.$$

For the rotational family with marginal source variances one, a Gaussian union bound gives $|Y_i(t) - \mathbb{E}Y_i(t)| \leq 16\|h_i\|_1$ outside an event of probability at most $6Ne^{-128}$. Under the recorder contract, projection then gives $\bar{Q}_i = (16\|h_i\|_1 + \eta_i)^2$. This bound applies to the post-sampling filters without added detector noise. It does not require independent time samples.

A bounded numerical implementation needs a further contract. Suppose an event E_C , with failure probability at most δ_C , ensures that the selected frequency calculation completes. Let deterministic vectors $\tilde{d}_\delta \geq d_\delta$ cover its computed upper edge radii on $E_Q \cap E_C$. Suppose its upper magnitude enclosures exceed the exact recorded magnitudes by at most ε_i , and its lower absolute-product bound falls below the exact absolute imaginary product by at most ε_J . Then

$$|J| > \mathcal{R}_\times(m, \tilde{d}_\beta) + \mathcal{R}_\times(m + \tilde{d}_\beta + \varepsilon, \tilde{d}_\alpha) + \varepsilon_J \quad (\text{S38})$$

implies implemented power at least $1 - \beta - \delta_Q - \delta_C$. The proof uses the same product expansion and the stated enclosure errors. At $\pi/2$, exact rational phase arithmetic gives $\varepsilon_J = 0$, and an 80-bit fixed-grid square root permits $\varepsilon_i = 2^{-80}$. At $\pi/4$, both magnitude and product errors must also cover the phase enclosures. These requirements do not prove numerical completion. A population check which fails any sufficient inequality above supplies no lower or upper bound on actual power. The retained negative population conditions therefore remain negative sufficient-condition results.

S4.4 A sharper bound for the two-time reflection matrix

The reflection statistic uses overlapping adjacent pairs, so its operator norm must retain their joint structure. The sum and difference operators are complementary. Bounding them together avoids the loss from bounding each operator separately. This improvement affects the sampling radius, while the calibrated population tolerance remains a separate part of the threshold.

For $N = n + 1$ with $n \geq 2$, let E_0 and E_1 select samples $0, \dots, n-1$ and $1, \dots, n$. Put $S_+ = E_0 + E_1$, $S_- = E_0 - E_1$ and $H_n = I_n - \mathbf{1}\mathbf{1}^\top/n$. The centered contrast is

$$s_n(y) = n^{-1}(S_+ y_1)^\top H_n (S_- y_2) = y^\top A_n y, \quad A_n = \frac{1}{2n} \begin{pmatrix} 0 & S_+^\top H_n S_- \\ S_-^\top H_n S_+ & 0 \end{pmatrix}. \quad (\text{S39})$$

Lemma S3 (Reflection matrix norms). *The matrix in (S39) satisfies*

$$\|A_n\|_{\text{op}} \leq \frac{1}{n}, \quad f_n^2 := \|A_n\|_F^2 = \frac{2n + 2 - 4/n - 4/n^2}{2n^2}. \quad (\text{S40})$$

The operator bound holds with any symmetric $0 \preceq H_n \preceq I_n$. Its constant is attained by the centered contrast when $n = 3$.

Proof. Let J_n have ones on its adjacent diagonals. Direct row products give $S_+ S_+^\top = 2I_n + J_n$ and $S_- S_-^\top = 2I_n - J_n$. Hence $R = [S_+, S_-]$ satisfies $RR^\top = 4I_n$. The matrix $R^\top H_n R$ lies between zero and $4I$. Subtracting $2I$ gives an operator of norm at most two. Selecting the rows of the first channel and the columns of the second gives the block $S_+^\top H_n S_-$. This restriction cannot increase the operator norm, so that block has norm at most two. The eigenvalues of the symmetric off-diagonal block matrix are its positive and negative singular values. Division by $2n$ proves the operator bound.

For the Frobenius norm, set $H_n = I_n - \mathbf{1}\mathbf{1}^\top/n$. Cyclicity of trace gives

$$\|S_+^\top H_n S_-\|_F^2 = \text{tr}\{H_n(2I_n + J_n)H_n(2I_n - J_n)\} = 2n + 2 - 4/n - 4/n^2.$$

Here $\text{tr } J_n^2 = 2(n-1)$, $\|J_n \mathbf{1}\|^2 = 4n - 6$, and $\mathbf{1}^\top J_n \mathbf{1} = 2(n-1)$. The two blocks contribute the factor $1/(2n^2)$. For sharpness at $n = 3$, take $v = (1, 0, -1)^\top$. Then $H_n v = v$ and $J_n v = 0$. The block maps $S_-^\top v$ to $2S_+^\top v$, and both vectors have the same norm. Its norm is therefore two. $\square$

This bound concerns the specified two-time statistic. A general pair of window projections still requires its own matrix norm bound.

S4.5 Separate variance ceilings in the calibrated test

Centering changes the expectation as well as the matrix norm. We must therefore check that the phase tolerance still applies before using the concentration result. Common centering introduces an even nonnegative spectral multiplier, which preserves the reversible-source structure needed for the phase bound. Normalizing each channel by its own variance then gives a geometric scale when the original units are restored.

Suppose the positive observed variances are v_1, v_2 and a supplied standardized covariance factor q satisfies

$$\Gamma_N \preceq q \text{diag}(v_1 I_N, v_2 I_N). \quad (\text{S41})$$

Let r bound $|\sin \omega \text{Im}(H_1 \overline{H_2})|/|H_1 H_2|$ over the source spectral support and every allowed response. The ratio is zero at a transfer-product zero. The common-centering expectation uses

$$g_n(\omega) = 1 - \left| n^{-1} \sum_{t=0}^{n-1} e^{it\omega} \right|^2, \quad 0 \leq g_n \leq 1.$$

Indeed, the uncentered cross moment contributes one in this multiplier and the covariance of the two sample means contributes the squared modulus. The multiplier is real and even. Let μ be the source spectral measure. Multiplication by g_n preserves its real symmetric structure under reversibility. The filtered cross moment is the integral of $2i \sin \omega H_1 \overline{H_2}$ against that measure. The phase bound and the positive-semidefinite cross-measure inequality give

$$|\mathbb{E} s_n| \leq 2r \sqrt{\int g_n |H_1|^2 d\mu_{11} \int g_n |H_2|^2 d\mu_{22}} \leq 2r \sqrt{v_1 v_2}. \quad (\text{S42})$$

Orthogonal errors contribute no cross term, and their variances can only increase v_i . Define

$$\mathfrak{r}_N(q, t) = 2 \min \left\{ q f_n, \frac{\sqrt{2qN}}{n} \right\} \sqrt{t} + \frac{2qt}{n}. \quad (\text{S43})$$

Theorem S7 (Geometric variance scale). *Suppose simultaneous upper bounds $D_i \geq v_i$ hold outside an event of probability δ_v . Suppose valid upper bounds on q and r hold outside an event of probability δ_c . With the covered upper values inserted, the rule*

$$|s_n| > \sqrt{D_1 D_2} \{2r + \mathbf{r}_N(q, t)\} \quad (\text{S44})$$

has size at most $\delta_c + \delta_v + 2e^{-t}$. For common positive denominator c_N , one can take

$$c_N = 1 - q/N - 2\sqrt{qu/N}, \quad D_i = Q_i(y)/c_N, \quad (\text{S45})$$

with variance-coverage failure at most $2e^{-u}$. If $c_N \leq 0$, the rule abstains. At the same q , r and c_N , its threshold is no larger than the threshold that uses $\max(D_1, D_2)$.

Proof. Divide channel i by $\sqrt{v_i}$. The normalized complete covariance is at most qI and has trace $2N$. Lemmas S2 and S3 give the two-sided radius (S43). Multiplication by $\sqrt{v_1 v_2}$ restores the original statistic and tolerance.

For channel i , centering removes at most qv_i/N from the expected energy. The energy's positive-semidefinite Gaussian quadratic form has trace at most v_i and operator norm at most qv_i/N . Its squared Frobenius norm is therefore at most qv_i^2/N . The lower tail in Lemma S2 has $a_- = 0$ and gives

$$\Pr\{Q_i(y) < v_i(1 - q/N - 2\sqrt{qu/N})\} \leq e^{-u}.$$

A union over the two channels proves the variance event. On that event, $\sqrt{D_1 D_2} \geq \sqrt{v_1 v_2}$. All radius terms increase with their nonnegative arguments. On the coverage events, the fitted threshold is therefore at least the threshold formed from the fixed true values. Rejection by the fitted rule then requires a tail event for that fixed rule or a coverage failure. The argument never conditions the Gaussian law on the fitted bounds. Finally, $\sqrt{D_1 D_2} \leq \max(D_1, D_2)$. $\square$

A zero true-variance channel is constant and gives zero ideal centered contrast. A zero recorded variance with a positive recording-error allowance does not establish zero true variance. For a fixed bank of contrasts, the probability allocations and logarithms retain the complete declared bank size.

Proposition S7 (Calibrated transfer through bounded recording errors). *Under (S35), put $U = S_+ z_1$, $V = S_- z_2$ and*

$$Q_i^+(z) = (\sqrt{Q_i(z)} + \eta_i)^2, \quad (\text{S46})$$

$$\tau_s(z) = \frac{2\eta_1 \|H_n V\| + 2\eta_2 \|H_n U\|}{\sqrt{n}} + 4\eta_1 \eta_2. \quad (\text{S47})$$

Use $D_i = Q_i^+(z)/c_N$ and add $\tau_s(z)$ to the right side of (S44). The level bound is unchanged for a supplied valid covariance envelope and covered phase tolerance.

Proof. Projection and the triangle inequality give $Q_i(y) \leq Q_i^+(z)$. The perturbations of U and V have norms at most $2\sqrt{n}\eta_1$ and $2\sqrt{n}\eta_2$. Expansion of their centered bilinear product gives $|s_n(y) - s_n(z)| \leq \tau_s(z)$. On the original variance and nuisance events, the enlarged threshold bounds the ideal tolerance and radius. A rejection from z therefore implies an ideal Gaussian tail event or a coverage failure. $\square$

Numerical evaluation uses upper square-root and threshold endpoints and a positive lower endpoint for c_N . This transfer concerns the supplied-envelope testing step. A source-confidence pivot and a fitted calibration require their own recording-error arguments. A finite Gaussian generator does not itself supply the deterministic contract (S35) relative to an ideal Gaussian path.

S4.6 Finite arithmetic in the calibrated step

The probability events above give covered bounds and a mathematical comparison. They do not bound the size of intermediate rational values or the work needed to evaluate that comparison. For the specified power example, bounded input magnitudes and a bounded source-endpoint height supply these missing controls. Source inversion is handled separately in Section S5.10.

Proposition S8 (Conditional arithmetic completion). *Use the population, calibration regression, offsets and bank of Section S5.9. The raw recording length is 524289, the bank has six members with selected coefficient $199/200$, and the recorder retains 20 fractional bits. Let the uniform raw recording and calibration allowances be 2^{-21} . Suppose source inversion completes with hull endpoints of rational height below 12000 bits, where height is the larger numerator or denominator bit length. Suppose the selected source-shape bound is below 335.920 and the covered calibration outputs satisfy*

$$r \leq 33/5000, \quad P_2 \leq 129/125.$$

Use 256 calibration rows with two orthogonal columns of signs. On the event that every demeaned ideal recording coordinate and every standardized calibration noise value has magnitude at most 16, the calibration and member calculations complete within a 262144-bit rational limit and 500000000 arithmetic operations per phase. This statement uses outward square roots at 80 bits and logarithm series with 48 terms. It assumes the positive calibration residual event and the population variance event used in the power calculation.

Proof. The recording offsets are 2 and -1 , so the raw magnitudes are below 32. Rounded recording integers have fewer than 26 bits. Calibration responses have magnitude below two. The Gram matrix is $256I_2$ and its inverse is $I_2/256$. A fitted coefficient has denominator dividing 2^{28} and numerator height at most 30 bits. The residual sum of squares has denominator dividing 2^{48} and numerator height at most 60 bits.

The logarithm series uses reduced parameters with numerator and denominator height at most eight bits. A common denominator for its 48 terms divides the 95th power of the parameter denominator times the product of the odd integers from 1 through 95. The remainder adds the factors 97 and $1 - z^2$. Including the multiple of $\log 2$ keeps these intermediates below 2048 bits. The residual event makes the lower residual norm exceed its recording perturbation. The resulting calibration intervals use fewer than 10000 bits, and the final phase and shape bounds use fewer than 512 bits each. The fit and recording correction each use fewer than 10000 arithmetic operations.

For a source endpoint of height below 12000, $(1+x)/(1-x)$ has height below 12002. Squared source ratios have height below 24040. The resulting standardized envelope has height below 25000. For the selected member, $q < 694$. At retained length 524288, put $u = \log 1200$. Since $q/524288 < 3/2000$ and the outward upper bound for u is below $71/10$, the computed lower endpoint of c_N exceeds $4/5$. Thus this member has a positive scale denominator.

The exact innovation-energy polynomials have coefficient height below 256 bits on the stated event. Evaluation followed by division by the retained length gives observed-variance height below 1024 bits and value at most 1025. The corrected variance has height below 256 bits. For a member with a positive computed denominator, its reciprocal is bounded by that denominator's rational height. A member with a nonpositive denominator returns unavailable. The successive upper

height bounds for available members are

quantity	height in bits
scale denominator	25200
variance scale	25400
covariance factor	50400
Frobenius branch	51000
mixed square-root argument	76000
sampling radius	104000
complete corrected threshold	140000

Multiplication and division add operand heights. Addition adds those heights and one bit. A fixed-grid square root has denominator at most 2^{80} , and its integer shift adds 160 bits. These rules give the displayed bounds. Each member evaluates three quadratic polynomials and two fixed logarithms, using fewer than 4000 operations. No member calculation repeats a loop over recording values. The limits therefore permit every available branch and the selected member in particular. $\square$

Section S5.9 defines the population event and the positive implemented margin for this model. Section S5.10 combines source completion with Proposition S8 and the magnitude-event probabilities. The threshold upper endpoints increase with q , r and the variance ceilings, while the scale-denominator lower endpoint decreases with q . These monotonicities hold for the outward square-root operations and for the minimum of the two radius branches. Thus the population upper threshold also bounds the implemented member threshold on the stated events.

S5 Source confidence and its propagation into detection

The calibrated test needs a dependence bound that holds for the unknown source. A reference confidence set can supply that bound only if it controls the entire range of accepted persistence values. Merely excluding persistence one leaves open a wide range and a correspondingly large threshold.

We first invert a pivot to cover the true persistence. We then use common bounds on the realized innovation energies to control every accepted candidate together. This converts coverage into a bound on the confidence hull's spectral ratio. A fixed common filter adds a placement cost. The final power argument compares an upper bound on the resulting threshold with a lower bound on the alternative statistic. Recording and arithmetic bounds preserve that comparison for the specified implementation.

The construction uses $n + 1$ scalar reference observations, n candidate innovations, and the operators P and K_B defined below. The matrix P centers the candidate innovations, and K_B forms their cumulative sums. The marginal variance is v , the persistence is ϕ , and $\delta = 1 - \phi$. A confidence hull means the smallest closed interval containing the accepted set.

Definitions and complete statements

Observe $Y_0, \dots, Y_n$, where $N = n + 1$ and $n \geq 2$. Assume

$$\text{Cov}(Y_s, Y_t) = v\phi^{|s-t|}, \quad 0 \leq \phi < 1, \quad v > 0.$$

At the true persistence, subtracting the candidate prediction leaves independent innovations before centering. For a wrong candidate, their cumulative sums can have a different amount of variation.

We compare that cumulative variation with the innovation energy. For a candidate $\psi \in [0, 1]$, let $e_\psi = (Y_t - \psi Y_{t-1})_{t=1}^n$, $P = I_n - \mathbf{1}\mathbf{1}^\top/n$, and let K_B form the first $n-1$ partial sums. Define

$$\begin{aligned} D(\psi) &= \|Pe_\psi\|^2, \quad Q(\psi) = \|K_B Pe_\psi\|^2, \\ R_n(\psi) &= Q(\psi)/\{nD(\psi)\}. \end{aligned} \tag{S48}$$

At the true parameter, the centered residuals are projected independent Gaussian innovations. Hence

$$R_n(\phi) \stackrel{d}{=} \frac{\sum_{k=1}^{n-1} \lambda_{k,n} Z_k^2}{n \sum_{k=1}^{n-1} Z_k^2}, \tag{S49}$$

where $\lambda_{k,n} = [4\sin^2(\pi k/(2n))]^{-1}$ and the Z_k are independent standard normal variables. The eigenvalues weight the independent Gaussian coordinates in the cumulative energy. Dividing by the innovation energy removes the common scale. The resulting law does not depend on the unknown mean, scale, or persistence, so the same reference distribution applies at every candidate. Such a statistic is called a pivot. The ratio has the constant-mean KPSS form [6], but we invert its finite Gaussian law at every candidate persistence.

Choose deterministic cutoffs with lower and upper tail probabilities at most $\alpha_B/2$. The confidence set is

$$\mathcal{C}_B = \left\{ \psi \in [0, 1] : \begin{aligned} &nc_{\text{lo}}D(\psi) - Q(\psi) \leq 0, \\ &Q(\psi) - nc_{\text{hi}}D(\psi) \leq 0 \end{aligned} \right\}. \tag{S50}$$

The set retains candidates whose normalized cumulative residual energy is neither too small nor too large under the innovation law. Both bounds matter because the test will use the whole accepted set. Writing the conditions as quadratics in ψ avoids division and retains zero-energy candidates. It gives $\Pr_\phi(\phi \in \mathcal{C}_B) \geq 1 - \alpha_B$. We use $\alpha_B = 0.005$ for the standalone set. Its intersection with a lag confidence set uses $\alpha_B = \alpha_L = 0.0025$, so the combined source error remains 0.005. A union bound suffices; independence is unnecessary. The split allocation widens both component sets. Thus the intersection need not be smaller than either standalone set computed at error 0.005 on the same record.

The cutoffs are computed by enclosing the distribution of the signed quadratic form

$$\sum_{k=1}^{n-1} \left(\frac{\lambda_{k,n}}{n^2} - \frac{c}{n} \right) Z_k^2.$$

Characteristic-function inversion follows the methods of Imhof and Davies [7, 8]. We bound numerical integration, omitted tails, and bisection error. The lower cutoff is rounded down and the upper cutoff up. These are certified finite- n quantile enclosures. At $n = 2$, the pivot is the constant $1/4$, which is handled separately.

Put $\delta = 1 - \phi$. For positive deterministic constants, suppose a common event of probability at least $1 - \beta$ gives

$$\begin{aligned} Q(\phi) &\geq v\delta(2 - \delta)n^2c_Q, \\ Q(1) &\leq vnb, \\ v\delta(2 - \delta)nd_- &\leq D(\phi) \leq v\delta(2 - \delta)nd_+, \\ \|P(Y_0, \dots, Y_{n-1})^\top\|^2 &\leq vnd_0. \end{aligned} \tag{S51}$$

These inequalities bound the quantities needed to compare a candidate with the true persistence and with the endpoint one. The lower bound on $Q(\phi)$ keeps the cumulative innovation energy from being too small. The upper bound on $Q(1)$ limits the endpoint-detrended path. The two bounds

on $D(\phi)$ control the random variance scale, while the last bound limits how much the residuals change when the candidate changes. Section S5.4 gives explicit finite- n constants for a common event on which all these bounds hold.

Theorem (Full source-envelope statement). *Define*

$$\begin{aligned} A_- &= \sqrt{(2-\delta)c_Q}, & b_e &= \sqrt{b/(n\delta)}, \\ E_0 &= \sqrt{\delta d_0}, & F_{\pm} &= \sqrt{(2-\delta)d_{\pm}}, \\ J_{\text{lo}} &= b_e + \sqrt{c_{\text{lo}}}E_0, & J_{\text{hi}} &= b_e + \sqrt{c_{\text{hi}}}E_0, \\ C_e &= \sqrt{c_{\text{lo}}}(F_- - E_0) - b_e, & H_e &= \sqrt{c_{\text{hi}}}(F_+ - E_0) - b_e. \end{aligned}$$

Assume $C_e > 0$, $F_- > E_0$, and $A_- > J_{\text{hi}}$. On the event in Eq. (S51) and source coverage, put $A = \sqrt{Q(\phi)/(v\delta n^2)}$. Every accepted candidate satisfies

$$\frac{C_e}{A - J_{\text{lo}}} \leq \frac{1 - \psi}{\delta} \leq \frac{H_e}{A - J_{\text{hi}}}. \quad (\text{S52})$$

The hull $[\ell, u]$ is nonempty, $u < 1$, and, for $\Lambda(x) = (1+x)/(1-x)$,

$$\frac{\Lambda(u)}{\Lambda(\ell)} \leq 2C_{\text{gap}}, \quad C_{\text{gap}} = \frac{H_e}{C_e} \frac{A_- - J_{\text{lo}}}{A_- - J_{\text{hi}}}. \quad (\text{S53})$$

If $\delta X_+ < 1$, with $X_+ = H_e/(A_- - J_{\text{hi}})$, the factor 2 can be replaced by $2/(2 - \delta X_+)$. The probability is at least $1 - \alpha_B - \beta$. For the lag intersection, subtract α_L as well.

Theorem (Complete specification of the implemented-power example). *Let X_t, Z_t be independent stationary Gaussian AR(1) processes with mean zero, variance one, and persistence 199/200. The ideal channels are*

$$Y_1(t) = 2 + X_t, \quad Y_2(t) = -1 + \frac{9}{10}X_{t-1} + \frac{\sqrt{19}}{10}Z_t.$$

Use $N = 524289$ observations, identity detector responses, and an independent calibration of the second response $(1, 0)$. Its 256 design rows have entries in $\{-1, 1\}$ and Gram matrix $256I_2$. Its errors are independent mean-zero Gaussian values with standard deviation 0.01. Suppose source and calibration recordings are multiples of 2^{-20} , each within 2^{-21} of its ideal value. Use the standalone two-sided set, bank $(0, 0.5, 0.8, 0.9, 0.97, 0.995)$, and the calibrated geometric-mean threshold with the recording corrections, outward cutoffs, precisions, and numerical limits specified in Secs. S5.9–S5.10. Then

$$\Pr(\text{reject}) \geq 0.91 - (4N + 512)e^{-128} > 0.90. \quad (\text{S54})$$

S5.1 Pivot law and two-sided coverage

At the true persistence, candidate residuals are innovations apart from a constant mean. This gives a known residual law even though persistence and variance are unknown. The cumulative operator weights slow residual variation through the eigenvalues below. Dividing its energy by innovation energy removes the unknown scale and produces a law that can be inverted over candidate persistence values. At $\psi = \phi$, centering removes the constant residual mean. The projected residual is σPZ , where $\sigma^2 = v(1 - \phi^2)$ and $Z \sim N(0, I_n)$. The nonzero eigenvalues of $PK_B^T K_B P$ are

$$\lambda_{k,n} = [4 \sin^2(\pi k/(2n))]^{-1}, \quad 1 \leq k < n. \quad (\text{S55})$$

One way to obtain them is to diagonalize $K_B P K_B^T$. Its (i, j) entry is $\min(i, j) - ij/n$. Its inverse is the tridiagonal matrix with diagonal 2 and adjacent entries -1 . The sine vectors give Eq. (S55).

Orthogonal Gaussian coordinates therefore give the pivot law in Eq. (S49). The scale cancels from numerator and denominator.

Choose $0 < c_{\text{lo}} \leq c_{\text{hi}}$ with both excluded tails at most $\alpha_B/2$. The two polynomial inequalities retain the true parameter except on their two tail events. They also retain every candidate with $D = Q = 0$. Since $D(\phi) > 0$ almost surely, this convention preserves coverage. Intersecting with any lag set of error α_L loses at most α_L more by a union bound. This argument uses no conditional Gaussian assumption after a confidence set is fitted.

For $n = 2$, the projected space has dimension one and $R_n = 1/4$. For $n = 3$, the ratio is an affine transform of a Beta(1/2, 1/2) variable, with support $[1/9, 1/3]$. These cases provide exact probability checks. For increasing n , $\lambda_{k,n}/n^2 \rightarrow 1/(\pi k)^2$ and the normalized innovation norm tends to one. The summable bound $\lambda_{k,n}/n^2 \leq 1/(4k^2)$ controls the remaining coordinates. Thus the limiting ratio is $\sum_{k \geq 1} Z_k^2/(\pi k)^2$, the Cramér-von Mises law. The confidence calculations below use the finite- n law.

S5.2 Lag confidence and one-sided comparators

The lag comparator uses adjacent residual covariance instead of cumulative energy. Sample centering makes its expected normalized lag slightly negative even for independent innovations. The correction below removes this shift before the Gaussian tail bound is applied. Only the true candidate is needed for coverage, so inversion does not require an error allocation over candidate values. Let A_L be the $n \times n$ symmetric matrix with adjacent entries 1/2 and all other entries zero. Put

$$L(\psi) = e_\psi^\top P A_L P e_\psi, \quad \mu_L = -1/n, \quad H_L = P A_L P - \mu_L P.$$

Direct multiplication gives

$$\text{tr } H_L = 0, \quad \|H_L\|_F^2 = (n-3)/2 + 2/n^2, \quad \|H_L\|_{\text{op}} \leq 1 + 1/n.$$

For lag error α_L , define $d = n - 1$, $t_0 = \log(2/\alpha_L)$, and $t_1 = \log(4/\alpha_L)$. If $d - 2\sqrt{dt_0} > 0$, use the outward upper radius

$$w_n = \frac{2\sqrt{\{(n-3)/2 + 2/n^2\}t_1} + 2(1 + 1/n)t_1}{d - 2\sqrt{dt_0}}. \quad (\text{S56})$$

The acceptance set is defined by

$$\begin{aligned} L(\psi) - (\mu_L + w_n)D(\psi) &\leq 0, \\ -L(\psi) + (\mu_L - w_n)D(\psi) &\leq 0. \end{aligned} \quad (\text{S57})$$

At the true persistence, $L - \mu_L D = \sigma^2 Z^\top H_L Z$. The absolute quadratic form exceeds the numerator of Eq. (S56), times σ^2 , with probability at most $2e^{-t_1}$ by the Gaussian quadratic bound [5]. Also D/σ^2 is chi-square with d degrees of freedom. Its lower bound $d - 2\sqrt{dt_0}$ fails with probability at most e^{-t_0} . The sum is α_L . This proves lag coverage without independence of the two events. If the denominator is not positive, use the full parameter range. Zero-energy candidates are retained. No further probability allocation over candidate values is needed.

The analytic cumulative comparators use one lower-tail inequality. For error α_B , put $t_B = \log(2/\alpha_B)$, $j_B = \lceil 9t_B \rceil$, and

$$V_n = n - 1 + 2\sqrt{(n-1)t_B} + 2t_B.$$

When $n \geq j_B + 1$, valid cutoffs are

$$c_n = \frac{n\{j_B - 2\sqrt{j_B t_B}\}}{\pi^2 j_B^2 V_n}, \quad c_* = \frac{3}{17\pi^2 j_B} \leq c_n. \quad (\text{S58})$$

Indeed, the first j_B bridge eigenvalues are at least $n^2/(\pi^2 j_B^2)$. Their Gaussian-square sum is at least $j_B - 2\sqrt{j_B t_B}$ except on probability e^{-t_B} . The denominator sum is at most V_n except on the same probability. Their ratio proves $\Pr\{R_n(\phi) < c_n\} \leq \alpha_B$. The inequalities $j_B \geq 9t_B$ and $V_n/n \leq 17/9$ give c_* . Either comparator retains $ncD(\psi) - Q(\psi) \leq 0$, with c rounded down. A failed length condition gives the full range. Standalone comparators use error 0.005; intersections use 0.0025 for each component. The recording correction below applies to all these polynomials with their original allocations.

S5.3 Certified finite-dimensional probabilities

The pivot law is known, but its two cutoffs still need numerical probability bounds. A cutoff error in the wrong direction could remove true parameters more often than the allocated tails permit. We therefore enclose the signed quadratic-form probability and retain outer quantile endpoints. The same probability calculation later evaluates endpoint exclusion and its converse. For real coefficients a_k and shift s , put $W = \sum_k a_k Z_k^2 - s$. Its characteristic function is

$$\chi(t) = e^{-its} \prod_k (1 - 2ita_k)^{-1/2}. \quad (\text{S59})$$

For a nondegenerate law, numerical inversion gives

$$\Pr(W \leq 0) = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\text{Im } \chi(t)}{t} dt. \quad (\text{S60})$$

The pivot calculation uses $a_k = \lambda_{k,n}/n^2 - c/n$ and $s = 0$. This is the quadratic-form inversion used in exact Gaussian inference [7, 8, 9].

We enclose the near-zero interval, a finite integration interval, and the infinite tail separately. Since $|e^{ix} - 1| \leq |x|$, the absolute integrand on $(0, \epsilon)$ is at most $|s| + \sum_k |a_k|$. The omitted integral is therefore at most $\epsilon(|s| + \sum_k |a_k|)$. For a truncation point $T > 0$, choose $0 \leq b_k \leq |a_k|$ and define

$$M_T = \prod_k (1 + 4T^2 b_k^2)^{-1/4}, \quad \kappa_T = \sum_k \frac{2T^2 b_k^2}{1 + 4T^2 b_k^2}. \quad (\text{S61})$$

The negative logarithmic slope of $|\chi(t)|$ is nondecreasing. If $\kappa_T > 0$, it follows that

$$|\chi(t)| \leq M_T (t/T)^{-\kappa_T}, \quad \int_T^\infty \frac{|\chi(t)|}{t} dt \leq M_T / \kappa_T.$$

Coefficient enclosures are included when selecting the b_k .

Complex interval arithmetic encloses the finite integral. Small factors can be grouped through

$$-\frac{1}{2} \log(1 - z) = \frac{1}{2} \sum_{j=1}^p \frac{z^j}{j} + E_p, \quad |E_p| \leq \frac{|z|^{p+1}}{2(p+1)(1 - |z|)}, \quad |z| < 1.$$

The remainder disks are added before exponentiation. Factors with a large disk are evaluated directly. Certified bisection moves an endpoint only when the entire probability enclosure is on the required side of the target probability. The lower confidence cutoff uses the lower endpoint of its quantile bracket; the upper cutoff uses the upper endpoint. Degenerate laws are evaluated directly. Failure to enclose a probability produces no cutoff certificate.

S5.4 Proof of the simultaneous source envelope

This subsection proves the full source-envelope statement above. Work on source coverage, $\phi \in \mathcal{C}_B$, and its common energy event, Eq. (S51). With $\delta = 1 - \phi$, that event is

$$\begin{aligned} Q(\phi) &\geq v\delta(2 - \delta)n^2c_Q, & Q(1) &\leq vnb, \\ v\delta(2 - \delta)nd_- &\leq D(\phi) \leq v\delta(2 - \delta)nd_+, \\ \|P(Y_0, \dots, Y_{n-1})^\top\|^2 &\leq vnd_0. \end{aligned}$$

The deterministic constants are positive, and the event has probability at least $1 - \beta$.

We bound the candidate gap $1 - \psi$ on both sides of the true gap δ . For candidates closer to one, lower-tail acceptance keeps the gap away from zero. For candidates farther from one, upper-tail acceptance bounds the gap from above. Both bounds retain the same realized numerator $Q(\phi)$. Taking their ratio before replacing this numerator by its lower bound controls the hull width.

Write $x = (1 - \psi)/\delta$ and

$$A = \sqrt{Q(\phi)/(v\delta n^2)}, \quad A_- = \sqrt{(2 - \delta)c_Q}, \quad b_e = \sqrt{b/(n\delta)}, \quad E_0 = \sqrt{\delta d_0}, \quad F_\pm = \sqrt{(2 - \delta)d_\pm}.$$

These normalizations put the energy comparisons on the same scale. The quantity A is the realized cumulative innovation norm and A_- is its lower bound. The endpoint contribution is bounded by b_e . The quantities F_- and F_+ bound the innovation norm, while E_0 bounds the change contributed by the centered observations. The remaining constants from the full source-envelope statement collect these terms with the two acceptance cutoffs:

$$\begin{aligned} J_{\text{lo}} &= b_e + \sqrt{c_{\text{lo}}}E_0, & J_{\text{hi}} &= b_e + \sqrt{c_{\text{hi}}}E_0, \\ C_e &= \sqrt{c_{\text{lo}}}(F_- - E_0) - b_e, & H_e &= \sqrt{c_{\text{hi}}}(F_+ - E_0) - b_e, \\ X_+ &= H_e/(A_- - J_{\text{hi}}). \end{aligned}$$

Assume $C_e > 0$, $F_- > E_0$, and $A_- > J_{\text{hi}}$. The transform is $\Lambda(x) = (1+x)/(1-x)$. The sharper hull factor also requires $\delta X_+ < 1$. These conditions ensure that the true innovation contribution exceeds the endpoint and level-energy terms needed in the exclusion bounds. The cumulative residual at a candidate is an affine combination of the true cumulative innovations and the endpoint-detrended path. Its innovation norm changes through the centered level vector. The exact identities

$$K_B P e_\psi = x K_B P e_\phi + (1 - x) K_B P e_1, \quad P e_\psi = P e_\phi + (\phi - \psi) P Y_-, \quad (\text{S62})$$

use $Y_- = (Y_0, \dots, Y_{n-1})^\top$. For $0 \leq x \leq 1$, triangle inequalities give

$$\begin{aligned} \sqrt{Q(\psi)/(v\delta n^2)} &\leq xA + (1 - x)b_e, \\ \sqrt{D(\psi)/(v\delta n)} &\geq F_- - (1 - x)E_0. \end{aligned}$$

The second right side is positive under $F_- > E_0$. Lower-tail acceptance implies

$$x \geq \frac{\sqrt{c_{\text{lo}}}(F_- - E_0) - b_e}{A - b_e - \sqrt{c_{\text{lo}}}E_0}.$$

For $x \geq 1$, reverse and direct triangle inequalities instead give

$$\sqrt{Q(\psi)/(v\delta n^2)} \geq xA - (x - 1)b_e, \quad \sqrt{D(\psi)/(v\delta n)} \leq F_+ + (x - 1)E_0.$$

The first right side is positive because $A \geq A_- > J_{\text{hi}} > b_e$. Upper-tail acceptance gives $x \leq H_e/(A - J_{\text{hi}})$.

On source coverage, $\sqrt{c_{\text{lo}}}F_- \leq A \leq \sqrt{c_{\text{hi}}}F_+$. Thus the lower gap bound is at most one and the upper gap bound is at least one. Each bound therefore also covers the candidates on the other side of $x = 1$, where its separate derivation was not needed. Both bounds apply to the whole accepted set. Taking their ratio while retaining the same A gives

$$\frac{H_e}{C_e} \frac{A - J_{\text{lo}}}{A - J_{\text{hi}}} \leq \frac{H_e}{C_e} \frac{A_- - J_{\text{lo}}}{A_- - J_{\text{hi}}},$$

because the ratio decreases with A . If x_- and x_+ are the smallest and largest accepted gap ratios, then

$$\frac{\Lambda(u)}{\Lambda(\ell)} = \frac{x_+}{x_-} \frac{2 - \delta x_-}{2 - \delta x_+}.$$

Physical candidates have $\delta x_+ \leq 1$, which gives the factor two. The absolute bound $x_+ \leq X_+$ gives the sharper factor. Coverage makes the hull nonempty. The same proof applies to a nonempty intersection on both coverage events.

Explicit five-event constants. Allocate $\beta/5$ to each energy event and put $t = \log(5/\beta)$. The innovation norm has a chi-square law with $n - 1$ degrees of freedom, so

$$d_{\pm} = \frac{n - 1 \pm 2\sqrt{(n-1)t} + 2t \mathbf{1}_{\{+\}}}{n}, \quad d_0 = 1 + 2\sqrt{\Lambda(\phi)t/n} + 2\Lambda(\phi)t/n. \quad (\text{S63})$$

Require $d_- > 0$. The bound for d_0 uses covariance norm at most $v\Lambda(\phi)$ and trace at most nv .

For $n \geq 129$, Eq. (S55) and $\sin y \leq y$ imply, on the same Gaussian-coordinate construction,

$$\frac{Q(\phi)}{v\delta(2-\delta)n^2} \geq \sum_{k=1}^{128} \frac{Z_k^2}{(\pi k)^2}.$$

Enclosing Eq. (S60) gives probability less than 0.008084 that the latter sum is below 0.023. Hence $c_Q = 0.023$ is valid for $\beta = 0.05$ at all these lengths.

At $\psi = 1$, the cumulative entry is $Y_j - (1 - j/n)Y_0 - (j/n)Y_n$. Put $r_\phi = \phi^n$ and

$$\begin{aligned} h_\phi &= \sum_{j=1}^{n-1} (1 - j/n)\phi^j, \\ \tau &= \frac{n-1}{n} + \frac{(n-1)(2n-1)}{3n^2} + \frac{r_\phi(n^2-1)}{3n^2} - \frac{4h_\phi}{n}, \\ g &= \left[\sqrt{\Lambda(\phi)} + \sqrt{(n-1)(1+r_\phi)/2} \right]^2. \end{aligned} \quad (\text{S64})$$

Its covariance Γ has trace $vn\tau$. The inner path has covariance norm at most $v\Lambda(\phi)$. The endpoint line has norm $v(n-1)(1+r_\phi)/2$. The triangle inequality for square roots of covariance norms therefore gives $\|\Gamma\| \leq vg$. Using $\text{tr}(\Gamma^2) \leq \|\Gamma\| \text{tr} \Gamma$ yields

$$b = \tau + 2\sqrt{(g/n)\tau}t + 2(g/n)t. \quad (\text{S65})$$

A union bound combines these five events. All concentration statements precede fitting.

S5.5 Uniform persistence-time order and bank coverage

The preceding constants can be evaluated at a specified persistence and length. For a uniform order statement, we use coarser events whose bounds depend on the product of length and the persistence gap. They first keep the upper hull endpoint below one and then bound the transformed hull span.

A length-dependent bank is still needed because a bounded span can move toward persistence one beyond every node of a fixed finite bank. For a uniform statement, let $c_{\text{lo}} > 0$ be a common lower bound on all lower cutoffs and $c_{\text{hi}} < \infty$ a common upper bound on all upper cutoffs over the lengths considered. Use six one-sided events with failure $\beta/6$ each. Set

$$t = \log(6/\beta), \quad m_0 = \lceil 9t \rceil, \quad H = 1 + 2\sqrt{t} + 2t, \quad c_{Q,0} = \frac{m_0 - 2\sqrt{m_0 t}}{\pi^2 m_0^2} > 0.$$

For $n \geq \max(4, 128t, m_0 + 1)$, these events give

$$\begin{aligned} v\delta(2 - \delta)n^2 c_{Q,0} &\leq Q(\phi) \leq v\delta(2 - \delta)n^2 H/6, \\ D(\phi) &\leq v\delta(2 - \delta)n(17/9), & \|PY_-\|^2 &\leq vnH, \\ D(1) &\geq vn\delta, & Q(1) &\leq 2vnH. \end{aligned}$$

The lower numerator bound uses the first m_0 bridge coordinates and a chi-square lower tail. The upper numerator and level-energy bounds use Gaussian trace tails. The innovation bound is chi-square. For the endpoint innovation bound, direct difference covariances give centered trace at least $2v\delta(n-1)$, uncentered trace $2vn\delta$, and norm at most $4v\delta$. Thus the squared Frobenius norm is at most $8v^2\delta^2 n$. The Gaussian lower tail gives $2v\delta(n-1-\sqrt{8nt}) \geq vn\delta$. The endpoint-line trace is at most $2v(n-1)$, which gives the last bound.

Choose

$$\begin{aligned} \eta_0 &= \min\{1/4, 1/(4\sqrt{H}), \sqrt{3c_{\text{lo}}/(32H)}\}, \\ \delta_0 &= \min\{1/2, c_{Q,0}/(64c_{\text{hi}}H)\}, \\ C_0 &= \max\{128t, 128H/c_{\text{lo}}, 128H/c_{Q,0}\}, \\ L_0 &= 1 + \frac{4\sqrt{34c_{\text{hi}}/9}}{3\sqrt{c_{Q,0}}}. \end{aligned} \tag{S66}$$

If $n\delta > C_0$, a candidate with $x \leq \eta_0$ has normalized cumulative norm at most $\eta_0\sqrt{H/3} + \sqrt{2H/(n\delta)}$. Its innovation norm is at least $1 - \eta_0\sqrt{H} \geq 3/4$, using $Pe_\psi = Pe_1 + (1-\psi)PY_-$. These constants make its ratio strictly less than c_{lo} , so it is excluded.

For $\delta \leq \delta_0$, the upper-tail exclusion has coefficient $\sqrt{(2-\delta)c_{Q,0}} - \sqrt{2H/(n\delta)} - \sqrt{c_{\text{hi}}\delta H} > 3\sqrt{c_{Q,0}}/4$ in front of x . The choice L_0 excludes $x \geq L_0$ when $L_0\delta \leq 1$. Otherwise only the physical lower endpoint $\ell \geq 0$ is used. For $\delta > \delta_0$, endpoint exclusion alone gives a uniform span. Consequently

$$\frac{\Lambda(u)}{\Lambda(\ell)} \leq C_{\text{span}} := \max\{2L_0/\eta_0, 2/(\eta_0\delta_0)\} \tag{S67}$$

with probability at least $1 - \alpha_B - \beta$. The constants establish the persistence-time order, not a short universal prescription.

Taking logarithms of the two terms defining $\mathcal{S}_a$ proves the main-text filter-placement identity. For $a_j = 1 - 2^{-j}$, $\Lambda(a_j) = 2^{j+1} - 1$, and adjacent ratios are at most three. A covered hull midpoint is within $(\log 3)/2$ of a bank node. If $1 - u \geq \eta_0\delta$ and $n\delta \geq 2/\eta_0$, then $\Lambda(u) \leq n$, while the last node is at least $2n - 1$. Thus the midpoint is covered and $\min_j \mathcal{S}_{a_j} \leq 3C_{\text{span}}$. The number of members remains $\lceil \log_2 n \rceil + 1$ in all testing bounds.

S5.6 Recording errors and continuous inversion

Source inversion must cover the ideal accepted set after recording errors change the residual energies. We enlarge the acceptance inequalities before solving them. This preserves coverage, but

can widen the returned hull and weaken its covariance bound. The relaxed-hull calculation below therefore checks usefulness separately from this coverage transfer. Suppose each recorded value differs from its ideal value by at most η . If z is this error sequence, its projected innovation error has norm at most

$$d_r(\psi) = (1 + \psi)\sqrt{n}\eta, \quad a_r(\psi) = 2\sqrt{n-1}\eta + \frac{n}{2}\sqrt{n}\eta(1 - \psi) \quad (\text{S68})$$

for the cumulative error. To prove the latter, at $\psi = 1$ its j th entry is $z_j - (1 - j/n)z_0 - (j/n)z_n$, of absolute value at most 2η . The remaining term is $(1 - \psi)K_BPz_-$, and $\|K_BP\| \leq n/2$.

Observed D and Q are convex squared norms. Their endpoint maxima $D_{\max}$ and $Q_{\max}$ bound them over $[0, 1]$. Hence

$$\Delta_D = 2\sqrt{D_{\max}}d_r + d_r^2, \quad \Delta_Q = 2\sqrt{Q_{\max}}a_r + a_r^2 \quad (\text{S69})$$

bound the ideal-to-observed moment changes. Subtract $\Delta_Q + nc\Delta_D$ from the corresponding observed cumulative acceptance polynomial. For a lag polynomial $L + kD$, subtract $(1 + |k|)\Delta_D$, since the symmetric lag operator has norm at most one. Every ideal accepted candidate remains accepted. Square-root constants are rounded up before the exact polynomial expansion; individual coefficients are not rounded separately. This proves the recording transfer with the original error allocation.

The continuous solver isolates roots of at most four quadratic polynomials. It partitions $[0, 1]$ at the root-interval endpoints and uses exact range or constant-sign tests. It retains isolated roots and unresolved intervals. After normalizing a nonzero quadratic leading coefficient to ± 1 , a root interval has width at most 2^{-r-1} at root precision r . There are at most eight intervals. Any retained cell crossing a strictly excluded region is within their connected union, of total width at most $8 \cdot 2^{-r-1}$. Thus an outward endpoint enlargement $8 \cdot 2^{-r}$ suffices. A retained component need not contain an exact accepted point for this bound to hold. Linear, constant, repeated-root and endpoint-root cases use the same sign convention. If a declared work limit prevents completion, the calculation returns the source fallback. Without a finite certificate, the bank abstains. An unexpected execution failure stops the run and is retained.

Bounding the relaxed hull. For the unit-variance population calculation, use the five-event constants above and source coverage. Put $A_+ = \sqrt{c_{\text{hi}}}F_+$. Let a_0, a_1, b_r be outward constants in $a_r(\psi) = a_0 + a_1(1 - \psi)$ and $d_r(\psi) = b_r(1 + \psi)$. With square-root error $\epsilon_s = 2^{-80}$, define

$$\begin{aligned} u_r(x) &= (a_0 + a_1\delta x)/(n\sqrt{\delta}), & w_r &= 2b_r/\sqrt{n\delta}, \\ C_Q &= (A_+ + b_e)/\delta + u_r(1/\delta) + \epsilon_s/(n\sqrt{\delta}), \\ C_D &= F_+ + \max(1, 1/\delta - 1)E_0 + w_r + \epsilon_s/\sqrt{n\delta}, \\ d_D &= 2C_Dw_r + w_r^2, & d_Q(x) &= 2C_Qu_r(x) + u_r(x)^2. \end{aligned} \quad (\text{S70})$$

These bounds include the observed endpoint maxima in Eq. (S69). To control the returned hull, we now work from recorded acceptance back to an ideal inequality. One error allowance covers the change in moments from ideal to recorded values. The other covers the relaxation already added to retain ideal candidates. Recorded acceptance therefore implies the ideal inequalities with twice the error allowance. Define

$$\begin{aligned} p_{\text{lo}}(x) &= [b_e + x(A_+ - b_e)]^2 - c_{\text{lo}}[F_- - E_0 + xE_0]^2 + 2[d_Q(x) + c_{\text{lo}}d_D], \\ p_{\text{hi}}(x) &= [b_e + x(A_- - b_e)]^2 - c_{\text{hi}}[F_+ - E_0 + xE_0]^2 - 2[d_Q(x) + c_{\text{hi}}d_D]. \end{aligned} \quad (\text{S71})$$

Convexity, $p_{\text{lo}}(0) < 0$ and $p_{\text{lo}}(\eta_*) < 0$ exclude $0 \leq x \leq \eta_* \leq 1$. Convexity, $p_{\text{hi}}(L_*) > 0$ and $p'_{\text{hi}}(L_*) > 0$ exclude $x \geq L_* \geq 1$. The numerator lower bound is positive under $A_- > J_{\text{hi}}$. If the upper exclusion fails, $L_* = 1/\delta$ supplies only the physical endpoint zero. Enlarge the resulting ψ interval by $8 \cdot 2^{-r}$ at both ends. All tests in these steps are enclosing inequalities.

S5.7 Endpoint probabilities and the converse

We now prove the endpoint comparison in the main-text discussion of endpoint exclusion. The sufficient calculation bounds the probability of a nonempty set whose upper endpoint is below one. The converse bounds that same probability over constant-shift-invariant rules with uniform coverage. Neither calculation is a detection-power bound.

Normalize the first differences by innovation standard deviation. Their covariance is A_ϕ , where

$$(A_\phi)_{ii} = \frac{2}{1 + \phi}, \quad (A_\phi)_{ij} = -\frac{1 - \phi}{1 + \phi} \phi^{|i-j|-1} \quad (i \neq j). \quad (\text{S72})$$

It tends to I_n as $\phi \uparrow 1$. Let $H_c = PK_B^\top K_B P - ncP$. Then $R_n(1) \leq c$ is the event that a quadratic form with eigenvalues of $A_\phi^{1/2} H_c A_\phi^{1/2} / n^2$ is nonpositive. Eigenvalue enclosures can be computed from the similar matrix $A_\phi H_c / n^2$. The exact zero direction is removed only when the other eigenvalue intervals exclude zero. Equations (S59)–(S60) then give a finite-dimensional probability enclosure.

A shift-invariant upper confidence rule is a fixed function of the differences. Define its value as one if its underlying set is empty. Assume coverage is uniform over all constant means, positive marginal variances, and persistence values. Fix an innovation standard deviation $\sigma > 0$. At persistence $\theta < 1$, choose marginal variance $v_\theta = \sigma^2 / (1 - \theta^2)$, so the difference law is $N(0, \sigma^2 A_\theta)$. For each $b < 1$ and $\theta > b$, coverage gives $\Pr_{\theta, \sigma}(U \leq b) \leq \alpha$. These laws converge in total variation to $N(0, \sigma^2 I_n)$ as $\theta \uparrow 1$. Passing to this limit, then letting $b \uparrow 1$, gives $\Pr_{1, \sigma}(U < 1) \leq \alpha$. Divide both comparison experiments by the same fixed σ and apply the original rule to σ times the normalized data. The rule itself is unchanged: normalized differences are multiplied back by the fixed σ before the rule is applied. Thus this comparison does not assume scale invariance. For any target alternative, choose $\sigma^2 = v(1 - \phi^2)$.

The Neyman–Pearson test between these two fixed-scale Gaussian laws therefore bounds every such endpoint rule. In normalized coordinates the laws are $N(0, I_n)$ and $N(0, A_\phi)$. If a_i are the positive eigenvalues of A_ϕ , its log-likelihood ratio, apart from a constant, is $\frac{1}{2} \sum_i (1 - a_i^{-1}) Z_i^2$ under the endpoint null. Under the alternative it has the same distribution as $\frac{1}{2} \sum_i (a_i - 1) Z_i^2$, apart from that constant. Enclose the $(1 - \alpha)$ null quantile and evaluate the alternative upper tail in the conservative direction. Giving the comparison test the innovation scale enlarges the admissible class. Thus this is an upper bound for unknown-scale invariant rules, not a claim of an attained envelope within that class [9, 10].

At $\phi = 0.8$, the cutoffs 0.019 and 1.1 at $n = 256$ have tails at most 0.0025 each. The probability of $R_n(1) < 0.019$ is at least 0.7767783157. Subtracting source noncoverage gives 0.7717783157 for a nonempty finite endpoint. At $n = 64$, the simple-law upper bound is below 0.297738472 for error 0.005. Every shorter difference vector is a marginal of this experiment, so a target probability 0.75 requires $n \geq 65$. The sufficient-to-necessary bracket is at most 256/65, or 257/66 in observation counts. This calculation has fixed ϕ , size and target probability; it is separate from detection power.

S5.8 Source hulls and the bank error allocation

Reference coverage is a marginal result. Using it to bound both testing channels requires their common persistence and the stated response model. Under those assumptions, each common filter has a source spectral ratio that can be bounded over the entire returned hull. Simultaneous testing events then cover every declared member, including a member selected through its observed rejection margin. The source-confidence argument uses only the noiseless scalar reference marginal. The detection model additionally assumes two jointly stationary Gaussian channels with the same autoregressive marginal persistence. The cross-spectrum can be arbitrary subject to positive semidefiniteness. Under the null it is real and even before the detector responses. The first detector is a nonzero scalar. The second has a nonzero finite impulse response, unchanged between the independent calibration and the test record. Testing errors are zero apart from a recorder with the stated deterministic error bound.

For a bank coefficient a , filtering the source spectrum gives a frequency-dependent factor proportional to

$$f_{a,\phi}(\omega) = \frac{1 + a^2 - 2a \cos \omega}{1 + \phi^2 - 2\phi \cos \omega}.$$

This ratio of affine functions of $\cos \omega$ is monotone or constant. Its extreme values occur at $\omega = 0, \pi$. Their ratio is $\max\{[\Lambda(\phi)/\Lambda(a)]^2, [\Lambda(a)/\Lambda(\phi)]^2\}$. Taking the supremum over a hull $[\ell, u]$ gives the expression for $\mathcal{S}_a$ in the main text. A response-shape certificate P_2 therefore gives the standardized covariance envelope $q_a = 2\mathcal{S}_a \max(1, P_2)$. The source and response gains cancel from this normalized bound.

A closed outer hull preserves source coverage. An empty hull, a hull reaching one, or a failed required numerical bound gives no finite source certificate. The bank then abstains. On the shared source and calibration coverage events, apply the variance and quadratic-tail inequalities simultaneously to every declared member. Source error 0.005, calibration error 0.01, variance error 0.01, and null contrast error 0.025 sum to 0.05. The last two allocations are divided over all M members. No independence across members or between fitting and the test statistic is used. The deterministic recording relaxations preserve the same event inclusion. This is why a fitted source set can be used in the complete threshold without a conditional Gaussian claim.

S5.9 From a source envelope to a power condition

For size control, a fitted variance ceiling must not be below the true variance. For power, we also need an upper bound on the fitted threshold. We therefore use a variance upper event in addition to the source and calibration events. The resulting sufficient condition contains separate terms for the null threshold, the alternative fluctuation, and centering bias. Fix a member a of a bank declared before observation. Suppose the source and calibration events give $\mathcal{S}_a \leq \mathcal{S}_0$, response shape at most P_0 , and phase tolerance at most r_0 . Set $q_0 = 2\mathcal{S}_0 \max(1, P_0)$. Let m be the retained length, $r = m - 1$, and let V_i be the two true filtered variances. Put $V_\star = \max(V_1, V_2)$. The full filtered covariance has norm at most $q_0 V_\star$ and trace at most $2mV_\star$.

With $t_v = \log(2/\beta_v)$, a common Gaussian upper event gives both empirical centered variances at most

$$V_+ = V_\star \{1 + 2\sqrt{q_0 t_v / m} + 2q_0 t_v / m\}.$$

Its failure probability is at most β_v . The centered norm is bounded by the norm centered at the population mean, so unknown constant means add no term. Require $c_m(q_0) > 0$ and put $D_+ = V_+ / c_m(q_0)$. The sharp two-time radius is

$$R(q, D, m, t) = D \left[2 \min \left\{ qF_r, \frac{\sqrt{2qm}}{r} \right\} \sqrt{t} + \frac{2qt}{r} \right], \quad F_r^2 = \frac{r+1}{r^2} - \frac{2}{r^3} - \frac{2}{r^4}. \quad (\text{S73})$$

It follows from the operator bound $1/r$, the exact Frobenius norm, and the trace bound. The geometric variance ceiling is no larger than the common ceiling D_+ .

Let κ_a be the true signed filtered contrast and M the bank size. Its centering bias is at most $b_0 = q_0 V_\star \sqrt{2(4r-2)}/r^2$. Therefore a sufficient ideal contrast condition is

$$|\kappa_a| > 2r_0 D_+ + R(q_0, D_+, m, \log(80M)) + R(q_0, V_\star, m, \log(2/\beta_s)) + b_0. \quad (\text{S74})$$

The power loss is bounded by the source-event failures, calibration-event failures, β_v , and β_s . These events are intersected by inclusion; the argument does not assume a Gaussian distribution conditional on fitted quantities. Bank placement, the filtered contrast, and variance-denominator positivity remain in Eq. (S74).

Calibration upper and lower events. For the implemented-power example, the response is $(1, 0)$, $m_c = 256$, $X_c^\top X_c = 256I_2$, and independent mean-zero Gaussian calibration errors have standard deviation $\sigma_c = 0.01$. Put $\nu = 254$, $t_c = \log 200$, and

$$C_c = 2 + 2\sqrt{2t_c} + 2t_c, \quad V_{c,+} = \nu + 2\sqrt{\nu t_c} + 2t_c, \quad V_{c,-} = \nu - 2\sqrt{\nu t_c}.$$

Three events each have failure at most 0.005: coefficient error at most $\sigma_c \sqrt{C_c/256}$, residual norm at most $\sigma_c \sqrt{V_{c,+}}$, and residual norm at least $\sigma_c \sqrt{V_{c,-}}$. The last event keeps the recorded residual positive.

Recording error η changes the fitted coefficient vector by at most η and the residual norm by at most 16η . The corrected radius uses the observed residual norm plus 16η , followed by the usual confidence factor and the coefficient-center allowance. Its upper population bound is

$$R_c \leq \left(\sigma_c \sqrt{V_{c,+}} + 32\eta + \epsilon_s \right) \sqrt{\frac{C_c}{256V_{c,-}}} + \eta, \quad (\text{S75})$$

with outward evaluation. Put $\zeta = \sigma_c \sqrt{C_c/256} + \eta + R_c + \epsilon_s$. The fitted coefficient ball lies in the radius- ζ ball about $(1, 0)$. The response bounds are then

$$\hat{r} \leq \frac{\zeta}{1-2\zeta}, \quad \hat{P} \leq \left(\frac{1+\sqrt{2}\zeta}{1-\zeta} \right)^2.$$

At $\eta = 2^{-21}$, enclosing evaluation gives

$$\hat{r} \leq \frac{33}{5000}, \quad \hat{P} \leq \frac{129}{125}, \quad \sigma_c \sqrt{V_{c,-}} > 16\eta. \quad (\text{S76})$$

Their event probability is at least 0.985. These upper bounds require the residual upper event as well as confidence-region coverage.

Recording correction in the contrast condition. For a filter $(-a, 1)$, each transformed recording error is at most $\eta_a = (1+a)\eta$. An observed variance $\hat{V}_i$ gives ideal variance at most $(\sqrt{\hat{V}_i} + \eta_a)^2$. If U_t is the adjacent sum in channel one and W_t the adjacent difference in channel two, their centered energies are bounded by $4m\hat{V}_1$ and $4m\hat{V}_2$. Expanding the centered product gives the deterministic contrast error bound

$$\Omega = 2\eta_a \sqrt{4m\hat{V}_2/r} + 2\eta_a \sqrt{4m\hat{V}_1/r} + 4\eta_a^2. \quad (\text{S77})$$

The recording error enters the power comparison on both sides. One Ω enlarges the implemented threshold. A second Ω allows the recorded statistic to be smaller than the ideal statistic. This explains the total correction used below. No independence of recording error is used.

Table S1: Population sufficient conditions for $\phi = 0.995$ and coupling 0.9, with the standalone finite- n cutoffs and bounded recording correction. Every length has $N = n + 1$ observations. A failed sufficient condition does not determine empirical power.

Innovations n	First failed condition or resulting margin
2048, 4096, 8192, 16384, 32768, 65536	Source-envelope inequality
131072	Positive variance denominator
262144	Contrast margin < -0.0157610
524288	Contrast margin > 0.0023006

For the specified source pair, both true filtered variances and the absolute contrast are

$$V_a = 1 + a^2 - 2a\phi, \quad |\kappa_a| = \frac{9}{10}(1 - \phi^2)(1 + a^2 - a\phi).$$

At $n = m = 524288$, $a = \phi = 199/200$, the standalone cutoffs are

$$c_{\text{lo}} = \frac{63669}{3276800}, \quad c_{\text{hi}} = \frac{130687}{131072}. \quad (\text{S78})$$

Use $\beta = 0.05$, $c_Q = 0.023$, and Eqs. (S63)–(S71). The rounded hull and root enlargement give $\mathcal{S}_a < 335.920$. Its corresponding q_0 is less than 694. Take $\beta_v = \beta_s = 0.01$. For each variance, first bound the recorded value by $(\sqrt{V_+} + \eta_a)^2$, then apply its outward variance correction again. Evaluate the null threshold at these upper inputs and at Eq. (S76). Subtract the alternative radius, centering bias, and 2Ω from $|\kappa_a| = 3591/400000$. The enclosing result is strictly greater than 0.0023006294. The calculation includes 80-bit outward square roots and 48-term logarithm enclosures. Monotonicity in q , phase, and both variance ceilings makes this an upper-threshold comparison for every record on the stated events.

The same source-envelope sufficient inequalities fail for all six pairs $\phi \in \{0.8, 0.97, 0.995\}$ and $n \in \{16384, 65536\}$. Table S1 gives the complete declared length grid at $\phi = 0.995$. Only its largest length is linked here to a complete numerical power theorem.

S5.10 Numerical completion and the probability bound

The positive contrast margin proves rejection only if the calculation reaches the comparison with enclosing inputs. Gaussian records are unbounded, whereas the implementation has finite input, work, and precision limits. We add bounded-magnitude events and show that the source solver, calibration, and selected member complete on their intersection with the statistical events. Their complement probabilities then enter the final rejection bound. The numerical rule uses 48 root bits, 80 square-root bits, and 48 terms for each reduced logarithm series. It allows 600000 observations, 5×10^8 charged arithmetic operations per phase, and 262144 bits per rational value. Input, regression-column, and bank limits are 4096 bits, 32 columns, and 64 members. Every source or calibration value is an integer divided by 2^{20} .

Add the event that all $2N$ population-demeaned source values have magnitude at most 16. Its complement has probability at most $4Ne^{-128}$. Add the event that all 256 standardized calibration errors have magnitude at most 16, with complement at most $512e^{-128}$. These union bounds require no independence between source observations. The specified means are 2 and -1 , so source input integers have magnitude below 2^{25} .

Source arithmetic. An exact dyadic compiler forms each moment at $\psi = -1, 0, 1$ and interpolates its quadratic coefficients. Prefix integers have magnitude below 2^{45} . A dot product is

evaluated in checked integer chunks with exact integer totals. If a product is too large, one split into base- 2^{26} limbs suffices; all four child products fit the direct integer range. A direct dot product charges twice its length, and a split dot product at most ten times its length. For each candidate value, residual construction, sums and norms charge $12n$ operations. Lag and reflection products and sums charge $8(n-1)$, and the two prefix products charge at most $20(n-1)$. Including input conversion gives at most

$$2N + 3\{40(N-1) - 28\} = 122N - 204 = 63963054$$

charged operations. Exact cumulative numerators have fewer than 151 bits before reduction; the common denominators and interpolation remain below the conservative 282-bit bound for compiled coefficients.

Define the height of a rational as the larger bit length of its reduced numerator and denominator. Products and quotients add heights; sums add heights and one bit. In the rounded source inequalities, coefficient heights are below 4096. Normalization gives 8192, discriminants 32769, and raw root endpoints 20000. Clipped roots lie in $[0, 1]$ and have denominator height at most $8192 + 49 + 2 = 8243$. Cell midpoints therefore have height below 16500. Horner evaluation has height below $3(4096) + 2(16500) + 3 < 45300$. Vertex and raw-root evaluations stay below 60000. The root calculation has at most four quadratics, eight root intervals, 18 endpoints, and 17 open cells. Including the recording relaxation, fewer than 5000 arithmetic calls suffice. Repeated or overlapping roots are retained and cause no refinement loop.

The rounded population event gives a nonempty source set with upper endpoint below one. The solver returns hull endpoints of height below 8243. These satisfy the calibrated arithmetic condition in Proposition S8. In particular, calibration and its recording correction each use fewer than 10000 operations and 10000 bits, while each member uses fewer than 4000 operations and 140000 bits. The selected member has an outward denominator lower endpoint greater than $4/5$, so it reaches the strict rejection comparison. Other members can return unavailable results without preventing its rejection.

Finally, the statistical event failures sum to

$$0.005 + 0.05 + 0.015 + 0.01 + 0.01 = 0.09,$$

for source coverage, the source envelope, calibration, the two variance events, and the alternative tail. The two bounded-input events add at most $(4N + 512)e^{-128}$. The positive margin and the completion bounds therefore prove

$$\Pr(\text{reject}) \geq 0.91 - (4N + 512)e^{-128} > 0.90.$$

This statement concerns the ideal Gaussian model followed by a recorder with the specified deterministic error bound. It does not identify the law of a finite-precision random generator with the ideal law, nor assert completion for every unbounded Gaussian input.

S6 Finite recording-cost bounds for a spectral class

This comparison isolates dependence by making the response known. The necessary bound uses one reversible law and one alternative fixed across recording lengths. Their difference is concentrated in narrow frequency bands. A finite trace-defect bound controls how much of that difference a record can resolve, and relative entropy converts it into a restriction on every test.

The sufficient bound instead covers the full spectral class for the reflection test. These scopes remain distinct when their lengths are compared. Finite-time and frequency concentration is

classical [11, 12]; the inequalities below retain finite-length control rather than replacing the concentration matrix by its asymptotic limit.

S6.1 The class and the fixed pair

To make a difficult pair with a fixed covariance contrast, we concentrate its imaginary cross spectrum near frequencies where the lag-one sine has large magnitude. The common marginal spectra retain a positive floor. Their ceiling limits the height of the bands, so increasing the allowed ratio narrows the bands while preserving unit variance. The pair remains fixed as the recording length changes.

Use the convention

$$C_{ij}(k) = \int_{-\pi}^{\pi} e^{ik\omega} f_{ij}(\omega) \frac{d\omega}{2\pi}.$$

Let $\mathcal{G}(\mathcal{S})$ contain real, zero-mean, stationary two-channel Gaussian processes with unit marginal variances and Hermitian positive-semidefinite densities satisfying

$$f(-\omega) = \overline{f(\omega)}, \quad \frac{1}{2} \leq f_{ii}(\omega) \leq \frac{\mathcal{S}}{2} \quad \text{almost everywhere.} \quad (\text{S79})$$

The response is known to be the identity and observation errors are zero. The null consists of reversible laws. The alternative has

$$|\kappa| \geq \gamma, \quad \kappa = C_{12}(1) - C_{12}(-1). \quad (\text{S80})$$

Assume $\mathcal{S} \geq 3$ and $0 < \gamma \leq 1/4$.

Put $\nu = \mathcal{S} - 1$, $w = \pi/(2\nu)$ and

$$\begin{aligned} E &= [\pi/2 - w, \pi/2 + w] \cup [-\pi/2 - w, -\pi/2 + w], \\ b(\omega) &= \frac{\nu}{2} \mathbf{1}_E(\omega), \quad g(\omega) = \frac{1}{2} + b(\omega), \quad \theta = \gamma \frac{w}{\sin w}. \end{aligned} \quad (\text{S81})$$

Define the two fixed spectra

$$f^{(0)} = gI_2, \quad f^{(1)} = \begin{pmatrix} g & i\theta \operatorname{sgn}(\omega)b \\ -i\theta \operatorname{sgn}(\omega)b & g \end{pmatrix}. \quad (\text{S82})$$

The normalized measure of E is $1/\nu$. Hence $\int b d\omega/(2\pi) = 1/2$ and both marginal variances are one. Their common marginal floor and ceiling are $1/2$ and $\mathcal{S}/2$.

Since $w \leq \pi/4$, monotonicity of $w/\sin w$ gives

$$0 < \theta \leq \frac{\pi\gamma}{2\sqrt{2}} \leq \frac{\pi}{8\sqrt{2}} < \frac{1}{2}.$$

The alternative eigenvalues are $1/2 + (1 \pm \theta)b > 0$. Both spectra satisfy real-process symmetry and define stationary Gaussian laws on all integer times. The null spectrum is real and even, so its law is reversible. Direct integration gives

$$C_{12}^{(1)}(1) = -\frac{\theta \sin w}{2w}, \quad C_{12}^{(1)}(-1) = \frac{\theta \sin w}{2w},$$

and therefore $\kappa = -\gamma$. The pair depends only on $\mathcal{S}$ and γ , not on the recording length.

S6.2 Concentration eigenvalues and their finite trace defect

The finite record sees each frequency band through a Toeplitz matrix. Its eigenvalues describe how strongly finite-record directions overlap the bands. The trace sums these overlaps. The trace defect instead sums each eigenvalue times one minus that eigenvalue, so it measures the contribution from directions that overlap the bands only partly. This extra quantity controls the remainder when the normalized cross-covariance energy is bounded by a multiple of the trace.

For a scalar symbol u , let

$$[T_N(u)]_{jk} = \int e^{i(j-k)\omega} u(\omega) \frac{d\omega}{2\pi}, \quad 0 \leq j, k < N.$$

Set

$$\begin{aligned} \mathbf{C}_N &= T_N(\mathbf{1}_E), & \mathbf{G}_N &= \frac{1}{2}(I_N + \nu \mathbf{C}_N), \\ \mathbf{A}_N &= T_N(i \operatorname{sgn}(\omega)b), & \mathbf{K}_N &= \nu \mathbf{C}_N(I_N + \nu \mathbf{C}_N)^{-1}. \end{aligned} \quad (\text{S83})$$

The eigenvalues of $\mathbf{C}_N$ lie in $[0, 1]$, because it is the compression of multiplication by an indicator. Its trace is N/ν .

Lemma S4 (A finite trace-defect bound). *For $N \geq 2$, define*

$$V_N^+ = \min \left\{ \frac{N}{\nu}, \frac{N}{4}, \frac{12 + 4 \log \max(1, \lfloor (N-1)/2 \rfloor)}{\pi^2} \right\}. \quad (\text{S84})$$

Then $\operatorname{tr}(\mathbf{C}_N - \mathbf{C}_N^2) \leq V_N^+$. With

$$c_\nu = \frac{\nu^2}{(1+\nu)^2}, \quad d_\nu = \frac{\nu^4}{4(1+\nu)^3}, \quad (\text{S85})$$

one also has

$$\operatorname{tr}(\mathbf{K}_N^2) \leq c_\nu N/\nu + d_\nu V_N^+. \quad (\text{S86})$$

Proof. The Fourier coefficients of $\mathbf{1}_E$ are

$$c_0 = 1/\nu, \quad c_{2m} = \frac{(-1)^m \sin(2wm)}{\pi m} \quad (m \neq 0), \quad c_{2m+1} = 0.$$

The Toeplitz trace identity and Parseval's identity give

$$\operatorname{tr}(\mathbf{C}_N - \mathbf{C}_N^2) = N \sum_{|k| \geq N} |c_k|^2 + \sum_{|k| < N} |k| |c_k|^2. \quad (\text{S87})$$

Let $M = \lceil N/2 \rceil$ and $H = \lfloor (N-1)/2 \rfloor$. Since x^{-2} is convex,

$$\sum_{m \geq M} \frac{1}{m^2} \leq \int_{M-1/2}^{\infty} \frac{dx}{x^2} = \frac{1}{M-1/2}.$$

The first term of (S87) is therefore at most $2N/[\pi^2(M-1/2)] \leq 8/\pi^2$. The second is at most $(4/\pi^2) \sum_{m=1}^H 1/m$. For $H \geq 1$, the harmonic sum is at most $1 + \log H$; for $H = 0$, it is zero. This proves the logarithmic bound in (S84). The other two bounds follow from $x(1-x) \leq x$ and $x(1-x) \leq 1/4$ for eigenvalues $x \in [0, 1]$.

For $0 < x < 1$,

$$\frac{(\nu x/(1+\nu x))^2 - c_\nu x}{x(1-x)} = \frac{\nu^2(\nu^2 x - 1)}{(1+\nu)^2(1+\nu x)^2}.$$

The derivative has the sign of $\nu + 2 - \nu^2 x$. Its maximum on $[0, 1]$ occurs at $x = (\nu + 2)/\nu^2$ for $\nu \geq 2$ and equals d_ν . The endpoint inequality follows by continuity. Thus

$$\left(\frac{\nu x}{1+\nu x} \right)^2 \leq c_\nu x + d_\nu x(1-x).$$

Sum this inequality over the eigenvalues of $\mathbf{C}_N$ to obtain (S86). □

For any $0 < \varepsilon < 1/2$, the number of concentration eigenvalues in $[\varepsilon, 1-\varepsilon]$ is at most $V_N^+ / [\varepsilon(1-\varepsilon)]$. Each such eigenvalue contributes at least $\varepsilon(1-\varepsilon)$ to the trace defect. This is a finite bound, not an asymptotic replacement for that defect.

S6.3 A necessary recording length

The two laws share their marginal covariance. After normalization by that covariance, only the skew cross block distinguishes them. Its singular values determine their Gaussian relative entropy. The preceding trace bound gives an upper bound on this information, while a test with the requested size and power requires a lower bound. The two bounds must be compatible at any admissible recording length.

Theorem S8 (Finite necessary condition). *Let a possibly randomized test have size at most α and power at least $1 - \beta$ for the fixed pair (S82), where $0 < \alpha, \beta < 1$ and $\alpha + \beta < 1$. For $N \geq 2$, define*

$$\mathcal{D}_N = \frac{-\log(1 - \theta^2 c_\nu)}{2c_\nu} \{c_\nu N / \nu + d_\nu V_N^+\}. \quad (\text{S88})$$

The test must satisfy

$$\mathcal{D}_N \geq (1 - \beta) \log \frac{1 - \beta}{\alpha} + \beta \log \frac{\beta}{1 - \alpha}. \quad (\text{S89})$$

At $\alpha = \beta = 0.05$, the right side is $0.9 \log 19$.

Proof. The real matrix $\mathbf{A}_N$ is skew-symmetric. Put

$$\mathbf{L}_N = \mathbf{G}_N^{-1/2} \mathbf{A}_N \mathbf{G}_N^{-1/2}.$$

The block matrix with diagonal blocks $T_N(b)$ and off-diagonal blocks $\mathbf{A}_N, -\mathbf{A}_N$ is positive semidefinite. Its spectral symbol has eigenvalues zero and $2b$. The matrix $T_N(b)$ is positive definite because a nonzero trigonometric polynomial cannot vanish on both bands almost everywhere. Normalization by this matrix gives a contraction U_N such that

$$\mathbf{L}_N = \mathbf{K}_N^{1/2} U_N \mathbf{K}_N^{1/2}, \quad \|U_N\|_{\text{op}} \leq 1.$$

It follows that $\|\mathbf{L}_N\|_{\text{op}}^2 \leq c_\nu$. In an eigenbasis of $\mathbf{K}_N$, the contraction makes each row and column sum of $|u_{jk}|^2$ at most one. Hence

$$\|\mathbf{L}_N\|_F^2 = \sum_{j,k} k_j k_k |u_{jk}|^2 \leq \frac{1}{2} \sum_{j,k} (k_j^2 + k_k^2) |u_{jk}|^2 \leq \text{tr}(\mathbf{K}_N^2).$$

Normalize the alternative record covariance by the null covariance. The resulting matrix is

$$\begin{pmatrix} I_N & \theta \mathbf{L}_N \\ -\theta \mathbf{L}_N & I_N \end{pmatrix},$$

with trace $2N$. The real Gaussian relative entropy is therefore

$$D_{\text{KL}}(P_1^{(N)} \| P_0^{(N)}) = -\frac{1}{2} \sum_{j=1}^N \log(1 - \theta^2 s_j(\mathbf{L}_N)^2). \quad (\text{S90})$$

All N singular values occur with their multiplicities. The factor $1/2$ is the real Gaussian factor and needs no further channel multiplier.

The exact divergence depends nonlinearly on the squared singular values. We need an upper bound that uses only their sum and their common ceiling. The function $x \mapsto -\log(1 - \theta^2 x)$ is convex and vanishes at zero. Its chord on $[0, c_\nu]$ gives

$$-\log(1 - \theta^2 x) \leq \frac{-\log(1 - \theta^2 c_\nu)}{c_\nu} x.$$

Apply this inequality in (S90), then use Lemma S4, to obtain the left side of (S89) as an upper bound for the divergence.

Data processing through the binary test bounds that divergence below by the Bernoulli divergence between its power and size. This is the two-hypothesis testing reduction [13]. For power at least $1 - \beta$ and size at most α , with $\alpha + \beta < 1$, the Bernoulli divergence is smallest at those two endpoints. Differentiating with respect to its two probabilities verifies the monotonicities. This gives the right side of (S89). Independent randomization has the same law under both hypotheses and adds no input divergence. $\square$

At $N = 1$, the pair has identical laws because its zero-lag cross covariance is zero. It cannot meet the stated size and power requirements. Independent response-calibration data with identical laws under both hypotheses also contribute no divergence. Equal calibration marginals without independence from the test record would not give that conclusion.

S6.4 A class-wide sufficient recording length

The difficult pair bounds every test from below in recording length. To obtain a sufficient length, we return to the reflection rule and ask whether it rejects every member of the alternative class. This requires an upper bound on its fitted threshold as well as a lower bound on its statistic. The variance upper event, alternative tail radius, and centering bias below provide those three controls.

Positive semidefiniteness and (S79) give $\lambda_{\max}(f) \leq \text{tr } f \leq \mathcal{S}$. Since the marginal variances are one, the complete standardized covariance envelope is $q = \mathcal{S}$. This conclusion uses the specified unit-variance class. It is not a general removal of the channel factor from an unknown marginal spectral-ratio bound.

For $N \geq 3$ and $n = N - 1$, use the reflection matrix and radius of Section S4. Set

$$\begin{aligned} u &= \log 200, & t_0 &= \log(200/3), & t_1 &= \log 50, & \zeta &= q/N, \\ c_N &= 1 - \zeta - 2\sqrt{\zeta u}, & d_N &= \frac{1 + 2\sqrt{\zeta u} + 2\zeta u}{c_N}. \end{aligned} \tag{S91}$$

When $c_N > 0$, define

$$\hat{D}_i = Q_i(y)/c_N, \quad \hat{D}_\times = \sqrt{\hat{D}_1 \hat{D}_2}.$$

The test rejects if $|s_n| > \hat{D}_\times \mathbf{r}_N(q, t_0)$. Identity responses give phase tolerance zero. The variance lower event fails with probability at most 0.01, and the null tail fails with probability at most 0.03. Thus size is at most 0.05.

A variance upper event of failure at most 0.01 gives $\hat{D}_\times \leq d_N$. Lemma S2, applied to each positive-semidefinite centered energy, bounds it above by $1 + 2\sqrt{\zeta u} + 2\zeta u$ on this event. The population reflection contrast and centered expectation differ by

$$|\mathbb{E}s_n - \kappa| \leq \frac{q\sqrt{2(4n-2)}}{n^2}. \tag{S92}$$

Indeed, the coefficient vectors of $n\bar{U}$ and $n\bar{V}$ are $(1, 2, \dots, 2, 1)$ and $(1, 0, \dots, 0, -1)$. Their squared norms are $4n - 2$ and two. The full covariance envelope bounds their covariance by their norm product times q .

It follows that

$$\gamma > d_N \mathfrak{r}_N(q, t_0) + \mathfrak{r}_N(q, t_1) + \frac{q\sqrt{2(4n-2)}}{n^2} \quad (\text{S93})$$

is sufficient for power at least 0.95. On the variance upper event and the one-sided alternative tail event, whose failure is at most 0.02, this strict inequality implies rejection. These events need not be independent. The stated 0.05 bounds retain unused calibration allowances and are conservative for known identity responses.

Theorem S9 (Uniform sufficient constant). *For $\mathcal{S} \geq 3$ and $0 < \gamma \leq 1/4$, the preceding test has size at most 0.05 over the reversible class and power at least 0.95 over the alternatives in (S80) whenever*

$$N \geq 1024 \mathcal{S} / \gamma^2. \quad (\text{S94})$$

Proof. The length condition gives $\zeta \leq \gamma^2/1024 \leq 1/16384$. Since $u < 6$,

$$2c_N - (1 + 2\sqrt{\zeta}u + 2\zeta u) = 1 - 6\sqrt{\zeta}u - 2\zeta(u + 1) > 0.$$

For example, the subtracted terms total less than $15/128 + 14/16384 < 1$. Hence $c_N > 0$ and $d_N < 2$.

Since $n \geq 2N/3$, the mixed branch in (S43) gives

$$\mathfrak{r}_N(q, t) \leq 3\sqrt{2\zeta t} + 3\zeta t.$$

The two radius terms in (S93), divided by γ , are therefore at most

$$\frac{3\sqrt{2}(2\sqrt{t_0} + \sqrt{t_1})}{32} + \frac{(3/4)(2t_0 + t_1)}{1024}.$$

The bounds $\sqrt{2} < 3/2$, $\sqrt{t_0} < 9/4$, $t_0 < 5$ and $t_1 < 4$ make this less than $29.25/32 + 10.5/1024 < 0.925$. The bias term divided by γ is at most

$$\frac{3\sqrt{6}\gamma^2}{1024^{3/2}\sqrt{\mathcal{S}}} \leq \frac{3\sqrt{2}}{16 \cdot 1024^{3/2}} < 0.00001.$$

Their sum is strictly less than one. Thus (S93) holds throughout the stated parameter range. $\square$

S6.5 A finite numerical comparison and its scope

At $\mathcal{S} = 9$ and $\gamma = 0.1$, outward interval evaluation gives the following finite comparison.

Quantity	Recording length	Scope
Necessary lower bound	5214	Every test for the fixed pair
Sufficient condition	129580	The stated test over the full class

At $N = 5213$, the divergence upper bound is below 2.649607, while $0.9 \log 19 > 2.649995$. The upper bound in (S89) is nondecreasing in N , so no $N \leq 5213$ can meet both error targets for this pair. The conclusion $N \geq 5214$ is a necessary condition, not an attainable detection length.

At $N = 129580$, the right side of (S93) is below 0.099999875. At $N = 129579$, it exceeds 0.100000280. These intervals locate the stated sufficient-condition crossing. They concern the ideal Gaussian recording and the mathematical test. They do not include a uniform finite-arithmetic completion claim.

Direct floating-point evaluation of the exact determinant expression (S90) gives approximately 2.62761 at $N = 5214$. This value is a numerical evaluation without a certified determinant enclosure. It does not strengthen the necessary integer bound. The two lengths compare a fixed-pair information bound with a sufficient condition for one class-wide procedure. Their separation does not establish either constant as optimal.

S7 Physical Gaussian models and sampling

The tests require covariance and tail bounds that cover the complete observed process. A specified physical model can supply them through relaxation, stationary covariance, and detector responses. We first derive this connection for a linear diffusion and then apply it to the Brownian and rotational families. The rotational model also separates two questions. Its sampled cycle can reveal asymmetry, but its continuous-time circulation can change without changing the sampled law.

S7.1 Bounds from an Ornstein–Uhlenbeck model

The drift must control the norm of the time-evolution matrix, not only its eigenvalues. The symmetric-part condition below gives exponential contraction in every direction. Integrating the contracted noise input bounds the stationary covariance, and summing the sampled covariance decay bounds the full spectral density. The same decay then supplies the cycle tail allowance after filtering.

Consider the stationary diffusion

$$dX(t) = -MX(t) dt + \sqrt{2D} dW(t), \quad \frac{M + M^\top}{2} \succeq a_d I, \quad 0 \preceq D \preceq d_* I, \quad (\text{S95})$$

where $a_d > 0$. The coordinates have even time-reversal parity. At sampling interval $\Delta > 0$, define $\rho = e^{-a_d \Delta} < 1$ and $s_* = d_*/a_d$.

Proposition S9 (Sampled covariance and filter bounds). *The stationary covariance and lag matrices satisfy*

$$\Sigma \preceq s_* I, \quad \|C_X(k)\|_{\text{op}} \leq s_* \rho^{|k|}. \quad (\text{S96})$$

The sampled spectral density and every complete-record covariance have scalar ceiling

$$K_X = s_* \frac{1 + \rho}{1 - \rho}. \quad (\text{S97})$$

Apply real channel filters after sampling. If their absolute coefficient sums satisfy $\|h_i\|_1 \leq g_i$, and mutually orthogonal white detector errors have variances at most ν_i , then

$$\Gamma_N \preceq \text{diag}(K_i I_N), \quad K_i = K_X g_i^2 + \nu_i. \quad (\text{S98})$$

Assume the errors are also source-orthogonal. If d_H bounds every difference between tap positions on a tested edge and $L \geq d_H$, a valid tail allowance is

$$B_{ij} = s_* g_i g_j \left\{ \frac{2\rho^{L+1-d_H}}{1-\rho} + \frac{1}{N} \sum_{|k| \leq L} |k| \rho^{\max(|k|-d_H, 0)} \right\}. \quad (\text{S99})$$

Proof. For $u(t) = e^{-Mt}u(0)$, the drift assumption gives

$$\frac{d}{dt} \|u(t)\|^2 = -u(t)^\top (M + M^\top) u(t) \leq -2a_d \|u(t)\|^2.$$

Thus $\|e^{-Mt}\|_{\text{op}} \leq e^{-a_d t}$. The stationary integral

$$\Sigma = 2 \int_0^\infty e^{-Mt} D e^{-M^\top t} dt$$

is at most $s_* I$. For $k \geq 0$, $C_X(k) = e^{-Mk\Delta} \Sigma$; negative lags give its transpose. Summing the absolute covariance series proves (S97). Fourier integration then gives the same ceiling for every finite recording.

Let $D_H(\omega) = \text{diag}(H_i(\omega))$. Spectral congruence and $|H_i| \leq g_i$ give $D_H F_X D_H^* \preceq K_X \text{diag}(g_i^2)$. The white errors add at most $\text{diag}(\nu_i)$. This proves the full observed envelope.

Filtering gives

$$|C_{ij}(k)| \leq s_* \sum_{u,v} |h_i(u)h_j(v)| \rho^{|k-u+v|} \leq s_* g_i g_j \rho^{\max(|k|-d_H, 0)}.$$

The two omitted tails sum to the first term of (S99). The finite-window weights give its second term. $\square$

Uniform family bounds can replace s_* and ρ . Signed filter coefficients are covered by their absolute sums. Colored detector errors need their own spectral ceilings. Cross-correlated detector errors also change the cycle null and need a separate argument. A deterministic zero channel can use any positive conservative ceiling when an interface requires $K_i > 0$.

Stable drift eigenvalues alone do not give (S96). For

$$M = \begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix}, \quad D = I_2,$$

the covariance $\Sigma = \begin{pmatrix} 9 & 2 \\ 2 & 1 \end{pmatrix}$ solves $M\Sigma + \Sigma M^T = 2I_2$. It exceeds I_2 , and $\|e^{-M}\|_{\text{op}} > e^{-1}$. A nonnormal drift therefore needs a semigroup bound, such as the symmetric-part condition in (S95).

S7.2 Brownian gyrator and a shared observed-process bound

For calibrated inference, the absolute covariance ceiling must be compared with each observed variance to give a standardized envelope. The Brownian family provides both an upper and a lower stationary covariance bound. The small delayed detector term affects this comparison as well as the null phase tolerance, so the bound must include the filtered observations.

Take

$$M_g = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}, \quad D_g = \text{diag}(T_1, T_2), \quad 1 \leq T_i \leq 4. \quad (\text{S100})$$

Solving $M_g \Sigma + \Sigma M_g = 2D_g$ gives

$$\Sigma_{11} = T_1 + \frac{T_1 + T_2}{6}, \quad \Sigma_{22} = T_2 + \frac{T_1 + T_2}{6}, \quad \Sigma_{12} = -\frac{T_1 + T_2}{3}. \quad (\text{S101})$$

Substitution verifies both diagonal equations and the zero off-diagonal equation. Equal temperatures give $M_g \Sigma = D_g$ and a reversible source. Unequal temperatures give nonzero stationary current.

The drift eigenvalues are 1/2 and 3/2. Since $I \preceq D_g \preceq 4I$, the stationary integral yields

$$M_g^{-1} \preceq \Sigma \preceq 4M_g^{-1}, \quad \frac{2}{3}I \preceq \Sigma \preceq 8I.$$

For $\Delta = 1$, the rational bound $e^{-1/2} < 61/100$ gives $K_X = 1288/39$. Apply the digital responses

$$H_1(\omega) = 1, \quad H_2(\omega) = 1 + \frac{3}{100}e^{-i\omega}.$$

Their full observed covariance is bounded by $K_X(103/100)^2 I$. The reverse triangle inequality in L^2 gives each observed variance a lower bound $(2/3)(97/100)^2$. Consequently

$$\Gamma_N \preceq 56 \text{diag}(v_1 I_N, v_2 I_N), \quad (\text{S102})$$

because

$$\frac{(1288/39)(103/100)^2}{(2/3)(97/100)^2} < 56.$$

The same $q = 56$ covers the full stated temperature family. It includes the detector responses and is not inferred from response calibration.

The exact sampled transition is $T_\Delta = e^{-M_g \Delta}$, with diagonal entries $(e^{-\Delta/2} + e^{-3\Delta/2})/2$ and off-diagonal entries $(e^{-3\Delta/2} - e^{-\Delta/2})/2$. Its independent innovation covariance is $\Sigma - T_\Delta \Sigma T_\Delta^\top$. These equations specify the stationary sampled Gaussian law without an Euler approximation. The response filters act after this sampling operation.

Cost of the common family bound. A known temperature pair permits a smaller sufficient envelope without changing the saved study. Since $\|C_X(k)\| \leq \lambda_{\max}(\Sigma)e^{-|k|/2}$, the same spectral argument gives

$$q_{\text{cell}} = \lambda_{\max}(\Sigma) \frac{1 + e^{-1/2}}{1 - e^{-1/2}} \max \left\{ \frac{1}{v_1}, \frac{1.03^2}{v_2} \right\},$$

where $v_1 = \Sigma_{11}$ and $v_2 = (1 + 0.03^2)\Sigma_{22} + 0.06[e^{-M_g \Sigma}]_{22}$ are the observed variances. Spectral congruence by the responses, followed by division by these variances, proves that q_{cell} is sufficient for every record length in that known model. Rounded-up values for temperature ratios one, two, and four are 6.30, 8.49, and 12.42. They are sufficient bounds, not minimum covariance eigenvalues. All empirical decisions in the study still use the common family bound 56; no performance curve is recomputed with these smaller values.

For $H_1 = 1$, $H_2 = 1 + ee^{-i\omega}$, $e = 0.03$, the exact-response phase integrand is

$$\frac{e \sin^2 \omega}{\sqrt{1 + 2e \cos \omega + e^2}}.$$

Its supremum is approximately 0.02999325. At $N = 524289$ and $q = 56$, the main-text radius divided by its variance scale is approximately 0.06080226, and $c_N(q) \simeq 0.95231503$. Thus the population-scale threshold reference is approximately 0.12683698, before calibration and rounding increments. The corresponding observed normalized contrasts are approximately 0.00698679, 0.09151650, and 0.18492382 at the three temperature ratios. These are analytic model comparisons, not empirical rejection fractions or a new power guarantee.

S7.3 A rotational model with a sampled cycle

The rotational family gives a sampled-law alternative with an explicit transition. Its stationary covariance and contraction factor also give the same envelope and tail bounds for the zero-rotation control. This common information prevents the comparison from changing its supplied bounds with the source regime. We then compute a nonzero cycle in the isotropic special case to show that the sampled witness need not vanish.

Let

$$R_0 = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \quad \Sigma_\xi = (1 - \xi)I_3 + \xi \mathbf{1}\mathbf{1}^\top, \quad (\text{S103})$$

$$M_r = a_d I_3 + \chi R_0, \quad D_r = a_d \Sigma_\xi.$$

Here $0 \leq \xi \leq 1/4$ and $a_d > 0$. The row and column sums of R_0 vanish, so it commutes with Σ_ξ . Its skew symmetry then gives $M_r \Sigma_\xi + \Sigma_\xi M_r^\top = 2D_r$.

Choose $\chi = 2\pi/(3\sqrt{3}\Delta)$. Rotation through one sample gives

$$e^{-M_r\Delta} = \rho\Pi, \quad \rho = e^{-a_d\Delta}, \quad \Pi = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}. \quad (\text{S104})$$

The innovations have covariance $(1 - \rho^2)\Sigma_\xi$. Since $\|\Sigma_\xi\|_{\text{op}} = 1 + 2\xi \leq 3/2$, the family $a_d\Delta \geq \log 4$ has $\rho \leq 1/4$ and

$$K_X = \frac{3}{2} \frac{1 + 1/4}{1 - 1/4} = \frac{5}{2}. \quad (\text{S105})$$

Equations (S98) and (S99), with $s_* = 3/2$ and $\rho = 1/4$, give common filtered bounds for this family and its zero-rotation reversible controls.

The sampled spectrum is

$$F_X(\omega) = (1 - \rho^2)(I - \rho\Pi e^{-i\omega})^{-1}\Sigma_\xi(I - \rho\Pi^\top e^{i\omega})^{-1}. \quad (\text{S106})$$

It follows by summing the moving-average representation of the stationary vector autoregression. In the isotropic case $\Sigma = sI_3$, put $A_0 = 1 + \rho^2$ and $D_0 = 1 - \rho^2 + \rho^4$. At $\omega = \pi/2$, all three directed entries equal

$$F_{12} = F_{23} = F_{31} = -\frac{s(1 - \rho^2)\rho^2}{A_0 D_0} - i\frac{s(1 - \rho^2)\rho}{D_0}. \quad (\text{S107})$$

To obtain this expression, use $\Pi^3 = I$ to write $(I - z\Pi)^{-1} = (I + z\Pi + z^2\Pi^2)/(1 - z^3)$, then multiply the two factors in (S106). Cubing (S107) gives

$$J(\pi/2) = \frac{s^3(1 - \rho^2)^3\rho^3}{A_0^2 D_0^2} > 0. \quad (\text{S108})$$

For $s = 1$ and $\rho = 1/4$, the entry is $-240/4097 - i60/241$ and the witness is $216000/16785409$. These identities provide a physical Gaussian model with a nonzero sampled cycle and supplied finite-record bounds. For a general ξ , the witness is computed from (S106). A stationary current alone does not guarantee a nonzero cycle at the chosen frequency.

S7.4 Sampling aliases and entropy production

The witness above concerns the sampled law. To relate it to physical irreversibility, we compute entropy production from the continuous-time probability current under the stated diffusion model. We then compare rotations with the same sampled transition. Equality of their stationary covariance and innovation covariance makes their entire sampled Gaussian laws equal, even though their entropy-production rates differ.

For positive-definite $\mathbf{D}$ and Σ , let p denote the stationary Gaussian density. Its probability-current velocity is

$$\frac{j(x)}{p(x)} = -(M - \mathbf{D}\Sigma^{-1})x.$$

In units with Boltzmann's constant equal to one, the entropy-production rate is the current integral [14]

$$\sigma = \int \frac{j(x)^\top \mathbf{D}^{-1} j(x)}{p(x)} dx = \text{tr}\{(M - \mathbf{D}\Sigma^{-1})^\top \mathbf{D}^{-1} (M - \mathbf{D}\Sigma^{-1}) \Sigma\}. \quad (\text{S109})$$

The Gaussian second moment gives the trace expression. It is a squared matrix norm. Thus $\sigma \geq 0$, with equality exactly when $M\Sigma = \mathbf{D}$. The rate is per unit of continuous time and uses the stated even reversal parity.

For the Brownian gyrator,

$$M_g \Sigma - D_g = \frac{T_2 - T_1}{4} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \det \Sigma = T_1 T_2 + \frac{(T_1 + T_2)^2}{12}.$$

Substitution in (S109) gives

$$\sigma_g = \frac{(T_2 - T_1)^2}{8T_1 T_2}. \quad (\text{S110})$$

Indeed, the remaining trace equals $(\Sigma_{22}/T_2 + \Sigma_{11}/T_1)/\det \Sigma = 2/(T_1 T_2)$. The temperature pairs $(1, 1)$, $(1, 2)$ and $(1, 4)$ have rates 0, $1/16$ and $9/32$.

For (S103), $M_r - D_r \Sigma_\xi^{-1} = \chi R_0$. Commutation with Σ_ξ gives

$$\sigma_r = \frac{\chi^2 \text{tr}(R_0^\top R_0)}{a_d} = \frac{6\chi^2}{a_d}. \quad (\text{S111})$$

At $\Delta = 1$, $a_d = \log 4$ and the rotation in (S104), this rate is $8\pi^2/(9 \log 4)$. If instead $\chi = 2\pi/\sqrt{3}$, the sampled transition is $(1/4)I_3$. Its stationary Gaussian law is reversible, while its continuous-time rate is $8\pi^2/\log 4$.

More generally, the rates

$$\chi_k = \frac{2\pi/3 + 2\pi k}{\sqrt{3}\Delta}, \quad \sigma_k = \frac{2(2\pi/3 + 2\pi k)^2}{a_d \Delta^2}, \quad k \in \mathbb{Z}, \quad (\text{S112})$$

give the same transition $\rho\Pi$ and the same stationary covariance. The rotation acts on the plane perpendicular to $\mathbf{1}$, where adding $2\pi k$ leaves its exponential unchanged. The innovation covariance is also unchanged. All these processes therefore have the same complete sampled Gaussian law, but σ_k is unbounded. A sampled cycle can exclude the stated reversible-source model. It cannot determine an unrestricted hidden continuous-time entropy-production rate without information that resolves this frequency ambiguity.

S7.5 Reversible controls near the phase tolerance

Let a stationary unit-variance scalar source obey

$$X_t = -bX_{t-2} + \sqrt{1-b^2} Z_t, \quad 0 \leq b < 1,$$

with standard Gaussian white innovations. Its covariance is zero at odd lags and $(-b)^{|k|/2}$ at even lags. This follows from the two independent stationary autoregressions on the even and odd times. The scalar Gaussian law is reversible.

Observe $Y_1(t) = X_t$ and $Y_2(t) = X_{t-1}$. The unit delay gives phase tolerance $r = 1$. The reflection contrast has magnitude $1+b$, and its ratio to $2r\sqrt{v_1 v_2}$ is $(1+b)/2$. The scalar density is

$$f_X(\omega) = \frac{1-b^2}{|1+be^{-2i\omega}|^2}, \quad \sup_\omega f_X(\omega) = \frac{1+b}{1-b}.$$

The delayed two-channel density has largest eigenvalue $2f_X$. Thus a full standardized envelope is $q = 2(1+b)/(1-b)$. For $b = 0, 4/5, 49/50$, the tolerance ratios are $1/2, 9/10, 99/100$ and the corresponding factors are 2, 18, 198. These are reversible-source controls. Their finite test still uses common centering and the supplied phase allowance.

S8 Synthetic study design and analysis

The comparisons test where the theoretical requirements become informative in the stated models. The source comparison changes how persistence is supplied while keeping the downstream calibrated rule shared. The rotational comparison changes frequency coverage, channel scaling, and supplied allowances. The Brownian controls compare source inference with recorded-signal asymmetry. Numerical and assumption-boundary comparisons address availability and the limits of the formal model. These comparisons have separate generating conditions and inference targets.

The study has 69 conditions and 20800 independent records. A condition fixes the generating law, recording length, and number of independent repetitions. All paired methods use the same record and, where required, the same independent calibration. Records at different lengths are independent. No comparison uses a prefix of a longer record as an additional independent repetition.

S8.1 Source-confidence comparison

This comparison follows the source proof through the complete decision. Coverage asks whether the set retains the true persistence. A finite endpoint asks whether it excludes one. The propagated hull and bank then determine whether a test is available and whether its threshold can be exceeded. Supplying the true coefficient gives a reference for the cost of source fitting while retaining the other uncertainties. Let X and Z be independent stationary Gaussian autoregressions with unit marginal variance and common coefficient ϕ . The two source coordinates are

$$X_1(t) = X(t), \quad X_2(t) = cX(t-d) + \sqrt{1-c^2}Z(t).$$

Both detector responses are identities. Recording means are 2 and -1 . The reference response is supplied, and the second response is estimated as a two-coefficient finite response. Thus the true zero coefficient is still fitted. The calibration uses 256 rows with two orthogonal sign columns and independent mean-zero Gaussian errors of standard deviation 0.01.

The core comparison crosses $\phi \in \{0.8, 0.97, 0.995\}$ with $N \in \{16385, 65537\}$. The reversible condition has $(d, c) = (0, 0.9)$. Weak and strong delays have $(d, c) = (1, 0.4)$ and $(1, 0.9)$. Each null condition has 400 records, except that $\phi = 0.97, N = 65537$ has 1000. Each alternative condition has 200 records. The strong-delay curve at $\phi = 0.995$ uses $N = 2^j + 1$, $j = 11, \dots, 19$, with 200 records per length. Its two core lengths are counted once.

Seven source-information methods share the downstream test. They use the lag set, the length-dependent analytic lower cutoff, the constant analytic lower cutoff, the lag/lower-tail intersection, the two-sided set, the lag/two-sided intersection, and the supplied true coefficient. The source coverage budget is 0.005. Each intersection assigns 0.0025 to each component. Two-sided cutoffs divide their component budget between the two tails. Section S5.1 gives the source sets and Section S5.8 gives the test allocation.

The primary bank is $(0, 0.5, 0.8, 0.9, 0.97, 0.995)$. The standalone two-sided rule also uses a paired bank with $1-2^{-j}$, for $j = 0, \dots, \lceil \log_2(N-1) \rceil$. The error allocation includes every member of each bank. For the standalone two-sided rule and its lag intersection, paired threshold comparisons replace the geometric variance scale by the maximum scale, or replace the reflection operator bound $1/n$ by $2/n$. The supplied-coefficient comparison retains the same calibration, bank, recorder correction, and downstream error allocation. A supplied coefficient does not remove calibration or sampling uncertainty. The standalone two-sided rule also has source-shape multipliers $2/3$ and $1/2$. These are deliberate underestimation comparisons.

S8.2 Rotational source and cycle comparisons

The cycle comparison retains a common physical family bound under both source regimes. The single-frequency comparison isolates the effect of separate channel ceilings because the estimates are shared. The two-frequency comparison changes the covered frequency set and pays its larger error allowance. Gain transport tests preservation under a change of units, whereas allowance inflation tests the loss from conservative information. The sampled rotational model has $\rho = 1/4$ and $\xi = 1/4$. The reversible transition is ρI ; the rotating transition is ρP , with the permutation orientation in Section S7. Both laws use the same family bounds $\rho \leq 1/4$ and $\xi \leq 1/4$. The detector responses are

$$H_1(z) = 1, \quad H_2(z) = 0.99z^{-1} + 0.01z^{-2}, \quad H_3(z) = 0.98z^{-2} + 0.02z^{-3}.$$

The means are 2, $-1, 1/2$. The test uses $L = 8$ and primary frequency $\pi/2$. The diagonal and common-envelope tests use the same estimates. The combined-frequency test uses $\{\pi/4, \pi/2\}$ and the corresponding two-frequency error allocation.

The recording lengths are $N = 2^j$, $j = 13, \dots, 21$. Each length has 400 reversible and 200 rotating records. The reversible condition at $N = 131072$ has 1000 records. Identity-detector conditions at $N = 2097152$ have 400 reversible and 200 rotating records.

The identity-detector records also supply the paired gain and bound comparisons. The third-channel gain takes the six values $1/16, 1/4, 1, 4, 16, -1$. Each change transports the covariance, bias, and recording-error bounds. The comparison includes the diagonal rule, the common scalar rule, and inverse transport back to the original units. All 16 pairs of covariance and bias multipliers in $\{1, 1.5, 2, 4\}^2$ are retained. The baseline bias allowance is positive. Separate comparisons reduce covariance or bias bounds by $2/3$ or $1/2$. These reduced bounds have no coverage guarantee.

Three additional reversible sampled-source conditions at $N = 131072$ have 400 records each. They use $\xi = 0.01$, four times the valid bias allowance, or the aliased continuous-time rotation with $\sqrt{3}w = 2\pi$. The aliased model has the same reversible sampled transition ρI . A 200-record rotating condition uses $H_1 = 1$, $H_2 = z^{-1}$, and $H_3 = (z^{-2} + z^{-4})/2$. Its third response vanishes at $\pi/2$.

S8.3 Brownian and phase-boundary conditions

The Brownian equilibrium source gives a control in which detector phase alone creates recorded asymmetry. The driven temperatures then ask whether the calibrated rule can detect source asymmetry beyond that allowance. The phase-boundary controls use a different reversible process to bring the detector-induced contrast close to the population tolerance. They test that restriction at large recorded contrasts, rather than adding evidence about the Brownian alternative. The Brownian model in Section S7 uses $(T_1, T_2) = (1, 1), (1, 2), (1, 4)$. The record lengths are $N = 32769, 131073, \text{ and } 524289$. Each equilibrium condition has 400 records. Each nonequilibrium condition has 200 records. The responses are $H_1 = 1$ and $H_2 = 1 + 0.03z^{-1}$. Every temperature uses the same supplied bound $q = 56$. Calibration has the same design and error scale as the source-confidence comparison. The first-order autoregressive (AR(1)) source-confidence method is not applied to this model.

The Brownian comparisons include the calibrated test, maximum-scale and operator-bound variants, and procedures with missing covariance or response information. The latter procedures abstain. A separate procedure sets the phase allowance to zero but retains the dependence bound. It tests an incorrect source-null restriction when detector phase is nonzero. It is not a substitute for the missing-response procedure. Brownian and phase-boundary comparisons also multiply the

supplied covariance envelope by $2/3$ or $1/2$, with a lower floor of one. These reduced envelopes have no coverage guarantee.

The boundary nulls use a unit-variance reversible scalar process

$$X_t = -bX_{t-2} + \sqrt{1-b^2}\varepsilon_t, \quad Y_1(t) = X_t + 2, \quad Y_2(t) = X_{t-1} - 1.$$

The innovations are independent standard Gaussian values. The even and odd stationary subsequences start independently. The ratio of the absolute population contrast to its phase tolerance is $(1+b)/2$. Thus $b = 0, 0.8, 0.98$ gives 50%, 90%, and 99%. The supplied standardized envelopes are $q = 2, 18, 198$, and the supplied phase bound is $r = 1$. Each condition has $N = 65537$ and 400 records. These are reversible-source controls despite their large recorded asymmetry.

S8.4 Assumption violations and recorder resolution

The Gaussian reference and error-orthogonality conditions support different steps of the proofs. Reference noise changes the residual law used by the innovation pivot, non-Gaussian innovations change its exact law, and cross-channel detector errors can contribute a cycle that the source null does not constrain. These conditions therefore test behavior outside the corresponding guarantees. The recorder comparisons address a different question, because bounded rounding can be covered by a deterministic correction within the ideal model. At $\phi = 0.97$, $N = 65537$, the reference-noise conditions add independent white Gaussian noise with signal-to-noise ratio 10 or 100. The ratio is the reference source variance divided by the added noise variance. The Student- t conditions replace the standardized Gaussian innovations by variance-one Student- t innovations with five degrees of freedom. They use 8192 initial values before the retained recording. Each of these three conditions has 400 null and 200 strong-delay records. The inference rule keeps its Gaussian, noiseless-reference assumptions. These comparisons therefore measure model violations.

Correlated-error conditions add an independent rotating three-channel error process to the filtered observations. The error has marginal variance $1/4$, persistence $1/4$, and channel correlation $1/4$. Its transition uses the same permutation orientation as the rotating source. It is independent of the source but violates the cross-channel error restriction. These conditions have $N = 131072$, with 400 reversible-source and 200 rotating-source records.

The primary recorder rounds to 20 fractional bits, with allowance 2^{-21} . Paired grids with 8 or 12 fractional bits isolate rounding without saturation. Separate finite 8-bit and 12-bit recorders divide $[-16, 16)$ into equal bins. They record bin centers and clip to the outer centers. A value outside the range counts as saturated. If any value saturates, the associated bound-based method abstains because no finite perturbation certificate is supplied. Calibration stays at 20 fractional bits. Recorder comparisons use the Gaussian source conditions at $\phi = 0.97$, $N = 65537$, the primary cycle conditions at $N = 131072$, Brownian conditions at $N = 131073$, and the 99% phase-boundary null.

The finite random generator and its floating-point recursion approximate the ideal Gaussian models. The saved pre-rounded arrays define the numerical experiment. A rounding allowance bounds the change from these arrays to the recorded grid. It does not prove that the pre-rounded arrays have exactly the ideal Gaussian law.

S8.5 Baseline definitions

The raw lag and bootstrap comparators target asymmetry of the observed signals. Unequal responses can make that null different from source reversibility. Phase-adjusted variants use the same supplied or calibrated phase upper bound as the corresponding calibrated test. Cycle conditions lack that certificate, so these variants abstain there.

Retain rows $1, \dots, N - 1$ and form the adjacent sum and difference with $n = N - 2$ pairs. Let $\widehat{v}_U, \widehat{v}_W$ be their centered empirical variances. The independent-window comparator uses radius

$$\frac{\sqrt{\widehat{v}_U \widehat{v}_W}}{n} \left\{ \sqrt{2(n-1) \log(2/\alpha)} + \log(2/\alpha) \right\}.$$

This radius deliberately ignores dependence between overlapping windows. Its phase-adjusted variant adds $2\widehat{r}\sqrt{\widehat{v}_1 \widehat{v}_2}$, using empirical marginal variances. These plug-in rules have no finite-record source-null guarantee here.

The moving-block comparator uses the centered product sequence. It resamples $\lceil n/b_n \rceil$ non-circular blocks with replacement, where $b_n = \max\{2, \lceil n^{1/3} \rceil\}$, and averages all $b_n \lceil n/b_n \rceil$ sampled values. This is a specified approximate bootstrap procedure, based on the block-bootstrap principle [15]. For 499 resamples, its probability estimate is $(1 + E)/500$, where E counts absolute resampled means at least as large as the comparison value. That value is the absolute observed contrast for the raw comparator. For the phase-adjusted comparator it is the positive part of the contrast minus the plug-in phase tolerance.

Imaginary coherency measures the imaginary part of a normalized cross spectrum [16]. The comparator uses nonoverlapping 256-sample segments, centered separately, at frequency $\pi/2$. It divides the mean imaginary cross product by the geometric mean spectral scale. The declared moving-block procedure acts on segment contributions. An incomplete final segment is omitted for this comparator only. Fewer than four complete segments or a zero spectral scale cause abstention.

Each record has a separate bootstrap stream. The raw lag bootstrap, the phase-adjusted bootstrap when available, and imaginary coherency consume successive parts of that stream in this order. No block length or frequency is selected from the results.

S8.6 Numerical limits and saved quantities

The primary source inversion encloses quadratic roots at 48 bits. Paired subdivision depths are 20, 28, and 36. Separate arithmetic comparisons use 80, 160, and 320 square-root bits with the same 48-bit root enclosure. They recompute calibration and the whole bank. Exact compiled moments can be reused because their values do not depend on these settings. The numerical comparisons use all Gaussian core source conditions at the two core lengths.

The source implementation permits at most 600000 observations, 64 bank members, 262144 rational bits, and 500000000 operations per phase. The normal configuration uses 80 square-root bits and 48 logarithm terms. The cycle implementation permits at most 8000000 record values, 300000000 lag products, 16384 integer bits, 8192 rational bits, and 1000000000 phase operations. Input, work, or precision failures cannot produce certified rejection. Arithmetic operation counts do not imply fixed wall time because integer operand sizes vary.

The saved outputs include all source-set components, hull endpoints, coverage indicators, bank members, covariance envelopes, scale-denominator endpoints, phase allowances, sampling and rounding radii, thresholds, and decisions. They also include all cycle edges, their radii, the product threshold, saturation counts, and the numerical failure reason. Coverage and finite-endpoint status use exact rational comparisons before plotting.

S8.7 Analysis and interpretation

Each method returns rejection, nonrejection, or abstention. Its unconditional rejection fraction uses every independent record in its condition. The descriptive rejection fraction among available tests divides the number of rejections by the number of rejections plus nonrejections. An all-

abstention condition gives no empirical evidence about the behavior of an available test. Execution failures are separate and cannot silently become nonrejections.

Pointwise two-sided 95% Clopper–Pearson intervals quantify uncertainty in rejection fractions [17]. For paired methods, let n_{10} and n_{01} count records rejected by only the first or only the second method. The difference is $(n_{10} - n_{01})/m$ for m independent records. Subtracting simultaneous 97.5% binomial intervals for the two discordant probabilities gives a conservative 95% interval. An exact sign-test probability conditions on $n_{10} + n_{01}$. These comparisons have no familywise guarantee across all method pairs.

Threshold summaries retain their available counts and report the minimum, quartiles, median, and maximum. A missing quantity is not assigned zero. The displayed bank member maximizes the exact statistic-minus-threshold margin over available members. This choice describes the fixed union decision; it does not change it. Source summaries distinguish empty sets, full fallback sets, hulls reaching one, finite envelopes, and available downstream tests. They also report persistence span and the loss from the finite bank.

The code, configurations, pre-rounded and recorded arrays, calibration data, random-stream states, per-record outputs, and numerical certificates are retained in the local reproducibility package. Exact saved input arrays, rather than seeds alone, define the arithmetic comparison. Floating-point generator outputs need not match across numerical library versions or platforms. The scientific timing disclosure is in the article’s Methods appendix.

S9 Synthetic study results

S9.1 Source confidence and whole-test availability

The source proof requires more than coverage before it supplies a usable threshold. We therefore compare the returned set, its finite endpoint, and the complete decision on the same records. The core conditions also vary coupling strength, so the long strong-delay curve does not determine which source procedure is most useful for weaker contrast.

Coverage, a finite source endpoint, and an available test are different events (Figs. S1 and S2). For persistence 0.995 and $N = 16,385$, the two-sided null set covers the true parameter in 396 of 400 records. Its pointwise 95% coverage interval is $[0.974, 0.998]$. It has finite endpoints in 383 records. The corresponding test gives 0 rejections, 373 nonrejections, and 27 abstentions. The intersection covers in 397 records, has finite endpoints in 376, and gives 0 rejections, 361 nonrejections, and 39 abstentions. The lag procedure abstains in all 400 records. Finite endpoints alone therefore do not determine whole-test availability.

The weak-delay conditions expose differences that the strong-delay curve cannot show. At persistence 0.8 and $N = 16,385$, lag and intersection each give 200 rejections among 200 records. The standalone two-sided rule gives 38 rejections, 162 nonrejections, and no abstentions. At persistence 0.97, the corresponding counts are lag 4/0/196, two-sided 92/108/0, and intersection 147/53/0. Here and below, triplets give rejection, nonrejection, and abstention counts. At persistence 0.995, the triplets are 0/0/200, 48/132/20, and 45/130/25. The useful source construction depends on persistence and signal strength.

The lower-tail comparison addresses the range left after endpoint exclusion. The source theorem uses both acceptance inequalities to bound that range. For the strong-delay alternative at persistence 0.995 and $N = 524,289$, the analytic-cutoff procedure gives 0/0/200. The finite-sample lower-tail procedure gives 0/54/146. The lower-tail intersection gives 152/26/22, and the two-sided procedure gives 200/0/0. These comparisons change the whole acceptance rule and its propagated test. They do not isolate the effect of one endpoint. The hull and threshold diagnostics below

show what information remains available for each procedure.

The rejection difference at $N = 8,193$ also depends on the uncertainty measure. The standalone rule rejects alone in 18 records. The intersection rejects alone in 4 records. The conservative paired 95% interval is $[-0.006, 0.142]$. The conditional sign-test probability is 0.00434. The sign test conditions on the number of rejection discordances. Its question differs from the conservative interval construction. Both use the same 200 paired records; status changes occur in 35 records.

S9.2 Gain transport and conservative bounds

These comparisons change the supplied scale information in two ways. Transporting all bounds with a gain preserves the mathematical decision of the diagonal rule. Replacing the separate ceilings by one common ceiling can lose that property. Inflating a valid allowance retains conservative coverage but can prevent rejection by increasing the threshold.

Figure S3 shows how an invertible gain can change the decision of the common scalar rule without removing source information. At gain $1/4$, that rule gives $0/200/0$ in the rotating identity-detector model. The diagonal rule and inverse transport each give $200/0/0$. The common-minus-diagonal paired interval is $[-1.000, -0.956]$. Gains $1/16$, 4 , and 16 give the same count pattern. At gains -1 and 1 , all three rules give $200/0/0$. Each corresponding null condition has $0/400/0$. These comparisons use the supplied gain and transported information.

Figure S4 shows the covariance and bias grid, which measures sensitivity to conservative allowances. At covariance factor 1 , every bias factor gives $200/0/0$ in the rotating identity-detector condition. At covariance factors 1.5 , 2 , and 4 , every bias factor gives $0/200/0$. Each null gives $0/400/0$. Equal decision counts across bias factors do not establish equal thresholds or identify a dominant radius term.

Underestimating either allowance is a different sensitivity question. The factors $2/3$ and $1/2$ give null counts $0/400/0$ and rotating counts $200/0/0$ in the identity-detector model. These are descriptive outcomes under invalid supplied coverage information. A zero observed null count does not validate the underestimated bound.

S9.3 Numerical and recorder settings

Numerical settings change how tightly the same input calculation is enclosed. Recorder settings change the input values and their required perturbation allowances. Their paired comparisons therefore address different parts of the implementation. In both cases, abstention reports an unavailable decision and stays in the full sample denominator.

The numerical comparisons use the same records within each condition. At persistence 0.97 and $N = 65,537$, all three subdivision depths and all three arithmetic precisions give the primary standalone two-sided counts. These are $0/1,000/0$ under the null and $199/1/0$ under strong delay. The strong-delay rejection interval is $[0.972, 1.000]$. Equal counts alone do not establish equal source sets, thresholds, or individual decisions.

Coarser recording can change both detection and availability. Fractional 8-bit rounding gives $0/65/135$ for the strong-delay condition. Fractional 12-bit rounding gives $199/1/0$. The finite 8-bit and 12-bit recorders each give $0/0/200$. Under the null, fractional 8-bit rounding gives $0/315/685$, while each finite recorder gives $0/0/1,000$. Abstention is kept separate from nonrejection because it indicates that the bound-based test did not return an available decision.

S9.4 Assumption boundaries and physical comparisons

The assumption-boundary conditions ask how the procedures behave when a step of their proof is unsupported. The physical controls ask a separate question about what the null permits. A reversible source can generate a large recorded contrast under unequal responses, so raw asymmetry and calibrated source rejection must be compared with their targets stated.

Reference noise and non-Gaussian innovations probe the source procedure outside its exact reference model. The two signal-to-noise ratios, 100 and 10, each give null counts 0/400/0. Their strong-delay counts are 200/0/0 and 199/1/0, respectively. Student-t innovations with 5 degrees of freedom give 0/400/0 under the null and 200/0/0 under strong delay. These empirical outcomes do not extend the noiseless Gaussian autoregressive pivot guarantee.

Correlated observation errors probe a separate cycle assumption. The diagonal rule gives 0/400/0 in the nominal null condition and 0/200/0 in the rotating condition. The source-null interpretation requires orthogonal cross-channel errors. These conditions provide sensitivity evidence under correlated errors.

The phase-boundary controls use population response-induced contrasts equal to 50%, 90%, and 99% of the phase tolerance. At each ratio, calibrated and phase-adjusted procedures give 0/400/0. Raw lag, raw bootstrap, phase-ignored, and imaginary-coherency procedures give 400/0/0. Their pointwise rejection interval is [0.990, 1.000]. The population ratio specifies the control. It is different from the observed statistic divided by the full threshold. The raw procedures target recorded-signal asymmetry.

The Brownian equilibrium comparison illustrates this target difference. Raw moving-block bootstrap counts rise from 87/313/0 to 271/129/0 and 397/3/0 as length increases. Its phase-adjusted variant gives 0/400/0 at all three lengths. Raw independent-window counts are 0/400/0, 9/391/0, and 269/131/0. Imaginary-coherency counts are 31/369/0, 24/376/0, and 27/373/0. A reversible source with unequal detector responses can have asymmetric observed signals, so these raw counts answer a different null question.

At temperature ratio 4 and $N = 524,289$, the calibrated rule gives 200/0/0. The maximum-scale rule and the rule with the doubled operator-bound factor each give 0/200/0. Each paired calibrated-minus-comparison interval is [0.956, 1.000]. At persistence 0.995 and $N = 16,385$, the sharper operator rule gives 153/29/18, while the rule with the doubled operator-bound factor gives 88/94/18. Their paired rejection-fraction difference is 0.325, with interval [0.230, 0.405]. These comparisons support improvements in the specified conditions.

The 22 rounded and unrounded cycle evaluations have negative sufficient margins. Empirical detection remains possible, but these margins do not establish the sufficient power condition. The separate ideal cycle bound applies only to its stated model and allowance. The estimated-source power result applies to its specified ideal Gaussian source, calibration, bounded recorder, and recording length. Its probability bound includes completion under the specified numerical settings and work limits. Empirical rejection intervals quantify uncertainty in the sampled cells.

Source confidence: reversible source

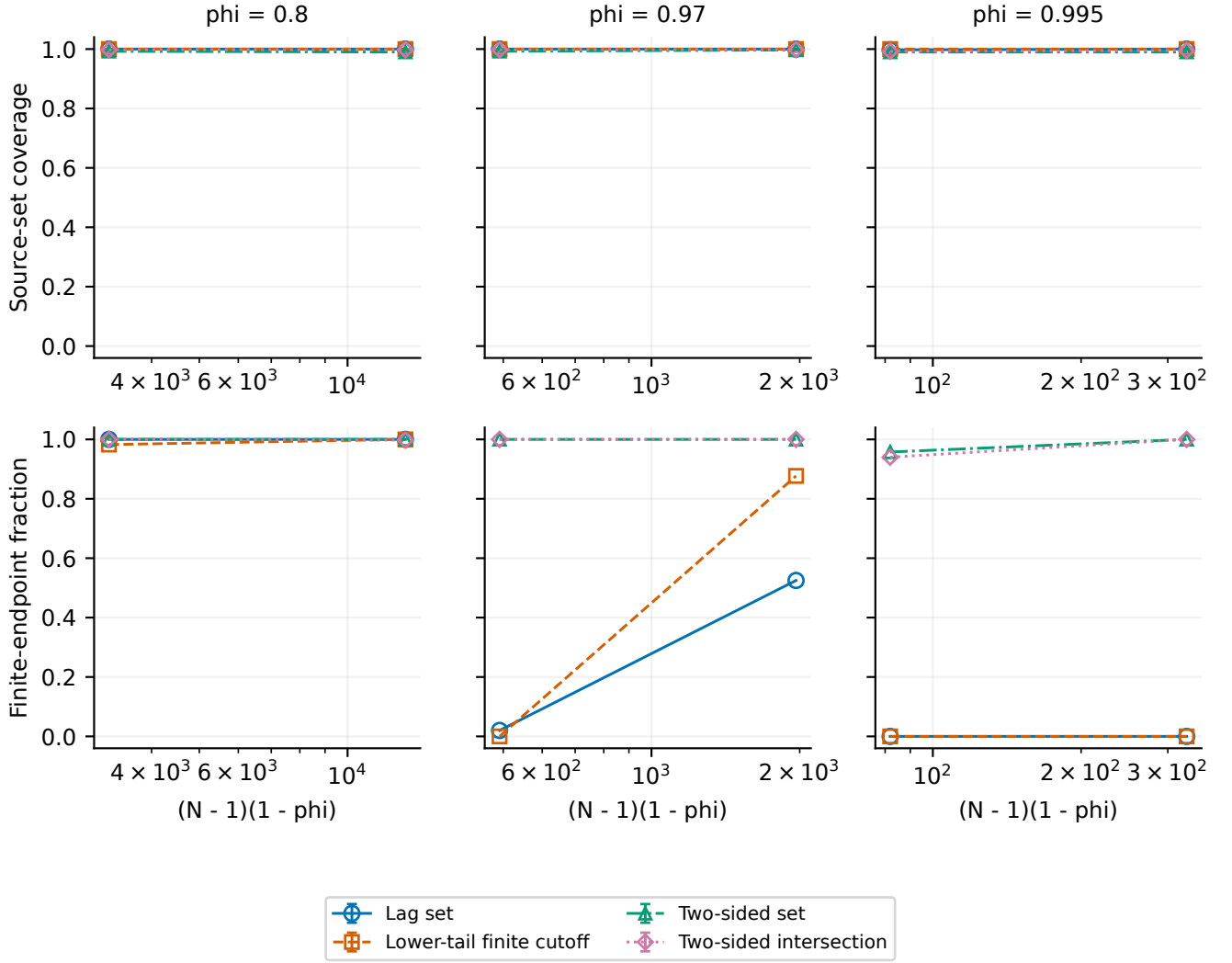


Figure S1: Coverage and finite endpoints differ across source constructions. Core null cells have 400 records, except $\phi = 0.97$, $N = 65537$, which has 1000. Coverage bars are pointwise 95% binomial intervals. All fractions use the complete cell. The source confidence error is 0.005, distinct from the display interval. Coincident curves have equal fractions; methods remain in the legend. Values are saved study results.

Source confidence: strong delay

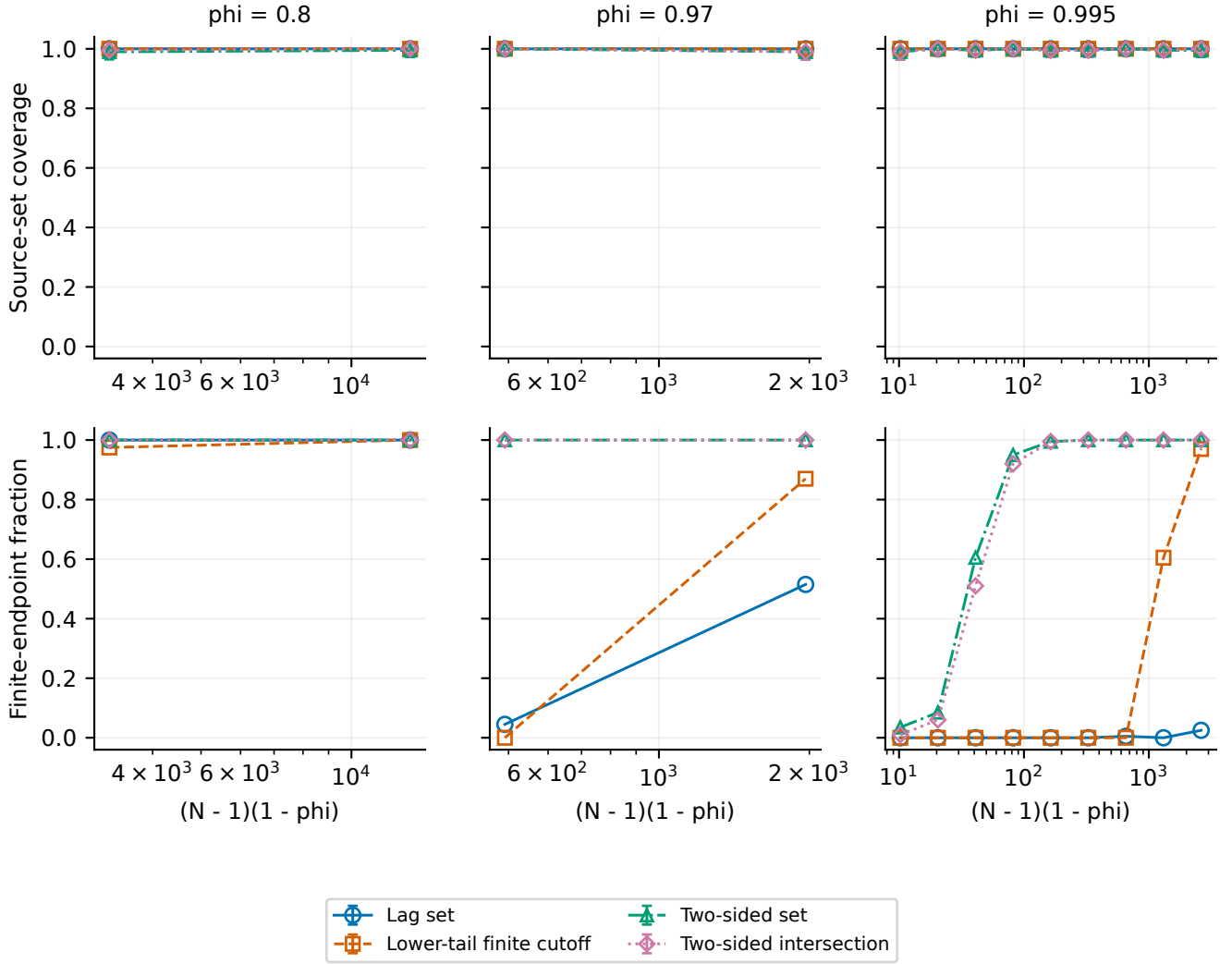


Figure S2: Strong-delay records can have covered source sets without finite upper endpoints. Each condition has 200 records. The display includes the nine-length $\phi = 0.995$ curve and both core lengths at other persistences. Coverage bars are pointwise 95% binomial intervals; endpoint fractions use all records. Equal curves indicate equal fractions. The source-set error allowance is 0.005. Values are saved study results.

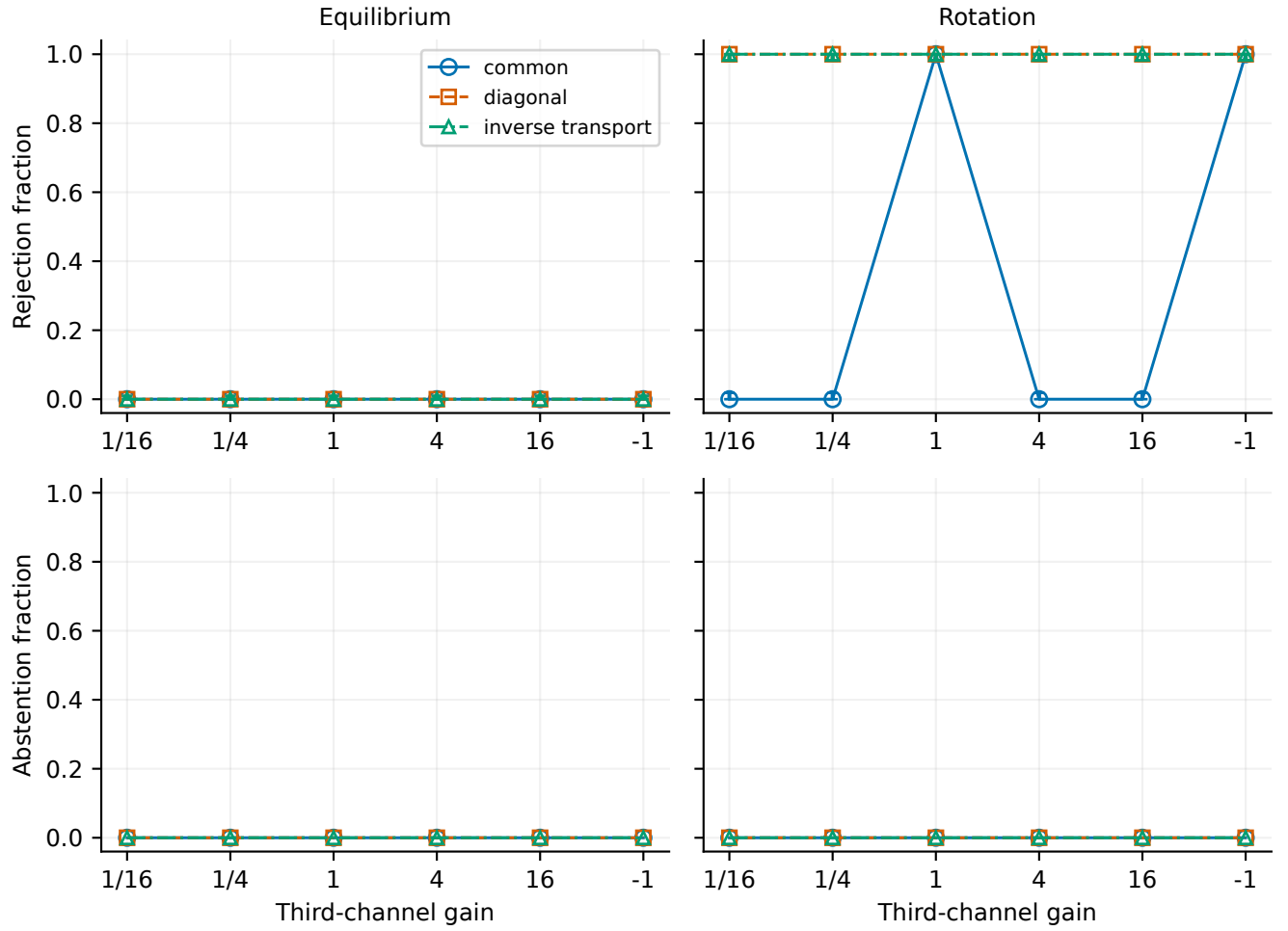


Figure S3: Diagonal bounds and inverse transport preserve all rotating-source rejections across six gains. The common scalar rule loses every rejection at four gains. Null and rotating cells have 400 and 200 records, respectively, with methods paired on each record. All abstention fractions and equilibrium rejection fractions are zero. These transformations change data and bounds together. Values are saved study results.

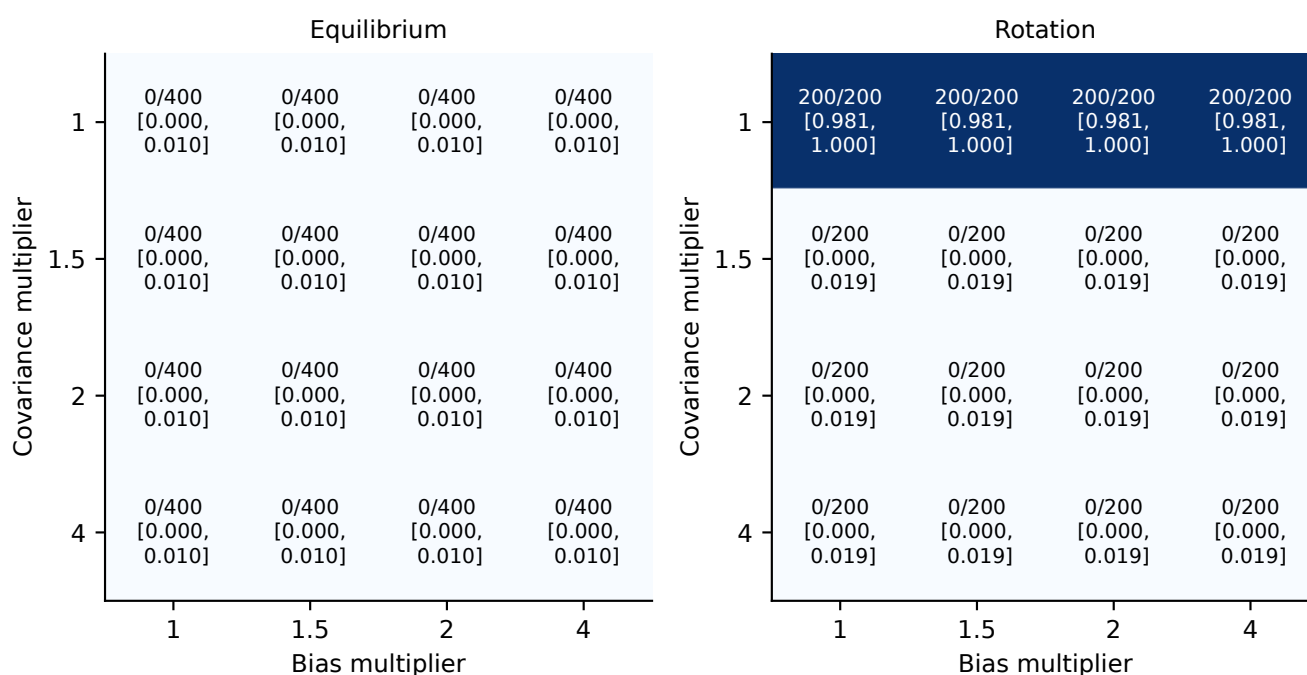


Figure S4: Conservative covariance inflation removes rotating-source rejections within this grid. All 16 covariance/bias factor pairs are shown for both regimes, with 400 null or 200 rotating records. Entries give rejection counts and pointwise 95% binomial intervals, rounded outward to 0.001. No cell has an abstention. Equal counts across bias factors do not establish equal thresholds. Values are saved study results.

S10 Source uncertainty and threshold diagnostics

S10.1 Diagnostic question and definitions

A covered source hull enters the covariance envelope through its transformed span. A fixed bank adds a placement cost when none of its filters is close to the hull center in that transform. These costs determine the dependence bound used by the threshold, but an available threshold can still exceed the statistic. The diagnostics separate the remaining source uncertainty, the additional bank cost, and the final statistic-to-threshold comparison.

The hull-span ratio is $\Lambda(u)/\Lambda(\ell)$, with $\Lambda(x) = (1+x)/(1-x)$. It measures the source uncertainty that remains even if a common filter could be placed at the transformed hull center. The bank-placement factor divides the smallest source spectral-ratio bound in the fixed bank by this hull-span ratio. It measures the additional cost of the available filter positions. The final statistic-to-threshold ratio compares the selected absolute statistic with the upper threshold endpoint. The selected member has the largest saved statistic-minus-threshold margin among available members. A ratio above one supports rejection; a zero ratio is valid.

The diagnostics use all nine strong-delay lengths at persistence 0.995. Each condition has 200 independent records. The four source constructions are lag, lower-tail finite-cutoff, two-sided, and two-sided intersection. Each numerical summary uses only its saved finite values. Thus the records can differ between quantities and methods. Availability is part of the result, and every median must be read with its finite, missing, and full counts.

Display conventions. Figure S5 shows medians and ranges from the 25th to the 75th percentile in its upper panels. Its lower panels show the fraction of records with a finite saved value. Figure S6 shows finite-value medians and the same percentile ranges for the four threshold quantities. These ranges are descriptive summaries, not confidence intervals. A missing quantity has no plotted summary point and is not replaced by zero. A legend entry does not imply that a method has finite points at every length. The captions specify the axes, reference line, and method styles.

S10.2 Geometry and selected threshold quantities

The two-sided geometry summaries use 7 of 200 records at $N = 2049$, 190 of 200 at $N = 16385$, and all 200 at $N = 524289$. The statistic-to-threshold ratio uses fewer records at the first two lengths. Table S2 keeps each median beside its own counts. These different subsets show why finite geometry and an available threshold must be reported separately. The changing finite groups are not trajectories of the same records.

The same summaries are much less available for lag and lower-tail inference. Both constructions have no finite plotted summaries at the first six lengths. At $N=524289$, the lag construction has only five finite values for each plotted quantity and 195 missing values. Its statistic-to-threshold median is 16.3. This median does not describe the other 195 records. The lower-tail construction has 194 finite geometry values and six missing values. Its hull-span median is $7.90e3$. Its selected statistic-to-threshold median is 0.0929, based on 54 finite values and 146 missing values. These finite counts limit comparisons between the curves.

Among the finite selected two-sided members, the sampling-radius median decreases from 0.0378 to 0.000503 across these example lengths. The recorder-allowance median stays between $7.55e-7$ and $7.60e-7$. Table S3 gives the intermediate values, phase allowances, and scale denominators with their counts. These summaries describe the saved selected members. They do not identify a cause for every decision.

Table S2: Two-sided geometry and selected statistic-to-threshold medians at three strong-delay lengths and persistence 0.995. Each median uses only the finite values in its row. Finite, missing, and full record counts are adjacent. Geometry and threshold summaries can use different records.

N	Quantity	Median	Finite	Missing	Full
2049	Hull-span ratio	67.2	7	193	200
2049	Bank-placement factor	2.22	7	193	200
2049	Statistic-to-threshold ratio	0.233	1	199	200
16385	Hull-span ratio	9.25	190	10	200
16385	Bank-placement factor	1.96	190	10	200
16385	Statistic-to-threshold ratio	1.95	182	18	200
524289	Hull-span ratio	7.45	200	0	200
524289	Bank-placement factor	1.62	200	0	200
524289	Statistic-to-threshold ratio	15.4	200	0	200

Table S3: Saved threshold quantities for the selected two-sided members at the same three strong-delay lengths and persistence 0.995. Allowances use saved upper endpoints. The scale denominator uses its saved lower endpoint. Each median uses only the finite values in its row.

N	Quantity	Median	Finite	Missing	Full
2049	Phase allowance	2.92e-4	1	199	200
2049	Sampling radius	0.0378	1	199	200
2049	Recorder allowance	7.55e-7	1	199	200
2049	Scale denominator	0.297	1	199	200
16385	Phase allowance	1.01e-4	182	18	200
16385	Sampling radius	0.00452	182	18	200
16385	Recorder allowance	7.59e-7	182	18	200
16385	Scale denominator	0.757	182	18	200
524289	Phase allowance	7.59e-5	200	0	200
524289	Sampling radius	0.000503	200	0	200
524289	Recorder allowance	7.60e-7	200	0	200
524289	Scale denominator	0.964	200	0	200

S10.3 Weak-delay source confidence

Figure S8 separates returned-set coverage from finite-endpoint availability. At persistence 0.995, lag and lower-tail sets cover the source value in 200/200 records at both fixed lengths. Neither has a finite endpoint in any record. The two-sided and intersection finite-endpoint counts are 185/200 and 180/200 at $N=16385$. Both are 200/200 at $N=65537$. At this longer length, their coverage counts are 198/200 and 199/200. The pointwise 95% coverage intervals, rounded outward, are [0.964, 0.999] and [0.972, 1.000]. Thus full finite-endpoint availability does not mean observed coverage is 200/200.

At persistence 0.97 and $N = 16385$, lag has 4/200 finite endpoints and lower-tail has 0/200. The two-sided and intersection constructions each have 200/200 finite endpoints. Their coverage counts are 199/200 for two-sided and 200/200 for intersection. At $N = 65537$, the finite-endpoint counts are 107/200 for lag and 172/200 for lower-tail. Both two-sided constructions still have 200/200 finite endpoints. Coverage is 198/200 for two-sided and 199/200 for intersection.

At persistence 0.8, all constructions have coverage 200/200 at both lengths. All finite-endpoint counts are 200/200 except the lower-tail count of 199/200 at $N=16385$. The source-set error allowance is 0.005. It is distinct from the 95% binomial display intervals. These empirical counts do not establish familywise coverage.

S10.4 First failed condition and available decisions

The finite-value medians omit records where an earlier requirement prevented a saved value. The stacked bars in Figure S7 restore those records to the comparison. A finite hull must still give a finite covariance bound and a positive scale denominator before the rule can compare its statistic with a threshold. The bars separate failures of these requirements from available nonrejections and rejections. The four panels compare lag, lower-tail finite-cutoff, two-sided, and two-sided intersection procedures across all nine strong-delay lengths at persistence 0.995. Every bar uses all 200 records in its condition.

Seven mutually exclusive categories cover the full sample: input or arithmetic failure, empty source set, a hull endpoint that reaches one, no finite covariance bound, no positive scale denominator, available nonrejection with insufficient margin, and rejection. For an unavailable decision, the category reports the first failed condition under the declared classification rule. It does not show that every other condition passed. Available nonrejection is distinct from abstention. Counts and full denominators remain in the complete tables. Category fractions are descriptive; the bars do not show confidence intervals. The caption specifies the visual styles.

Figure S7 retains all seven categories in all 36 method-condition groups. At $N=2049$, lag and lower-tail procedures both have 200 reported hull-endpoint first failures. The two-sided procedure has 193 such failures, six reported nonpositive scale-denominator first failures, and one available nonrejection. The intersection procedure has 198 hull-endpoint first failures, one nonpositive scale-denominator first failure, and one available nonrejection. None rejects in this condition.

At $N=16385$, the two-sided procedure has 153 rejections, 29 available nonrejections, and 18 abstentions. The abstentions comprise ten hull-endpoint first failures and eight nonpositive scale-denominator first failures. The intersection has 152 rejections, 22 available nonrejections, and 26 abstentions. Its abstentions comprise 16 hull-endpoint first failures and ten nonpositive scale-denominator first failures. Both procedures reject in all 200 records at each of the four longest lengths. At $N=524289$, lag has five rejections and 195 hull-endpoint first failures. Lower-tail has 54 available nonrejections and 146 abstentions. Those abstentions comprise six hull-endpoint first failures and 140 nonpositive scale-denominator first failures.

Input or arithmetic failure, empty source sets, and no finite covariance bound have zero reported first-failure counts in these 36 selected groups. The figure retains these categories. A reported nonpositive scale denominator does not establish saturation or a recording defect.

S10.5 Cycle radius terms

The cycle threshold propagates complete edge radii through a product. Equal rejection counts under bias inflation therefore do not show that bias is absent or determine which term limits each record. The component summaries compare the terms entering those radii on the selected conditions, with all their finite-value counts retained.

The complete cycle-component CSV identified in Sec. S11 retains the primary reversible and rotating conditions at $N=131072$. The file also retains both identity-detector conditions at $N=2097152$. It shows all three methods and all three edges. All 216 component-summary rows have finite values for the full condition sample. Their missing counts are zero. The full counts are

1000 for the primary reversible condition, 400 for the identity reversible condition, and 200 for each rotating condition.

Across all 36 selected cell-method-edge groups, the square-root sampling term has a larger median than each other non-total term. Table S4 shows this comparison for the diagonal method and edge 1–2. It retains all six component medians in each of the four conditions, with each condition’s counts.

Table S4: Diagonal-method cycle-component medians for edge 1–2. Primary conditions use $N = 131072$. Identity-detector conditions use $N = 2097152$. Each median uses all records in its condition. The identity rotating medians round to the same displayed values as the identity reversible medians. Exact strings remain in the associated comma-separated-value (CSV) files.

Condition	Quantity	Median	Finite	Missing	Full
Primary reversible	Tail bias	0.00115	1000	0	1000
Primary reversible	Centering	0.000973	1000	0	1000
Primary reversible	Square-root sampling	0.133	1000	0	1000
Primary reversible	Linear sampling	0.00251	1000	0	1000
Primary reversible	Recording	1.62e-5	1000	0	1000
Primary reversible	Total	0.138	1000	0	1000
Primary rotating	Tail bias	0.00115	200	0	200
Primary rotating	Centering	0.000973	200	0	200
Primary rotating	Square-root sampling	0.133	200	0	200
Primary rotating	Linear sampling	0.00251	200	0	200
Primary rotating	Recording	1.61e-5	200	0	200
Primary rotating	Total	0.138	200	0	200
Identity reversible	Tail bias	1.59e-5	400	0	400
Identity reversible	Centering	6.08e-5	400	0	400
Identity reversible	Square-root sampling	0.0333	400	0	400
Identity reversible	Linear sampling	0.000157	400	0	400
Identity reversible	Recording	1.62e-5	400	0	400
Identity reversible	Total	0.0336	400	0	400
Identity rotating	Tail bias	1.59e-5	200	0	200
Identity rotating	Centering	6.08e-5	200	0	200
Identity rotating	Square-root sampling	0.0333	200	0	200
Identity rotating	Linear sampling	0.000157	200	0	200
Identity rotating	Recording	1.62e-5	200	0	200
Identity rotating	Total	0.0336	200	0	200

The tail-bias medians remain positive even in the identity-detector conditions. The sampling-term comparison concerns descriptive medians. It does not establish a cause for each rejection or nonrejection. The component values were calculated from the declared formulas with Decimal50 arithmetic. They are not saved component enclosures. A median total need not equal the sum of component medians. These tables do not replace the separate cycle power result or turn its negative proof margins into a positive guarantee.

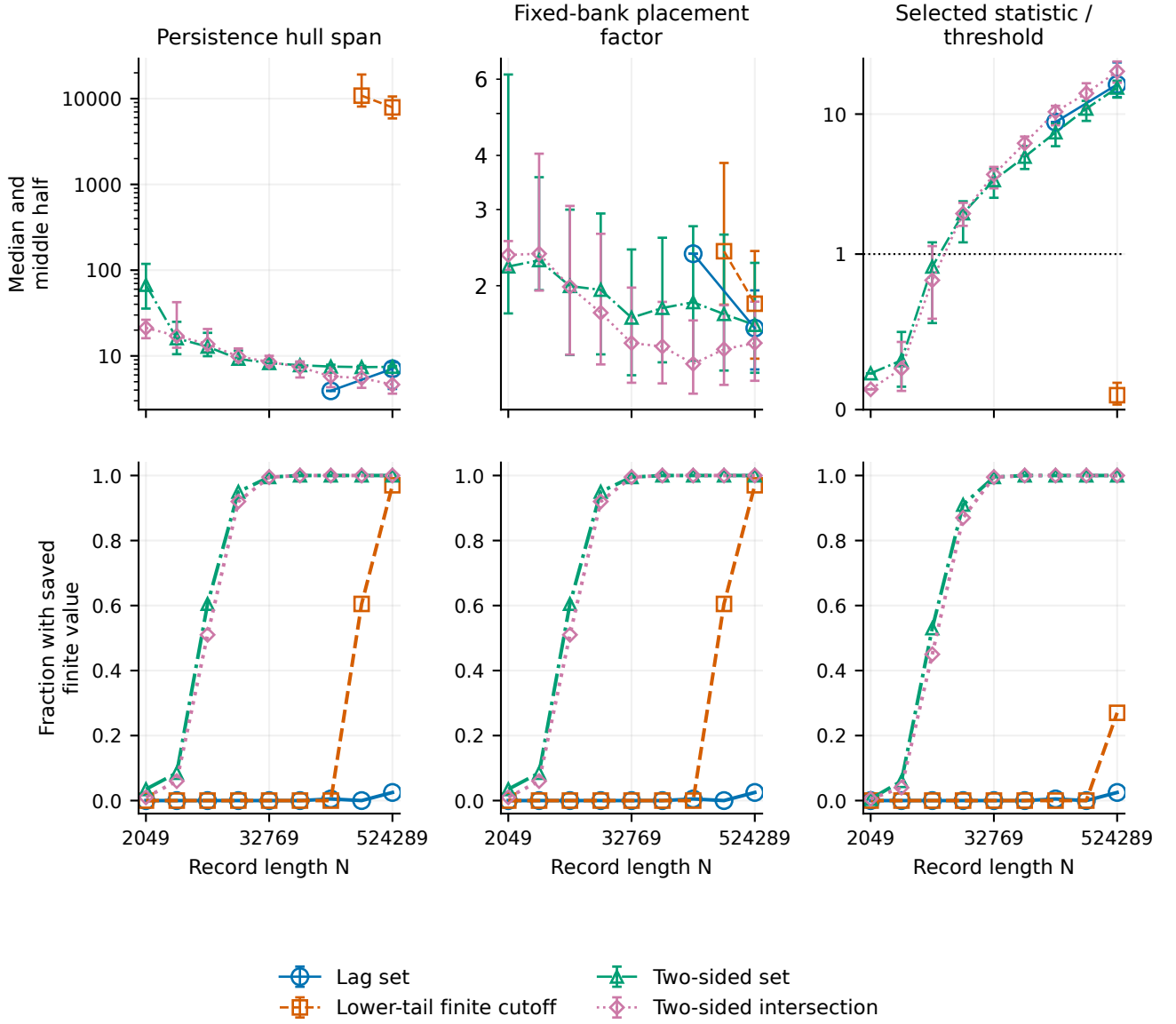


Figure S5: Finite source geometry and an available threshold occur on different record subsets. At $\phi = 0.995$, each length has 200 records. Upper panels show finite-value medians and interquartile ranges; lower panels show finite-value fractions. The first two upper axes are logarithmic; the third is linear through one and logarithmic above it. Missing values are omitted, not zero. Section S10.1 defines the summaries.

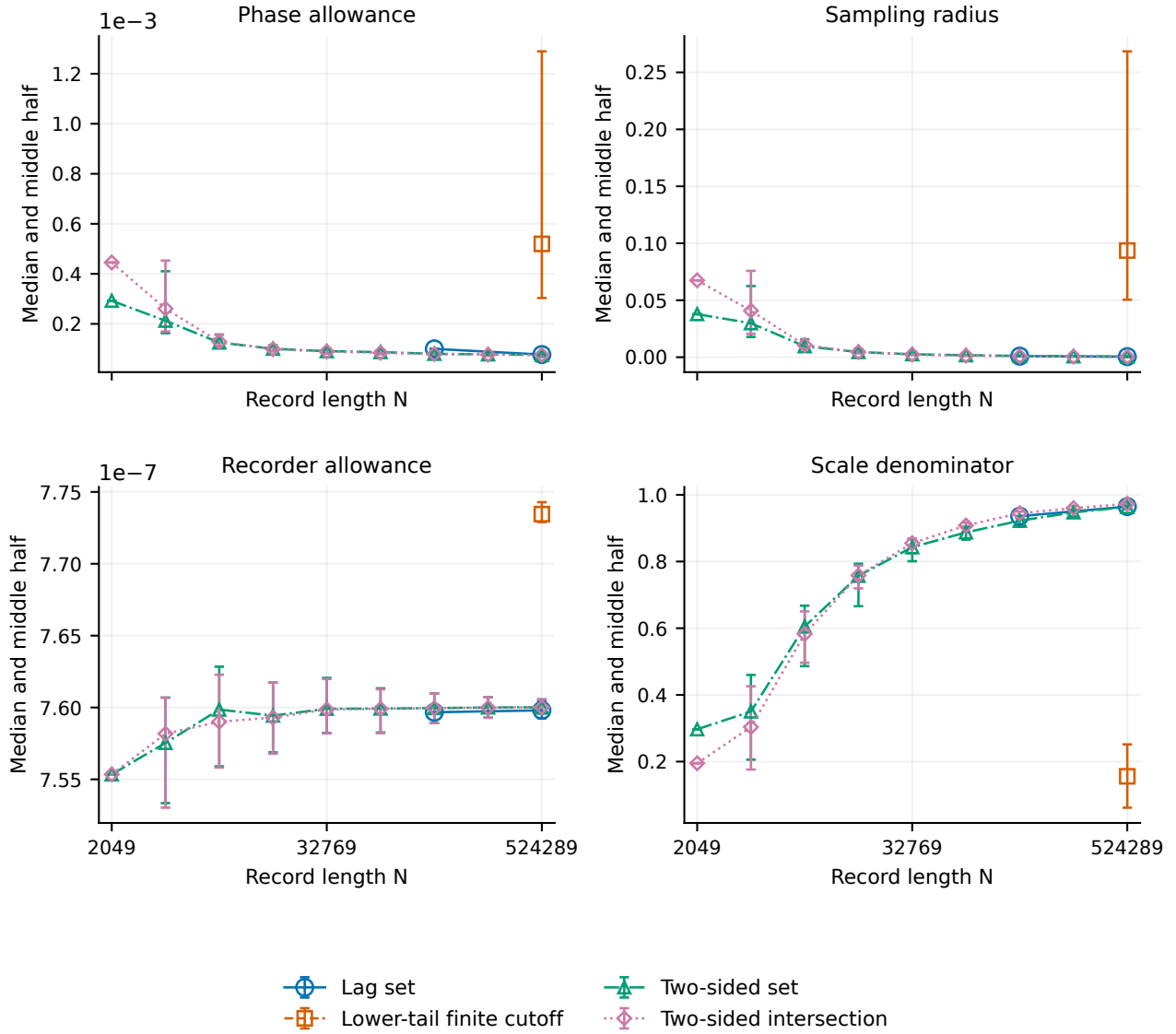


Figure S6: Median sampling allowances remain larger than median phase allowances in the selected two-sided members. At $\phi = 0.995$, each length has 200 records. Curves give medians and interquartile ranges of saved finite values for the member with largest statistic-minus-threshold margin. Allowances use upper endpoints and denominators use lower endpoints. These ranges are descriptive, not confidence intervals. Missing values are omitted; a median total need not equal summed medians.

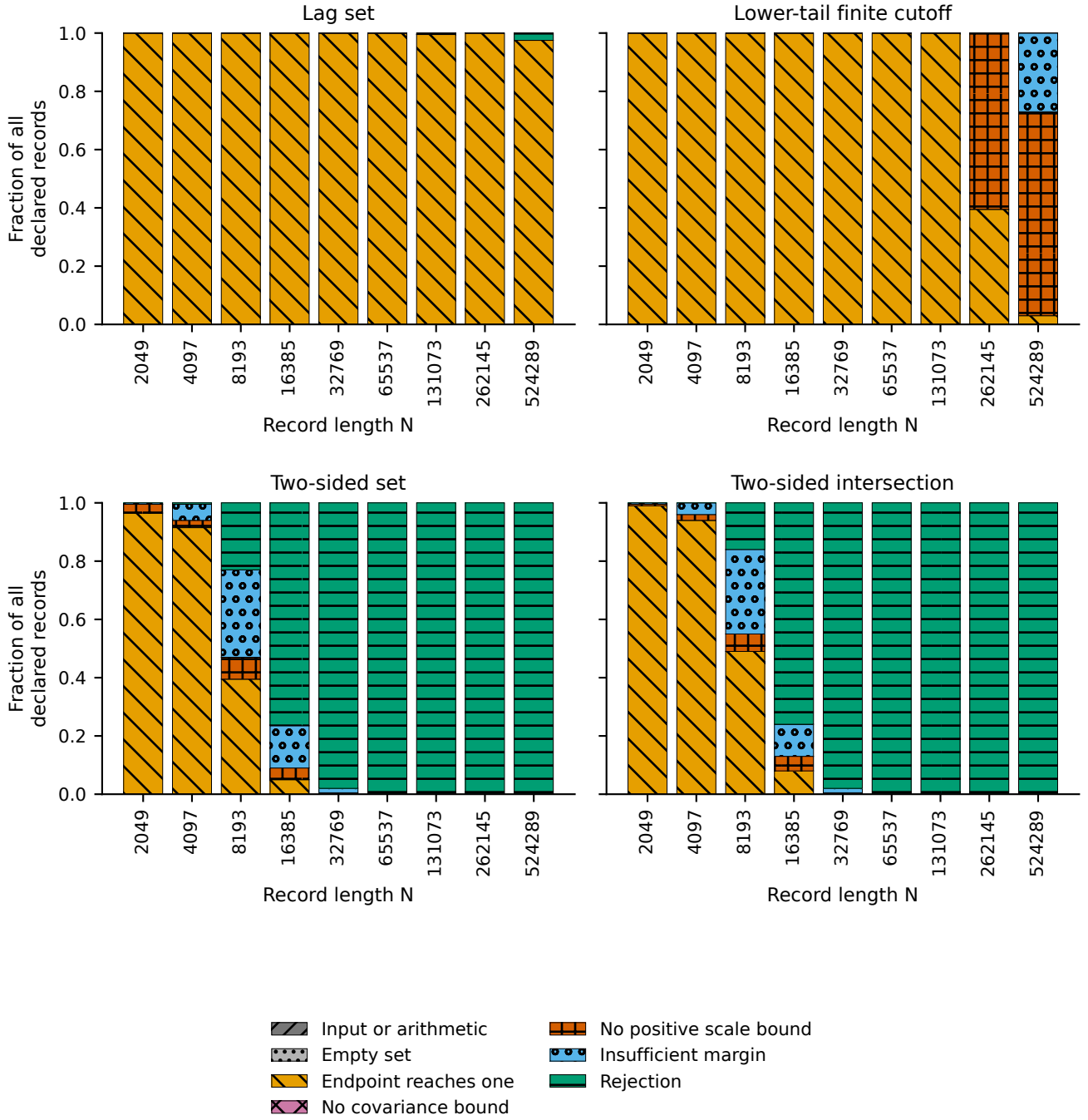


Figure S7: The first failed condition separates unavailable tests from available nonrejections. Each bar uses all 200 records at one strong-delay length, with $\phi = 0.995$. Seven mutually exclusive categories cover failures, nonrejections, and rejections, as defined in Sec. S10.4. A first failure does not imply that later requirements passed. Fractions are descriptive and carry no confidence intervals.

Source confidence: weak delay

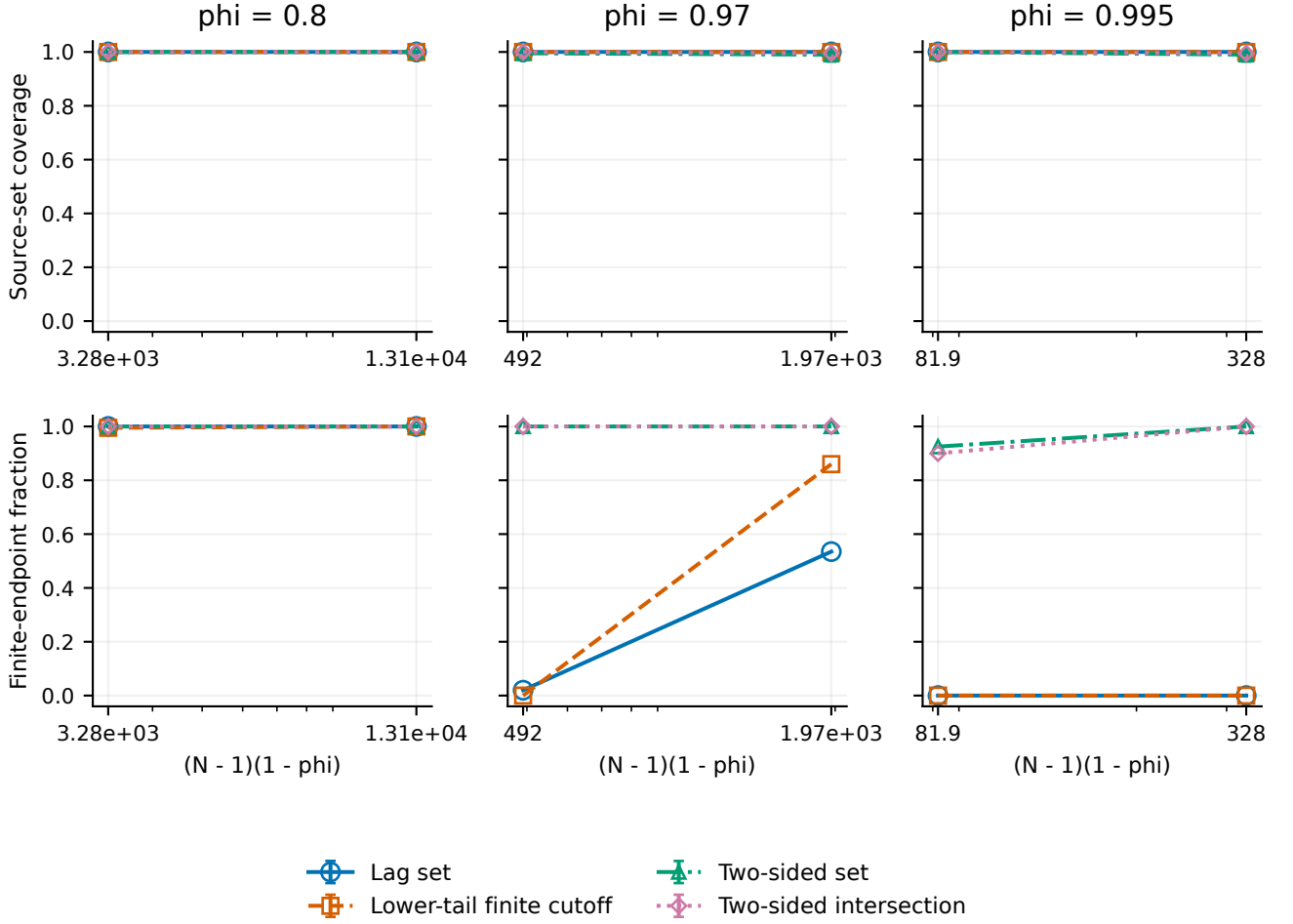


Figure S8: Weak-delay source coverage and endpoint availability can differ sharply. Each cell has 200 records. Coverage bars are pointwise 95% binomial intervals. Coverage and finite-endpoint fractions use the complete sample, while the confidence-set error allowance is 0.005. At $\phi = 0.995$, lag and lower-tail methods cover every source value but have no finite endpoint at either length. Coincident curves indicate equal fractions.

S11 Complete empirical tables and retained comparisons

The main comparisons are summarized in Secs. S9–S10. The complete per-condition tables remain in the accompanying package under `results/current/tables/`. Table S5 identifies the exact files. They retain full denominators, rejection/nonrejection/abstention counts, uncertainty intervals, and available-value counts. Paired differences use their separate companion files. No condition or adverse result is removed from those data.

Table S5: Complete empirical tables in the reproducibility package. File names are relative to `results/current/tables/`.

Comparison	Files
Covariance and bias inflation	<code>inflation_counts.csv</code> , <code>inflation_pairs.csv</code>
Transported gains	<code>gain_counts.csv</code> , <code>gain_pairs.csv</code>
Numerical settings and recorders	<code>settings_counts.csv</code> , <code>settings_pairs.csv</code>
Assumption violations	<code>violations_counts.csv</code>
Phase-boundary controls	<code>near_boundary_counts.csv</code>
All cycle-radius components	<code>cycle_components.csv</code>

S11.1 Decision patterns retained by the complete tables

The gain and inflation tables support different conclusions. Transporting a gain preserves the diagonal rule’s mathematical decision, while assigning one common scalar ceiling to every channel can inflate the smaller-channel edges. All six diagonal gain variants retain 200/200 rotating-source rejections; the common rule loses all rejections at four gains. Inflation of a valid bound remains conservative, but increasing its covariance factor from one to 1.5 removes all rotating-source detections throughout the tested bias grid. Zero detections under an underestimated bound, by contrast, cannot validate that bound.

The settings table retains recorder failures alongside successful numerical refinements. At $\phi = 0.97$, $N = 65537$, all tested subdivision depths and arithmetic precisions give two-sided counts 0/1000/0 under the null and 199/1/0 under strong delay. Fractional eight-bit recording changes the strong-delay counts to 0/65/135. Both finite-range recorders give 0/0/200, because a valid perturbation certificate is unavailable. These triplets mean rejection/nonrejection/abstention and retain every record.

The assumption-violation table concerns procedures used outside their corresponding guarantees. Reference noise and Student- t innovations produce no null rejections in these cells, but this observation does not extend Gaussian reference coverage. Correlated detector errors remove the source interpretation of the cycle. The phase-boundary table instead stays within a reversible-source model: calibrated and phase-adjusted procedures return 0/400/0, while raw procedures return 400/0/0 at each tested fraction of the phase tolerance. The raw procedures concern recorded asymmetry.

S11.2 Cycle component summaries

The complete component table contains 216 rows with full finite-value counts and zero missing values in the selected conditions. The largest median non-total term in every cell-method-edge group is the square-root sampling term. Table S4 retains representative diagonal-rule values in

Sec. S10.5. These are descriptive Decimal50 calculations from the formulas, not saved component enclosures. Their medians do not establish the cause of every decision, and a median total need not equal the sum of component medians.

S12 Earlier comparisons and the limits they identify

Earlier comparisons support two requirements of the present tests: control the whole source-confidence set, and preserve separate channel scales in cycle bounds. They use different designs and must not be combined with the 69-condition study. Complete results, including unsuccessful cases, remain in the full local reproducibility package. The public release includes the comparison tables under `earlier_studies/`. The Methods timing disclosure states which outcomes were available when later methods were developed.

S12.1 Finite source bounds can still prevent a decision

An earlier six-member-bank comparison used a noiseless AR(1) reference at persistence 0.8, 0.97, or 0.995. Each model cell had 32 records at nested lengths 16385 and 65537. A cumulative confidence method with only a lower-tail cutoff gave no rejections. It produced 274 finite upper endpoints but only 136 available tests across 576 outcomes. Its confidence hulls retained zero, leaving a large spectral range even after excluding persistence one. At $\phi = 0.97$, $N = 65537$, the median smallest bank source-ratio bound exceeded 4000 in each regime. Thus a finite bound need not give a positive variance denominator or a useful threshold.

Intersecting confidence sets did not uniformly improve detection. In the strong-delay cell at $\phi = 0.97$, $N = 65537$, two records rejected only with the intersection and three only with lag confidence. The paired 95% difference interval was $[-0.429, 0.374]$. At $\phi = 0.995$, all estimated methods abstained at both lengths, while the supplied-persistence control rejected every delayed alternative. All 18 population endpoint sufficient conditions failed. A separate comparison changed both the bank and calibration, so it could not isolate either change. All 144 sufficient power conditions were unavailable, and no record rejected only with estimated persistence. These results support controlling the full confidence hull without promising that an intersection always improves detection.

Complete outcomes and diagnostic quantities are in `earlier_studies/source/analysis/`, including `method_outcomes.csv`, `paired_outcomes.csv`, and `member_diagnostics.csv`. The earlier one-sided results are distinct from the later two-sided thresholds.

S12.2 Dependence and channel scale limit detection

In a supplied-source comparison, all 30 reversible-source rate cells had zero rejections. At $\phi = 0.5$ with identity responses, a supplied-marginal test rose from 2/400 to 400/400 detections between lengths 8192 and 32768; a floor-and-ceiling test detected none. At $\phi = 0.9$, the supplied-marginal test detected no alternatives at the tested lengths. Filtering with the true persistence gave 400/400 in every alternative cell. These differences concern the bounds supplied to that comparison.

An earlier cycle rule used one common radius for all channel pairs. Long rotating records with identity or unequal-delay detectors gave 64/64 detections. Scaling only channel 3 to one quarter while retaining the common ceiling gave 0/32; added noise gave 9/32, and a response zero gave 0/32. Short records gave no rejection. These results motivate separate channel bounds but do not evaluate the later common-family comparison. In a separate short-memory model, a certified population margin near 0.0332 at length 131072 implied at least 80% detection for the

ideal mathematical rule. The other ten model-length conditions failed the sufficient inequality. This result did not establish completion of the bounded implementation. Model definitions and outcome tables are supplied under `earlier_studies/mechanisms/study/`. Detailed per-record numerical files remain in the full local package.

S12.3 Planning bounds are conditional

Earlier planning calculations checked sufficient inequalities without generating recordings. Of 360 white-marginal designs, joint phase-and-shape calibration with a Frobenius-only radius gave 208 sufficient cases. Including a trace bound gave 221, while 37 positive-gap cases still had no passing length. In 360 autoregressive cases, joint calibration gave 163 sufficient cases; 192 lacked a positive gap and five more lacked a passing length. The largest analytic length was 2147483648; no such recording was generated. Complete designs and unsuccessful rows remain under `earlier_studies/base/analysis/`. These calculations give neither physical recording durations nor performance guarantees for the present sharper tests.

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